

CHAPTER 8

INFINITE SERIES

People who don't count won't count.

Anatole France
W. F. Osgood (1907)

PREVIEW

Is it possible for the sum of infinitely many nonzero numbers to be finite? This concept, which may seem paradoxical at first, plays a central role in mathematics and has a variety of important applications. The goal of this chapter is to examine the theory and applications of infinite sums, which will be referred to as *infinite series*. Geometric series, introduced in Section 8.2, are among the simplest infinite series we will encounter and, in some ways, the most important. In Sections 8.3–8.6, we will develop *convergence tests*, which provide ways of determining quickly whether or not certain infinite series have a finite sum. Next, we will turn our attention to series in which the individual terms are functions instead of numbers. We will be especially interested in the properties of *power series*, which may be thought of as polynomials of infinite degree, although some of their properties are quite different from those of polynomials. We will find that many common functions, such as e^x , $\ln(x+1)$, $\sin x$, $\cos x$, and $\tan^{-1}x$, can be represented as power series, and we will discuss some important theoretical and computational aspects of this kind of representation.

PERSPECTIVE

Series, or sums, arise in many different ways. For example, suppose it is known that a certain pollutant is released into the atmosphere at weekly intervals and is dissipated at the rate of 2% per week. If m grams of pollutant are released each week, then at the beginning of the first week, there will be $S_1 = m$ grams in the atmosphere, and at the beginning of the second, there will be $0.98m$ grams of “old” pollutant left plus m grams of “new” pollutant, to yield a total of $S_2 = m + 0.98m$ grams. Continuing, at the beginning of the n th week there will be $S_n = m + 0.98m + (0.98)^2m + \cdots + (0.98)^{n-1}m$ grams. It is natural to wonder how much pollutant will accumulate in the “long run” (as $n \rightarrow \infty$). But just exactly what do we mean by such a sum, and if the total is a finite number, how can we compute its value? We seek answers to these questions in this chapter.

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8.1 SEQUENCES AND THEIR LIMITS

IN THIS SECTION: Sequences; limits of sequences; bounded, monotonic sequences

Most phenomena we have considered occur continuously, but in practically every field of inquiry, there are situations that can be described by cataloging individual items in a numerical listing. In this section, we define a mathematical tool, called a *sequence*, in order to do this cataloging, and then we define what we mean by *the limit of a sequence*.



16-mm movie film

The making of a motion picture is a complex process, and editing all the film into a movie requires that all the frames of the action be labeled in chronological order. For example, R21-435 might signify the 435th frame of the 21st reel. A mathematician might refer to the movie editor's labeling procedure by saying the frames are arranged in a *sequence*.

Sequences

A sequence is a succession of numbers that are listed according to a given prescription or rule. Specifically, if n is a positive integer, the sequence whose n th term is the number a_n can be written as

$$a_1, a_2, \dots, a_n, \dots$$

or, more simply,

$$\{a_n\}$$

The number a_n is called the **general term** of the sequence. We will deal only with infinite sequences, so each term a_n has a **successor** a_{n+1} and for $n > 1$, a **predecessor** a_{n-1} . For example, by associating each positive integer n with its reciprocal $\frac{1}{n}$, we obtain the sequence denoted by $\{\frac{1}{n}\}$, which represents the succession of numbers $1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}, \dots$. The general term is denoted by $a_n = \frac{1}{n}$. The following examples illustrate the notation and terminology used in connection with sequences.

Example 1 Given the general term, find particular terms of a sequence

Find the 1st, 2nd, and 15th terms of the sequence $\{a_n\}$, where the general term is

$$a_n = \left(\frac{1}{2}\right)^{n-1}$$

Solution If $n = 1$, then $a_1 = \left(\frac{1}{2}\right)^{1-1} = \left(\frac{1}{2}\right)^0 = 1$. Similarly,

$$\begin{aligned} a_2 &= \left(\frac{1}{2}\right)^{2-1} = \frac{1}{2} \\ a_{15} &= \left(\frac{1}{2}\right)^{15-1} = \left(\frac{1}{2}\right)^{14} = 2^{-14} \end{aligned}$$

The reverse question, that of finding a general term given certain terms of a sequence, is a more difficult task, and even if we find a general term, we have no assurance that the general term is unique. For example, consider the sequence

$$2, 4, 6, 8, \dots$$

This seems to have a general term $a_n = 2n$. However, the general term

$$a_n = (n-1)(n-2)(n-3)(n-4) + 2n$$

has the same first four terms, but $a_5 = 34$ (not 10, as we would expect from the sequence 2, 4, 6, 8).

It is sometimes useful to start a sequence with a_0 instead of a_1 ; that is, to have a sequence of the form

$$a_0, a_1, a_2, \dots$$

So far, we have been discussing the concept of a sequence informally, without a definition. We have observed that a sequence $\{a_n\}$ associates the number a_n with the positive (or possibly, nonnegative) integer n . Hence, a sequence is really a special kind of function, one whose domain is a set of positive (or possibly, nonnegative) integers.

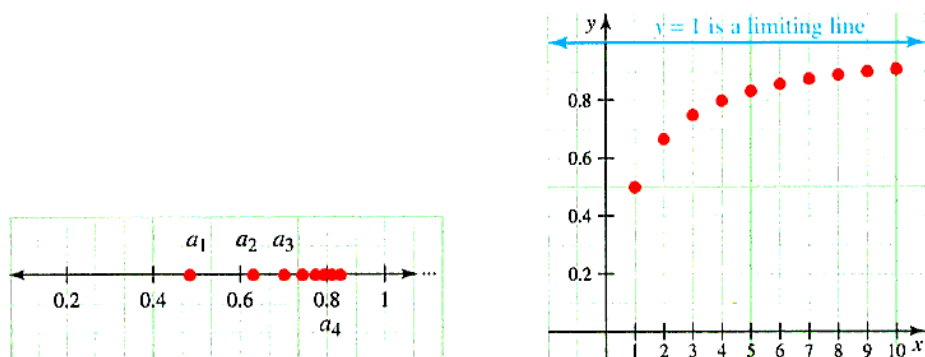
SEQUENCE A sequence $\{a_n\}$ is a function whose domain is a set of nonnegative integers and whose range is a subset of the real numbers. The functional values $a_1, a_2, a_3, \dots$ are called the **terms** of the sequence, and a_n is called the **n th term**, or **general term**, of the sequence.

Limits of Sequences

It is often desirable to examine the behavior of a given sequence $\{a_n\}$ as n gets arbitrarily large. For example, consider the sequence

$$a_n = \frac{n}{n+1} \quad \text{Even though we write } a_n, \text{ remember this is a function, } f(n) = \frac{n}{n+1}, \text{ where the domain is the set of positive integers.}$$

Because $a_1 = \frac{1}{2}$, $a_2 = \frac{2}{3}$, $a_3 = \frac{3}{4}$, $\dots$, we can plot the terms of this sequence on a number line, as shown in Figure 8.1a, or the sequence can be plotted in two dimensions, as shown in Figure 8.1b.



a. Graphing a sequence in one dimension

b. Graphing a sequence in two dimensions

Figure 8.1 Graphing the sequence $a_n = \frac{n}{n+1}$

By looking at either graph in Figure 8.1, we see that it appears the terms of the sequence are approaching 1. In general, if the terms of the sequence approach the number L as n increases without bound, we say that the sequence *converges to the limit* L and write

$$L = \lim_{n \rightarrow \infty} a_n$$

By looking at Figure 8.1, we would expect

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{n}{n+1} = 1$$

This limiting behavior is analogous to the continuous case (discussed in Section 4.4), and may be defined formally as follows.

CONVERGENT SEQUENCE The sequence $\{a_n\}$ **converges** to the number L , and we write

$$L = \lim_{n \rightarrow \infty} a_n$$

if for every $\epsilon > 0$, there is an integer N such that

$$|a_n - L| < \epsilon \quad \text{whenever} \quad n > N$$

Otherwise, the sequence **diverges**.

■ **What this says:** The notation $L = \lim_{n \rightarrow \infty} a_n$ means that eventually the terms of the sequence $\{a_n\}$ can be made as close to L as may be desired by taking n sufficiently large.

A geometric interpretation of this definition is shown in Figure 8.2.

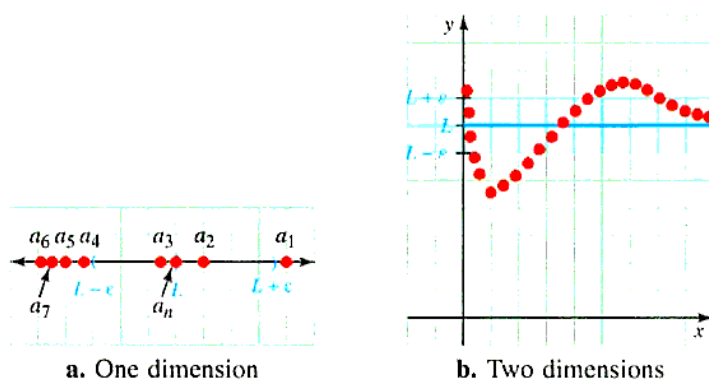


Figure 8.2 Geometric interpretation of a convergent sequence

Note that the numbers a_n may be practically anywhere at first (that is, for “small” n), but eventually, the a_n must “cluster” near the limiting value L .

The theorem on limits of functions carries over to sequences. We have the following useful result.

Theorem 8.1 Limit theorem for sequences

If $\lim_{n \rightarrow \infty} a_n = L$ and $\lim_{n \rightarrow \infty} b_n = M$, then

Linearity rule for sequences $\lim_{n \rightarrow \infty} (ra_n + sb_n) = rL + sM$

Product rule for sequences $\lim_{n \rightarrow \infty} (a_n b_n) = LM$

Quotient rule for sequences $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{L}{M}$ provided $M \neq 0$

Root rule for sequences $\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \sqrt[n]{L}$ provided $\sqrt[n]{a_n}$ is defined for all n and $\sqrt[n]{L}$ exists.

Proof: The proof of these rules is similar to the proof of the limit rules that were stated in Section 4.4. ♦

Example 2 Convergent sequences

Find the limit of each of these convergent sequences:

a. $\left\{ \frac{100}{n} \right\}$ b. $\left\{ \frac{2n^2 + 5n - 7}{n^3} \right\}$ c. $\left\{ \frac{3n^4 + n - 1}{5n^4 + 2n^2 + 1} \right\}$

Solution

- a. As n grows arbitrarily large, $\frac{100}{n}$ gets smaller and smaller. Thus,

$$\lim_{n \rightarrow \infty} \frac{100}{n} = 0$$

A graphical representation is shown in Figure 8.3.

- b. We cannot use the quotient rule of Theorem 8.1 because neither the limit in the numerator nor the one in the denominator exists. However,

$$\frac{2n^2 + 5n - 7}{n^3} = \frac{2}{n} + \frac{5}{n^2} - \frac{7}{n^3}$$

and by using the linearity rule, we find that

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{2n^2 + 5n - 7}{n^3} &= 2 \lim_{n \rightarrow \infty} \frac{1}{n} + 5 \lim_{n \rightarrow \infty} \frac{1}{n^2} - 7 \lim_{n \rightarrow \infty} \frac{1}{n^3} \\ &= 2 \cdot 0 + 5 \cdot 0 - 7 \cdot 0 \\ &= 0 \end{aligned}$$

A graph is shown in Figure 8.4.

- c. Divide the numerator and denominator by n^4 , the highest power of n that occurs in the expression, to obtain

$$\lim_{n \rightarrow \infty} \frac{3n^4 + n - 1}{5n^4 + 2n^2 + 1} = \lim_{n \rightarrow \infty} \frac{3 + \frac{1}{n^3} - \frac{1}{n^4}}{5 + \frac{2}{n^2} + \frac{1}{n^4}} = \frac{3}{5}$$

Note that this procedure is very similar to the one we used in Section 4.4 when we evaluated limits of functions to infinity.

A graph of this sequence is shown in Figure 8.5.

Example 3 Divergent sequences

Show that the following sequences diverge:

a. $\{(-1)^n\}$ b. $\left\{ \frac{n^5 + n^3 + 2}{7n^4 + n^2 + 3} \right\}$

Solution

- a. The sequence defined by $\{(-1)^n\}$ is $-1, 1, -1, 1, \dots$ and this sequence diverges by oscillation because the n th term is always either 1 or -1 . Thus a_n cannot approach one specific number L as n grows large.

b.
$$\lim_{n \rightarrow \infty} \frac{n^5 + n^3 + 2}{7n^4 + n^2 + 3} = \lim_{n \rightarrow \infty} \frac{1 + \frac{1}{n^2} + \frac{2}{n^5}}{\frac{7}{n} + \frac{1}{n^3} + \frac{3}{n^5}}$$

The numerator tends toward 1 as $n \rightarrow \infty$, and the denominator approaches 0. Hence the quotient increases without bound, and the sequence must diverge.

If $\lim_{n \rightarrow \infty} a_n$ does not exist because the numbers a_n become arbitrarily large as $n \rightarrow \infty$, we write $\lim_{n \rightarrow \infty} a_n = \infty$. We summarize this more precisely in the following box.

LIMIT NOTATION

$\lim_{n \rightarrow \infty} a_n = \infty$ means that for any real number A , we have $a_n > A$ for all sufficiently large n .

$\lim_{n \rightarrow \infty} b_n = -\infty$ means that for any real number B , we have $b_n < B$ for all sufficiently large n .

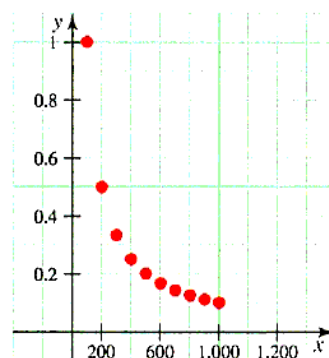


Figure 8.3 Graphical representation of $a_n = \frac{100}{n}$

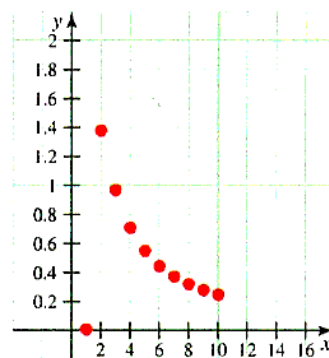


Figure 8.4 Graph of $a_n = \frac{2n^2 + 5n - 7}{n^3}$

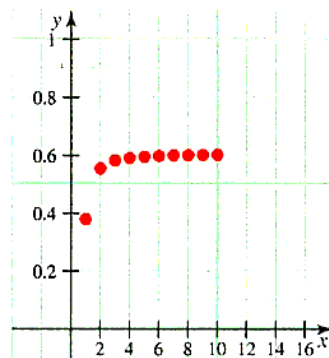


Figure 8.5 Graph of $a_n = \frac{3n^4 + n - 1}{5n^4 + 2n^2 + 1}$

Rewriting the answer to Example 3b in this notation,

$$\lim_{n \rightarrow \infty} \frac{n^5 + n^3 + 2}{7n^4 + n^2 + 3} = \infty$$

Example 4 Determining the convergence or divergence of a sequence

Determine the convergence or divergence of the sequence $\{\sqrt{n^2 + 3n} - n\}$.

Solution It would not be correct to apply the linearity property for sequences (because neither $\lim_{n \rightarrow \infty} \sqrt{n^2 + 3n}$ nor $\lim_{n \rightarrow \infty} n$ exists). It is also not correct to use this as a reason to say that the limit does not exist since infinity minus infinity is indeterminate. You might try some values of n (shown in Figure 8.6) to guess that there is some limit.

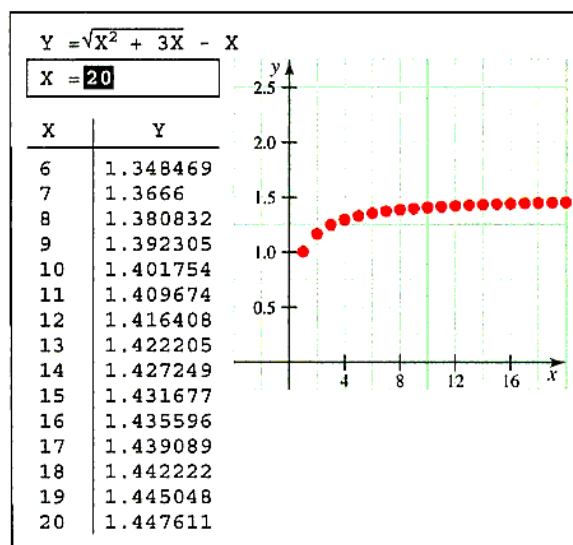


Figure 8.6 Interactive Is this the graph of a convergent or divergent series?

To find the limit, however, we will rewrite the general term algebraically as follows:

$$\begin{aligned}
 \sqrt{n^2 + 3n} - n &= (\sqrt{n^2 + 3n} - n) \frac{\sqrt{n^2 + 3n} + n}{\sqrt{n^2 + 3n} + n} \\
 &= \frac{n^2 + 3n - n^2}{\sqrt{n^2 + 3n} + n} \\
 &= \frac{n^2 + 3n - n^2}{\sqrt{n^2 + 3n} + n} \cdot \frac{\frac{1}{n}}{\frac{1}{n}} \\
 &= \frac{3}{\sqrt{n^2 + 3n} + n} \\
 &= \frac{3}{\frac{\sqrt{n^2 + 3n}}{n} + 1} \\
 &= \frac{3}{\sqrt{1 + \frac{3}{n}} + 1}
 \end{aligned}$$

Hence,

$$\lim_{n \rightarrow \infty} (\sqrt{n^2 + 3n} - n) = \lim_{n \rightarrow \infty} \frac{3}{\sqrt{1 + \frac{3}{n}} + 1} = \frac{3}{2}$$

Note the graph of the sequence in Example 4 in Figure 8.6. The graph of a sequence consists of a succession of isolated points. This can be compared with the graph of $y = \sqrt{x^2 + 3x} - x$, $x \geq 1$, which is a continuous curve (shown in Figure 8.7).

The only difference between $\lim_{n \rightarrow \infty} a_n = L$ and $\lim_{x \rightarrow \infty} f(x) = L$ is that n is required to be an integer. This is stated in the hypothesis of the following theorem.

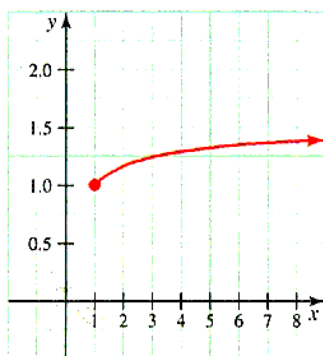


Figure 8.7 Graph of the function $f(x) = \sqrt{x^2 + 3x} - x$, $x \geq 1$

Theorem 8.2 Limit of a sequence from the limit of a continuous function

Given the sequence $\{a_n\}$, let f be a continuous function such that $a_n = f(n)$ for $n = 1, 2, \dots$. If $\lim_{x \rightarrow \infty} f(x)$ exists and $\lim_{x \rightarrow \infty} f(x) = L$, the sequence $\{a_n\}$ converges and $\lim_{n \rightarrow \infty} a_n = L$.

Proof: Let $\epsilon > 0$ be given. Because $\lim_{x \rightarrow \infty} f(x) = L$, there exists a number $N > 0$ such that

$$|f(x) - L| < \epsilon \text{ whenever } x > N$$

In particular, if $n > N$, it follows that $|f(n) - L| = |a_n - L| < \epsilon$. $\blacklozenge$

Be sure you read this theorem correctly. In particular, note that it *does not* say that if $\lim_{n \rightarrow \infty} a_n = L$, then $\lim_{x \rightarrow \infty} f(x) = L$ (see Figure 8.8b).

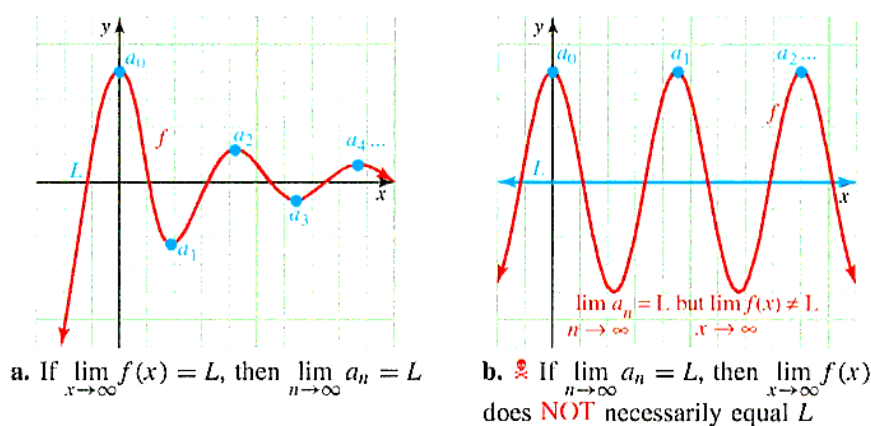


Figure 8.8 Comparison of the graphs of $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{n \rightarrow \infty} a_n$ where $f(n) = a_n$ for $n = 1, 2, \dots$

Example 5 Evaluating a limit using l'Hôpital's rule

Given that the sequence $\left\{ \frac{n^2}{1 - e^n} \right\}$ converges, evaluate $\lim_{n \rightarrow \infty} \frac{n^2}{1 - e^n}$.

Solution Let $L = \lim_{x \rightarrow \infty} f(x)$, where $f(x) = \frac{x^2}{1 - e^x}$. Because $f(n) = a_n$ for $n = 1, 2, \dots$, Theorem 8.2 tells us that $\lim_{n \rightarrow \infty} \frac{n^2}{1 - e^n}$ is the same as $\lim_{x \rightarrow \infty} f(x)$, provided this latter limit exists. Since the function f is continuous for all $x \neq 0$, we use l'Hôpital's rule.

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{x^2}{1 - e^x} &= \lim_{x \rightarrow \infty} \frac{2x}{-e^x} && \text{l'Hôpital's rule} \\ &= \lim_{x \rightarrow \infty} \frac{2}{-e^x} && \text{l'Hôpital's rule again} \\ &= 0 \end{aligned}$$

Thus, by Theorem 8.2, $\lim_{n \rightarrow \infty} \frac{n^2}{1 - e^n} = \lim_{x \rightarrow \infty} f(x) = L = 0$.

 The squeeze rule (Section 2.2) can be reformulated in terms of sequences.

Theorem 8.3 Squeeze theorem for sequences

If $a_n \leq b_n \leq c_n$ for all $n > N$, and $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$, then

$$\lim_{n \rightarrow \infty} b_n = L$$

Proof: The proof of this theorem follows the same steps as the proof of the squeeze rule given in Appendix A. $\blacklozenge$

Example 6 Using l'Hôpital's rule and the squeeze theorem

Show that the following sequences converge, and find their limits.

a. $\lim_{n \rightarrow \infty} n^{1/n}$ b. $\lim_{n \rightarrow \infty} \frac{n!}{n^n}$

Solution

- a. Let $L = \lim_{n \rightarrow \infty} n^{1/n}$; then we let $f(x) = x^{1/x}$. The function f is continuous for all $x > 0$, so we can apply l'Hôpital's rule.

$$\ln L = \lim_{n \rightarrow \infty} \frac{\ln n}{n} = \lim_{x \rightarrow \infty} \frac{\ln x}{x} = \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1} = 0 \quad \text{l'Hôpital's rule}$$

Thus, $L = e^0 = 1$.

- b. We cannot use l'Hôpital's rule because $x!$ is not defined as an elementary function when x is not an integer. Instead, we use the squeeze theorem for sequences. Accordingly, note that if we let $a_n = 0$ and $c_n = \frac{1}{n}$, for all n , we have

$$\underbrace{0}_{a_n} \leq \underbrace{\frac{n!}{n^n}}_{b_n} \leq \underbrace{\frac{1}{n}}_{c_n}$$

The right inequality is true because

$$\begin{aligned} \frac{n!}{n^n} &= \frac{n(n-1)(n-2) \cdots 1}{n \cdot n \cdot n \cdots n} \\ &= \left(\frac{n}{n}\right) \left(\frac{n-1}{n}\right) \left(\frac{n-2}{n}\right) \cdots \left(\frac{1}{n}\right) \\ &< (1) \left(\frac{n}{n}\right) \left(\frac{n}{n}\right) \cdots \left(\frac{n}{n}\right) \left(\frac{1}{n}\right) \\ &= \frac{1}{n} \end{aligned}$$

Because $\lim_{n \rightarrow \infty} a_n = 0$ and $\lim_{n \rightarrow \infty} c_n = 0$, it follows from the squeeze theorem that $\lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0$. $\blacksquare$

Bounded, Monotonic Sequences

We introduce, along with a simple example, some terminology associated with sequences $\{a_n\}$.

Name	Condition	Example
strictly increasing	$a_1 < a_2 < \cdots < a_{k-1} < a_k < \cdots$	$1, 2, 3, 4, 5, \dots$
increasing	$a_1 \leq a_2 \leq \cdots \leq a_{k-1} \leq a_k \leq \cdots$	$1, 1, 2, 2, 3, 3, \dots$
strictly decreasing	$a_1 > a_2 > \cdots > a_{k-1} > a_k > \cdots$	$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$
decreasing	$a_1 \geq a_2 \geq \cdots \geq a_{k-1} \geq a_k \geq \cdots$	$1, 1, \frac{1}{2}, \frac{1}{2}, \frac{1}{3}, \frac{1}{3}, \dots$
bounded above by M	$a_n \leq M$ for $1, 2, 3, \dots$	$M = 1$ for $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$ Note $M = 2, M = 3$ are other choices.
bounded below by m	$m \leq a_n$ for $1, 2, 3, \dots$	$m = 1$ for $1, 2, 3, 4, \dots$ Note $m = 0, m = -1$ are other choices.
bounded	It is bounded both above and below.	$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$ $m = 0, M = 1$, so sequence is bounded.

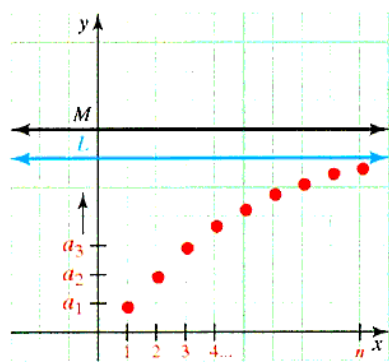
We also say that a series is **monotonic** if it is increasing or decreasing or **strictly monotonic** if it is strictly increasing or strictly decreasing.

In general, it is difficult to tell whether a given sequence converges or diverges, but because of the following theorem, it is easy to make this determination if we know the sequence is monotonic.

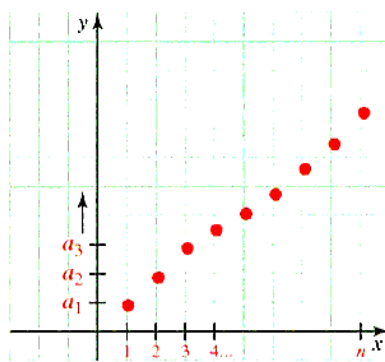
Theorem 8.4 BMCT: The bounded, monotonic, convergence theorem

A monotonic sequence $\{a_n\}$ converges if it is bounded and diverges otherwise.

Proof: A formal proof of the BMCT is outlined in Problems 55–56. For the following informal argument, we will assume that $\{a_n\}$ is an increasing sequence. You might wish to see whether you can give a similar informal argument for the decreasing case. Because the terms of the sequence satisfy $a_1 \leq a_2 \leq \cdots \leq a_{k-1} \leq a_k \leq \cdots$, we know that the sequence is bounded from below by a_1 and that the graph of the corresponding points (n, a_n) will be rising in the plane. Two possible cases are shown in Figure 8.9.



a. If $a_n < M$ for $n = 1, 2, \dots$, the graph of the points (n, a_n) will approach a horizontal “barrier” line $y = L$



b. If $\{a_n\}$ is not bounded from above, the graph rises indefinitely

Figure 8.9 Graphical possibilities for the BMCT

Suppose the sequence $\{a_n\}$ is bounded from above by a number M , so that $a_1 \leq a_n < M$ for $n = 1, 2, \dots$. Then the graph of the points (n, a_n) must continually rise (because the sequence is monotonic) and yet it must stay below the line $y = M$. The only way this can happen is for the graph to approach a “barrier” line $y = L$ (where $L \leq M$), and we have $\lim_{n \rightarrow \infty} a_n = L$, as shown in Figure 8.9a. However, if the sequence is not bounded from above, the graph will rise indefinitely (Figure 8.9b), and the terms in the sequence $\{a_n\}$ cannot approach any finite number L in the limit. ♦

Example 7 Convergence using the BMCT

Show that the sequence $\left\{ \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)} \right\}$ converges.

Solution The first few terms of this sequence are

$$a_1 = \frac{1}{2} \quad a_2 = \frac{1 \cdot 3}{2 \cdot 4} = \frac{3}{8} \quad a_3 = \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} = \frac{5}{16}$$

Because $\frac{1}{2} > \frac{3}{8} > \frac{5}{16}$, it appears that the sequence is decreasing (that is, it is monotonic). We can prove this by showing that $a_{n+1} < a_n$ for all $n > 0$, or equivalently, $\frac{a_{n+1}}{a_n} < 1$. (Note that $a_n \neq 0$ for all n .)

$$\begin{aligned}
 \frac{a_{n+1}}{a_n} &= \frac{\frac{1 \cdot 3 \cdot 5 \cdots [2(n+1)-1]}{2 \cdot 4 \cdot 6 \cdots [2(n+1)]}}{\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)}} \\
 &= \frac{1 \cdot 3 \cdot 5 \cdots [2(n+1)-1]}{2 \cdot 4 \cdot 6 \cdots [2(n+1)]} \cdot \frac{2 \cdot 4 \cdot 6 \cdots (2n)}{1 \cdot 3 \cdot 5 \cdots (2n-1)} \\
 &= \frac{2n+1}{2n+2} \\
 &< 1
 \end{aligned}$$

for any $n > 0$. Hence $a_{n+1} < a_n$ for all n , and $\{a_n\}$ is a decreasing sequence. Because $a_n > 0$ for all n , it follows that $\{a_n\}$ is bounded below by 0. Applying the BMCT, we see that $\{a_n\}$ converges, but the BMCT tells us nothing about the limit. ■

The BMCT also applies to sequences whose terms are *eventually* monotonic. That is, it can be shown that the sequence $\{a_n\}$ converges if it is bounded and there exists an integer N such that $\{a_n\}$ is monotonic for all $n > N$. This modified form of the BMCT is illustrated in the following example.

Example 8 Convergence of a sequence that is eventually monotonic

Show that the sequence $\left\{ \frac{\ln n}{\sqrt{n}} \right\}$ converges.

Solution We will apply the BMCT. A few values are shown in Figure 8.10.

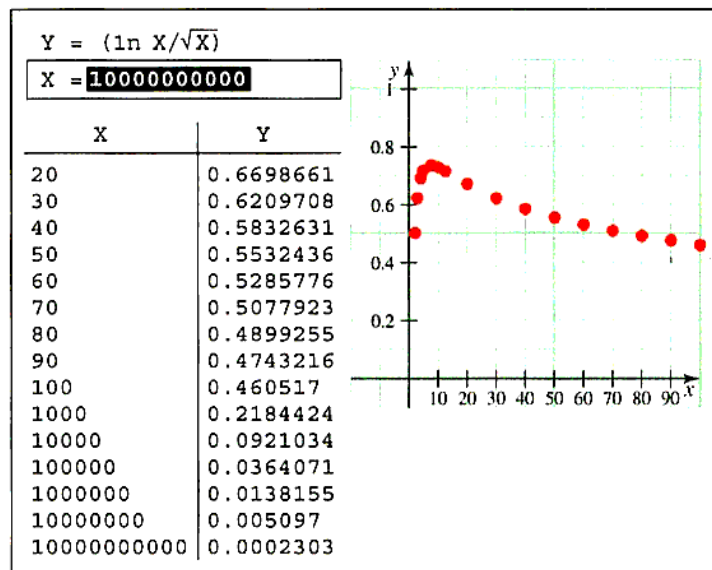


Figure 8.10 Interactive Sequence $a_n = \frac{\ln n}{\sqrt{n}}$

The succession of numbers suggests that the sequence increases at first and then gradually begins to decrease. To verify this behavior, we let

$$f(x) = \frac{\ln x}{\sqrt{x}}$$

and find that

$$f'(x) = \frac{\sqrt{x} \left(\frac{1}{x} \right) - (\ln x) \left(\frac{1}{2} x^{-1/2} \right)}{x} = \frac{2\sqrt{x} - \sqrt{x} \ln x}{2x^2}$$

Find the critical values:

$$\begin{aligned}\frac{2\sqrt{x} - \sqrt{x} \ln x}{2x^2} &= 0 \\ 2\sqrt{x} &= \sqrt{x} \ln x \\ \ln x &= 2 \\ x &= e^2\end{aligned}$$

Thus, $x = e^2$ is the only critical value, and you can show that $f'(x) < 0$ for $x > e^2$ since $2 - \ln x < 0$ for $x > e^2$. This means that f is a decreasing function for $x > e^2$. Thus, the sequence $\left\{\frac{\ln n}{\sqrt{n}}\right\}$ must be decreasing for $n > 8$ (because e^2 is between 7 and 8).

We see that the sequence $\left\{\frac{\ln n}{\sqrt{n}}\right\}$ is bounded from below because

$$0 < \left\{\frac{\ln n}{\sqrt{n}}\right\} \quad \text{for all } n \geq 2$$

Therefore, the given sequence is bounded from below and is eventually decreasing, so it must converge. ■

The BMCT is an extremely valuable theoretical tool. For example, in Section 2.4, we defined the number e by the limit

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$$

but to do so we assumed that this limit exists. We can now show this assumption is warranted, because it turns out that the sequence $\left\{\left(1 + \frac{1}{n}\right)^n\right\}$ is increasing and bounded from above by 3 (see Problems 57–59). Thus, the BMCT assures us that the sequence converges, and this in turn guarantees the existence of the limit. We end this section with a result that will be useful in our subsequent work. The proof makes use of the formal definition of BMCT.

Theorem 8.5 Convergence of a power sequence

If r is a fixed number such that $|r| < 1$, then $\lim_{n \rightarrow \infty} r^n = 0$.

Proof: The case where $r = 0$ is trivial. We will prove the theorem for the case $0 < r < 1$ and leave the case $-1 < r < 0$ as an exercise (Problem 60).

$$0 \leq r^{n+1} = r^n r < r^n$$

The sequence $\{r^n\}$ is monotonic and bounded, so the limit exists. Either $\lim_{n \rightarrow \infty} r^n = 0$ or $\lim_{n \rightarrow \infty} r^n = L > 0$. Suppose

$$\begin{aligned}\lim_{n \rightarrow \infty} r^n &= L \\ r \lim_{n \rightarrow \infty} r^n &= rL \quad \text{Multiply both sides by } r. \\ \lim_{n \rightarrow \infty} r r^n &= rL \quad \text{Properties of limits} \\ \lim_{n \rightarrow \infty} r^{n+1} &= rL \quad \text{Properties of exponents} \\ &< L \quad \text{Since } 0 < r < 1 \text{ and } L > 0\end{aligned}$$

But $\lim_{n \rightarrow \infty} r^{n+1} = \lim_{n \rightarrow \infty} r^n = L$ by the definition of limit. Hence $L = rL < L$ which is impossible, so $\lim_{n \rightarrow \infty} r^n = 0$.

PROBLEM SET 8.1

Level 1

1. ■ **What does this say?** What do we mean by the limit of a sequence?
2. ■ **What does this say?** What is meant by the bounded, monotonic, convergence theorem?

Write out the first five terms (beginning with $n = 1$) of the sequences given in Problems 3-12.

3. $\{1 + (-1)^n\}$
4. $\left\{\left(\frac{-1}{2}\right)^{n+2}\right\}$
5. $\left\{\frac{\cos 2n\pi}{n}\right\}$
6. $\left\{n \sin \frac{n\pi}{2}\right\}$
7. $\left\{\frac{3n+1}{n+2}\right\}$
8. $\left\{\frac{n^2-n}{n^2+n}\right\}$
9. $\{a_n\}$ where $a_1 = 256$ and $a_n = \sqrt{a_{n-1}}$ for $n \geq 2$
10. $\{a_n\}$ where $a_1 = 512$ and $a_n = \sqrt[3]{a_{n-1}}$ for $n \geq 2$
11. $\{a_n\}$ where $a_1 = 1$ and $a_n = (a_{n-1})^2 + a_{n-1} + 1$ for $n \geq 2$
12. $\{a_n\}$ where $a_1 = -1$ and $a_n = na_{n-1}$ for $n \geq 2$.

Compute the limit of the convergent sequences in Problems 13-36.

13. $\left\{\frac{5n}{n+7}\right\}$
14. $\left\{\frac{5n+8}{n}\right\}$
15. $\left\{\frac{4-7n}{8+n}\right\}$
16. $\left\{\frac{2n+1}{3n-4}\right\}$
17. $\left\{\frac{100n+7,000}{n^2-n-1}\right\}$
18. $\left\{\frac{8n^2+800n+5,000}{2n^2-1,000n+2}\right\}$
19. $\left\{\frac{n^3-6n^2+85}{2n^3-5n+170}\right\}$
20. $\left\{\frac{8n^2+6n+4,000}{n^3+1}\right\}$
21. $\left\{\frac{8n-500\sqrt{n}}{2n+800\sqrt{n}}\right\}$
22. $\left\{\frac{2n}{n+7\sqrt{n}}\right\}$
23. $\left\{\frac{\ln n}{n^2}\right\}$
24. $\left\{\frac{3\sqrt{n}}{5\sqrt{n}+\sqrt[4]{n}}\right\}$
25. $\{n^{3/n}\}$
26. $\{2^{5/n}\}$
27. $\{(n+4)^{1/n}\}$
28. $\left\{\left(1+\frac{3}{n}\right)^n\right\}$
29. $\{(\ln n)^{1/n}\}$
30. $\{n^{1/(n+2)}\}$
31. $\{\sqrt{n^2+n}-n\}$
32. $\{\int_0^\infty e^{-nx} dx\}$
33. $\{\sqrt[n]{n}\}$
34. $\{\sqrt{n+5\sqrt{n}}-\sqrt{n}\}$

35. $\{(an+b)^{1/n}\}$ a and b positive constants
36. $\{\ln n - \ln(n+1)\}$

Level 2

Show that each sequence given in Problems 37-42 converges either by showing it is increasing with an upper bound or decreasing with a lower bound.

37. $\left\{\ln\left(\frac{n+1}{n}\right)\right\}$
38. $\left\{\frac{n}{2^n}\right\}$
39. $\left\{\frac{4n+5}{n}\right\}$
40. $\left\{\frac{3n-2}{n}\right\}$
41. $\{\sqrt[n]{n}\}$
42. $\left\{\frac{3n-7}{2^n}\right\}$

Explain why each sequence in Problems 43-46 diverges.

43. $\{\cos n\pi\}$
44. $\{1 + (-1)^n\}$
45. $\{\sqrt[n]{n}\}$
46. $\left\{\frac{n^3-7n+5}{100n^2+219}\right\}$
47. a. Does the sequence $\{\frac{1}{n}(\sin n)\}$ converge, as $n \rightarrow \infty$? Give reasons.
b. How about the sequence $\{n(\sin(\frac{1}{n}))\}$?
c. Explain the relationship between the two sequences in this problem. *Hint:* Let $m = \frac{1}{n}$ in part a.
48. **Modeling Problem** A drug is administered into the body. At the end of each hour, the amount of drug present is half what it was at the end of the previous hour. What percent of the drug is present at the end of 4 hours? At the end of n hours?
49. Does the sequence $\left\{\sum_{k=1}^n \frac{n}{n^2+k^2}\right\}$ converge? If so, to what? *Hint:* The sequence is related to an integral from 0 to 1, where k/n can be thought of as x_n .

50. Historical Quest

Leonardo de Pisa, also known as Fibonacci, was one of the best mathematicians of the Middle Ages. He played an important role in reviving ancient mathematics and introduced the Hindu-Arabic place-value decimal system to Europe. His book, *Liber Abaci*, published in 1202, introduced the Arabic numerals, as well as the famous rabbit problem, for which Fibonacci is best remembered today.



Fibonacci
(1170-1250)

To describe Fibonacci's rabbit problem, we consider a sequence whose n th term is defined by a *recursion formula* — that is, a formula in which the n th term is given in terms of previous terms in the sequence. Suppose rabbits breed in such a way that each pair of adult rabbits produces a pair of baby rabbits each month.

Number of Months	Number of Pairs	Pairs of Rabbits (the pairs shown in color are ready to reproduce in the next month)
Start	1	
1	1	
2	2	
3	3	
4	5	
5	8	
⋮	⋮	⋮

Same pair (rabbits never die)

The first month after birth, the rabbits are adolescents and produce no offspring. However, beginning with the second month the rabbits are adults, and each pair produces a pair of offspring every month. The sequence of numbers describing the number of rabbits is called the *Fibonacci sequence*, and it has applications in many areas, including biology and botany.

In this **Quest** you are to examine some properties of the Fibonacci sequence. Let a_n denote the number of pairs of rabbits in the "colony" at the end of n months.

- a. Explain why $a_1 = 1, a_2 = 1, a_3 = 2, a_4 = 3$, and, in general,

$$a_{n+1} = a_n + a_{n-1}$$

for $n = 2, 3, 4, \dots$. The equation $a_{n+1} = a_n + a_{n-1}$ that specifies each term of the sequence $\{a_n\}$ for $n = 2, 3, \dots$ is called a *recursion formula* because it defines later terms from those that precede them.

- b. The *growth rate* of the colony during the $(n + 1)$ th month is

$$r_n = \frac{a_{n+1}}{a_n}$$

Compute r_n for $n = 1, 2, 3, \dots, 10$.

- c. Assume that the growth rate sequence $\{r_n\}$ defined in part b converges. Find

$$L = \lim_{n \rightarrow \infty} r_n$$

Hint: Use the recursion formula in part a.

Level 3

51. The claim is made that

$$\lim_{n \rightarrow \infty} \frac{n}{n+1} = 1$$

If $\epsilon = 0.01$ is chosen, find the smallest integer N so that

$$\left| \frac{n}{n+1} - 1 \right| < 0.01$$

if $n > N$.

52. The claim is made that

$$\lim_{n \rightarrow \infty} \frac{2n+1}{n-3} = 2$$

If $\epsilon = 0.01$ is chosen, find the smallest integer N so that

$$\left| \frac{2n+1}{n-3} - 2 \right| < 0.01$$

if $n > N$.

53. The claim is made that

$$\lim_{n \rightarrow \infty} \frac{n^2+1}{n^3} = 0$$

If $\epsilon = 0.001$ is chosen, find the smallest integer N so that

$$\left| \frac{n^2+1}{n^3} \right| < 0.001$$

if $n > N$.

54. The claim is made that

$$\lim_{n \rightarrow \infty} e^{-n} = 0$$

If $\epsilon = 0.001$ is chosen, find the smallest integer N so that

$$\frac{1}{e^n} < 0.001$$

if $n > N$.

In Problems 55–56, we prove the BMCT. Suppose there exist a number M and a positive integer N so that

$$a_n \leq a_{n+1} \leq M$$

for all $n > N$.

55. It is a fundamental property of the real numbers that a sequence with an upper bound must have a *least* upper bound; that is, there must be a number A with the property that $a_n \leq A$ for all n but if c is a number such that $c < A$, then $a_m > c$ for at least one m . Use this property to show that if $\epsilon > 0$ is given, there exists a positive integer N so that

$$A - \epsilon < a_n < A$$

for all $n > N$.

56. Show that $|a_n - A| < \epsilon$ for all $n > N$ and conclude that $\lim_{n \rightarrow \infty} a_n = A$.

Problems 57-59 can be used to prove the convergence of the sequence

$$\left\{ \left(1 + \frac{1}{n} \right)^n \right\}$$

57. Suppose n is a positive integer. Use the binomial theorem to show that

$$\begin{aligned} \left(1 + \frac{1}{n} \right)^n &= 1 + 1 + \frac{1}{2!} \left(1 - \frac{1}{n} \right) \\ &\quad + \frac{1}{3!} \left(1 - \frac{1}{n} \right) \left(1 - \frac{2}{n} \right) \\ &\quad + \cdots \\ &\quad + \frac{1}{n!} \left(1 - \frac{1}{n} \right) \left(1 - \frac{2}{n} \right) \cdots \left(1 - \frac{n-1}{n} \right) \end{aligned}$$

58. Use Problem 57 to show that

$$\left(1 + \frac{1}{n} \right)^n < 3$$

for all positive integers n .

59. Use Problem 57 to show that

$$\left\{ \left(1 + \frac{1}{n} \right)^n \right\}$$

is an increasing sequence. Then use the BMCT to show that this sequence converges.

60. **Proof of the convergence power theorem** In the text, we proved the case where $0 < r < 1$. Prove the remaining parts.

a. $r = 0$

b. $-1 < r < 0$

8.2 INTRODUCTION TO INFINITE SERIES; GEOMETRIC SERIES

IN THIS SECTION: *Definition of infinite series, general properties of infinite series, geometric series, applications of geometric series*

We define an infinite series (sum) as the limit of a special type of sequence. We then study a few basic properties of infinite series and examine *geometric series*, a particular kind of series with numerous applications.

Definition of Infinite Series

One way to add a list of numbers is to form subtotals until the end of the list is reached. Similarly, to give meaning to the infinite sum

$$S = a_1 + a_2 + a_3 + a_4 + \cdots$$

it is natural to examine the “partial sums”

$$a_1, \quad a_1 + a_2, \quad a_1 + a_2 + a_3, \quad a_1 + a_2 + a_3 + a_4, \quad \cdots$$

If it is possible to attach a numerical value to the infinite sum, we would expect the partial sums $S = a_1 + a_2 + a_3 + a_4 + \cdots + a_n$ to approach that value as n increases without bound. These ideas lead us to the following definition.

INFINITE SERIES An **infinite series** is an expression of the form

$$a_1 + a_2 + a_3 + \cdots = \sum_{k=1}^{\infty} a_k$$

and the **n th partial sum** of the series is

$$S_n = a_1 + a_2 + a_3 + a_4 + \cdots + a_n = \sum_{k=1}^n a_k$$

The *series* is said to **converge with sum S** if the *sequence* of partial sums $\{S_n\}$ converges to S . In this case, we write

$$\sum_{k=1}^{\infty} a_k = \lim_{n \rightarrow \infty} S_n = S$$

If the sequence $\{S_n\}$ does not converge, the series $\sum_{k=1}^{\infty} a_k$ **diverges** and has no sum.

■ **What this says** An infinite series converges if its sequence of partial sums converges and diverges otherwise. If it converges, its sum is defined to be the limit of the sequence of partial sums.

We will use the symbol $\sum_{k=1}^{\infty} a_k$ to denote the series $a_1 + a_2 + a_3 + \cdots$ regardless of whether this series converges or diverges. If the sequence of partial sums $\{S_n\}$ converges, then

$$\sum_{k=1}^{\infty} a_k = \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n a_k \right)$$

and the symbol $\sum_{k=1}^{\infty} a_k$ is used to represent *both* the series and its sum.

We will also consider certain series in which the starting point is not 1; for example, the series

$$\frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \cdots \text{ can be denoted by } \sum_{k=3}^{\infty} \frac{1}{k} \text{ or } \sum_{k=2}^{\infty} \frac{1}{k+1}$$

Example 1 Convergent series

Show that the series $\sum_{k=1}^{\infty} \frac{1}{2^k}$ converges.

Solution This series has the following partial sums.

$$\begin{array}{ll} S_1 = \frac{1}{2} & \text{Note: } 1 - \frac{1}{2^1} = \frac{1}{2} \\ S_2 = \frac{1}{2} + \frac{1}{4} = \frac{3}{4} & 1 - \frac{1}{2^2} = \frac{3}{4} \\ S_3 = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} = \frac{7}{8} & 1 - \frac{1}{2^3} = \frac{7}{8} \\ \vdots & \\ S_n = \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n} & \end{array}$$

The sequence of partial sums is $\frac{1}{2}, \frac{3}{4}, \frac{7}{8}, \frac{15}{16}, \frac{31}{32}, \frac{63}{64}, \frac{127}{128}, \dots$, and in general (by mathematical induction)

$$S_n = 1 - \frac{1}{2^n}$$

Because $\lim_{n \rightarrow \infty} (1 - \frac{1}{2^n}) = 1$, we conclude that the series converges and its sum is $S = 1$. 

Example 2 Divergent series

Show that the series $\sum_{k=1}^{\infty} (-1)^k$ diverges.

Solution The series can be **expanded** (written out) as

$$\sum_{k=1}^{\infty} (-1)^k = -1 + 1 - 1 + 1 - 1 + \dots$$

and we see that the n th partial sum is

$$S_n = \begin{cases} -1 & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$$

Because the sequence $\{S_n\}$ has no limit, the given series must diverge. 

A series is called a **telescoping series** (or collapsing series) if there is internal cancellation in the partial sums, as illustrated by the following example.

Example 3 A telescoping series

Show that the series $\sum_{k=1}^{\infty} \frac{1}{k^2 + k}$ converges and find its sum.

Solution Using partial fractions we find that

$$\frac{1}{k^2 + k} = \frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$$

Thus, the n th partial sum of the given series can be represented as follows:

$$\begin{aligned} S_n &= \sum_{k=1}^n \frac{1}{k^2 + k} \\ &= \sum_{k=1}^n \left[\frac{1}{k} - \frac{1}{k+1} \right] \\ &= \left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \dots + \left(\frac{1}{n} - \frac{1}{n+1} \right) \\ &= 1 + \left(-\frac{1}{2} + \frac{1}{2} \right) + \left(-\frac{1}{3} + \frac{1}{3} \right) + \dots + \left(-\frac{1}{n} + \frac{1}{n} \right) - \frac{1}{n+1} \\ &= 1 - \frac{1}{n+1} \end{aligned}$$

The limit of the sequence of partial sums is

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left[1 - \frac{1}{n+1} \right] = 1$$

so the series converges, with sum $S = 1$.

must diverge because even though $\sum_{k=1}^{\infty} \frac{1}{k^2 + k}$ converges (Example 3), the series $\sum_{k=1}^{\infty} (-1)^k$ diverges (Example 2).

Geometric Series

We are still very limited in the available techniques for determining whether a given series is convergent or divergent. In fact, the purpose of a great part of this chapter is to develop efficient techniques for making this determination. We begin this quest by considering an important special kind of series.

GEOMETRIC SERIES A **geometric series** is an infinite series in which the ratio of successive terms in the series is constant. If this constant ratio is r , then the series has the form

$$\sum_{k=0}^{\infty} ar^k = a + ar + ar^2 + ar^3 + \cdots + ar^n + \cdots \quad a \neq 0$$

For example, $3 + \frac{3}{2} + \frac{3}{4} + \frac{3}{8} + \cdots$ is a geometric series because each term is one-half the preceding term. The ratio of a geometric series may be positive or negative. For example,

$$\sum_{k=0}^{\infty} \frac{2}{(-3)^k} = 2 - \frac{2}{3} + \frac{2}{9} - \frac{2}{27} + \cdots$$

is a geometric series with $r = -\frac{1}{3}$. The following theorem tells us how to determine whether a given geometric series converges or diverges and, if it does converge, what its sum must be.

Theorem 8.8 Geometric series theorem

The geometric series $\sum_{k=0}^{\infty} ar^k$ with $a \neq 0$ diverges if $|r| \geq 1$ and converges if $|r| < 1$ with sum

$$\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}$$

Proof: Note that the n th partial sum of the geometric series is

$$S_n = a + ar + ar^2 + \cdots + ar^{n-1}$$

Definition of a geometric series

$$rS_n = ar + ar^2 + ar^3 + \cdots + ar^n$$

Multiply both sides by r .

$$rS_n - S_n = (ar + ar^2 + ar^3 + \cdots + ar^n) - (a + ar + ar^2 + \cdots + ar^{n-1})$$

Subtract.

$$(r-1)S_n = ar^n - a$$

Simplify.

$$S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{where } r \neq 1$$

If $|r| > 1$, the sequence of partial sums $\{S_n\}$ has no limit since r^n is unbounded in this case, so the geometric series must diverge. However, if $|r| < 1$, Theorem 8.5 tells

us that $r^n \rightarrow 0$ as $n \rightarrow \infty$, so we have

$$\sum_{k=0}^{\infty} ar^k = \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} a \left(\frac{r^n - 1}{r - 1} \right) = \frac{a(0 - 1)}{r - 1} = \frac{a}{1 - r}$$

To complete the proof, it must be shown that the geometric series diverges when $|r| = 1$, and we leave this final step as a problem. ♦

Example 5 Testing for convergence in a geometric series

Determine whether each of the following geometric series converges or diverges. If the series converges, find its sum.

a. $\sum_{k=0}^{\infty} \frac{1}{7} \left(\frac{3}{2} \right)^k$ b. $\sum_{k=2}^{\infty} 3 \left(-\frac{1}{5} \right)^k$

Solution

a. Because $r = \frac{3}{2}$ satisfies $|r| \geq 1$, the series diverges.

b. We have $r = -\frac{1}{5}$ so $|r| < 1$, and the geometric series converges. The first value of k is 2 (not 0), so the value a (the first value) is

$$a = 3 \left(-\frac{1}{5} \right)^2 = \frac{3}{25}$$

and

$$\sum_{k=2}^{\infty} 3 \left(-\frac{1}{5} \right)^k = \frac{a}{1 - r} = \frac{\frac{3}{25}}{1 - \left(-\frac{1}{5} \right)} = \frac{1}{10}$$

Applications of Geometric Series

Geometric series can be used in many different ways. Our next three examples illustrate several applications involving geometric series.

Recall that a rational number r is one that can be written as $r = p/q$ for integer p and nonzero integer q . It can be shown that any such number has a decimal representation in which a pattern of numbers repeats. For example,

$$\frac{5}{10} = 0.5 = 0.5\overline{0} \quad \frac{5}{11} = 0.454545 \dots = 0.4\overline{5}$$

where the bar indicates that the pattern repeats indefinitely. Example 6 shows how geometric series can be used to reverse this process by writing a given repeating decimal as a rational number.

Example 6 Expressing a repeating decimal as a rational number

Write $15.4\overline{23}$ as a rational number $\frac{p}{q}$.

Solution The bar over the 23 indicates that this block of digits is to be repeated, that is, $15.4\overline{23} = 15.423232323 \dots$. The repeating part of the decimal can be written as a geometric series as follows.

$$15.4\overline{23} = 15 + \frac{4}{10} + \frac{23}{10^3} + \frac{23}{10^5} + \frac{23}{10^7} + \dots$$

$$= 15 + \frac{4}{10} + \frac{23}{10^3} \left[1 + \frac{1}{10^2} + \frac{1}{10^4} + \dots \right]$$

$$= 15 + \frac{4}{10} + \frac{23}{10^3} \left[\frac{1}{1 - \frac{1}{100}} \right]$$

$$= 15 + \frac{4}{10} + \frac{23}{10^3} \left[\frac{100}{99} \right]$$

Sum of a geometric series with
 $a = 1, r = 0.01$.

$$\begin{aligned}
 &= 15 + \frac{4}{10} + \frac{23}{990} \\
 &= \frac{15,269}{990}
 \end{aligned}$$

A tax rebate that returns a certain amount of money to taxpayers can result in spending that is many times this amount. This phenomenon is known in economics as the *multiplier effect*. It occurs because the portion of the rebate that is spent by one individual becomes income for one or more others, who, in turn, spend some of it again, creating income for yet other individuals to spend. If the fraction of income that is saved remains constant as this process continues indefinitely, the total amount spent as a result of the rebate is the sum of a geometric series.*

Example 7 Modeling Problem: Multiplier effect in economics

Suppose that nationwide approximately 90% of all income is spent and 10% is saved. How much total spending will be generated by a \$40 billion tax rebate if savings habits do not change?

Solution The amount (in billions) spent by original recipients of the rebate is 36. This becomes new income, of which 90%, or $0.9(36)$, is spent. This, in turn, generates additional spending of $0.9[0.9(36)]$, and so on. The total amount spent if this process continues indefinitely is

$$\begin{aligned}
 36 + 0.9(36) + 0.9^2(36) + 0.9^3(36) + \cdots &= 36[1 + 0.9 + 0.9^2 + \cdots] \\
 &= 36 \sum_{k=0}^{\infty} 0.9^k && \text{Geometric series with } a = 1, r = 0.9 \\
 &= 36 \left(\frac{1}{1 - 0.9} \right) \\
 &= 360
 \end{aligned}$$

The total spending generated by the \$40 billion tax rebate is \$360 billion.

Example 8 Modeling Problem: Accumulation of medication in a body

A patient is given an injection of 10 units of a certain drug every 24 hours. The drug is eliminated exponentially so that the proportion that remains in the patient's body after t days is $f(t) = e^{-t/5}$. If the treatment is continued indefinitely, approximately how many units of the drug will eventually be in the patient's body just prior to an injection?

Solution Of the original dose of 10 units, only $10e^{-1/5}$ are left in the patient's body after the first day (just prior to the second injection). That is,

$$S_1 = 10e^{-1/5}$$

The medication in the patient's body after 2 days consists of what remains from the first two doses. Of the original dose, only $10e^{-2/5}$ units are left (because 2 days have elapsed), and of the second dose, $10e^{-1/5}$ units remain:

$$S_2 = 10e^{-1/5} + 10e^{-2/5}$$

Similarly, for n days,

$$S_n = 10e^{-1/5} + 10e^{-2/5} + \cdots + 10e^{-n/5} = \sum_{k=1}^n 10e^{-k/5}$$

*The fraction s of income that is saved is called the *marginal propensity to save* and $c = 1 - s$ is the *marginal propensity to consume*. In Example 7, we have $s = 0.1$ and $c = 0.9$.

The amount S of medication in the patient's body in the long run is the limit of S_n as $n \rightarrow \infty$. That is,

$$\begin{aligned} S &= \lim_{n \rightarrow \infty} S_n \\ &= \sum_{k=1}^{\infty} 10e^{-k/5} \quad \text{Geometric series with } a = 10e^{-1/5} \text{ and } r = e^{-1/5}. \\ &= \frac{10e^{-1/5}}{1 - e^{-1/5}} \\ &\approx 45.166566 \end{aligned}$$

We see that about 45 units remain in the patient's body. ■



As a final note, remember that a sequence is a succession of terms, whereas a series is a sum of such terms. Do not confuse the two concepts because they have very different properties. For example, a sequence of terms may converge, but the series of the same terms may diverge:

$\left\{1 + \frac{1}{2^n}\right\}$ is the **sequence** $\frac{3}{2}, \frac{5}{4}, \frac{9}{8}, \frac{17}{16}$ which **converges** to 1.

$\sum_{n=1}^{\infty} \left(1 + \frac{1}{2^n}\right)$ is the **series** $\frac{3}{2} + \frac{5}{4} + \frac{9}{8} + \dots$, which **diverges**.

PROBLEM SET 8.2

Level 1

- What does this say? Explain the difference between sequences and series.
- What does this say? What is a geometric series?

Determine whether the geometric series given in Problems 3–22 converges or diverges, and find the sum of each convergent series.

3. $\sum_{k=0}^{\infty} \left(\frac{4}{5}\right)^k$

4. $\sum_{k=0}^{\infty} \left(-\frac{4}{5}\right)^k$

5. $\sum_{k=0}^{\infty} \frac{2}{3^k}$

6. $\sum_{k=0}^{\infty} \frac{2}{(-3)^k}$

7. $\sum_{k=1}^{\infty} \left(\frac{3}{2}\right)^k$

8. $\sum_{k=1}^{\infty} \frac{3}{2^k}$

9. $\sum_{k=0}^{\infty} \frac{(-2)^k}{5^{2k+1}}$

10. $\sum_{k=1}^{\infty} 5(0.9)^k$

11. $\sum_{k=1}^{\infty} e^{-0.2k}$

12. $\sum_{k=1}^{\infty} \frac{3^k}{4^{k+2}}$

13. $\sum_{k=2}^{\infty} \frac{(-2)^{k-1}}{3^{k+1}}$

14. $\sum_{k=2}^{\infty} (-1)^k \frac{2^{k+1}}{3^{k-3}}$

15. $\frac{1}{2} - \frac{1}{2^2} + \frac{1}{2^3} - \frac{1}{2^4} + \dots$

16. $1 + \pi + \pi^2 + \pi^3 + \dots$

17. $\frac{1}{4} + \left(\frac{1}{4}\right)^4 + \left(\frac{1}{4}\right)^7 + \left(\frac{1}{4}\right)^{10} + \dots$

18. $\frac{2}{3} - \left(\frac{2}{3}\right)^3 + \left(\frac{2}{3}\right)^5 - \left(\frac{2}{3}\right)^7 + \dots$

19. $2 + \sqrt{2} + 1 + \frac{1}{\sqrt{2}} + \dots$

20. $3 - \sqrt{3} + 1 - \frac{1}{\sqrt{3}} + \dots$

21. $(1 + \sqrt{2}) + 1 + (-1 + \sqrt{2}) + (3 - 2\sqrt{2}) + \dots$

22. $(\sqrt{2} - 1) + 1 + (\sqrt{2} + 1) + (2\sqrt{2} + 3) + \dots$

In Problems 23–30, each series telescopes. In each case, express the n th partial sum S_n in terms of n and determine whether the series converges or diverges by examining $\lim_{n \rightarrow \infty} S_n$. Notice that some of these problems may require partial fraction decomposition.

23. $\sum_{k=1}^{\infty} \left[\frac{1}{k^{0.1}} - \frac{1}{(k+1)^{0.1}} \right]$
24. $\sum_{k=1}^{\infty} \left[\frac{1}{2k+1} - \frac{1}{2k+3} \right]$
25. $\sum_{k=0}^{\infty} \frac{1}{(k+1)(k+2)}$
26. $\sum_{k=2}^{\infty} \frac{1}{k(k+1)}$
27. $\sum_{k=1}^{\infty} \ln \left(1 + \frac{1}{k} \right)$
28. $\sum_{k=1}^{\infty} \frac{1}{(2k-1)(2k+1)}$
29. $\sum_{k=1}^{\infty} \frac{2k+1}{k^2(k+1)^2}$
30. $\sum_{k=1}^{\infty} \frac{\sqrt{k+1} - \sqrt{k}}{\sqrt{k^2+k}}$

Express each repeating decimal given in Problems 31–34 as a common (reduced) fraction.

31. $0.\overline{01}$ 32. $2.23\overline{1}$ 33. $1.\overline{405}$ 34. $41.20\overline{10}$

Level 2

35. a. Find numbers A and B such that

$$\frac{k-1}{2^{k+1}} = \frac{Ak}{2^k} - \frac{B(k+1)}{2^{k+1}}$$

- b. Evaluate

$$\sum_{k=1}^{\infty} \frac{k-1}{2^{k+1}}$$

as a telescoping series.

36. Evaluate

$$\sum_{k=1}^{\infty} \frac{2k-1}{3^{k+1}}$$

Hint: Follow the procedure in Problem 35.

37. Evaluate

$$\sum_{n=1}^{\infty} \frac{\ln \left[\frac{n^{n+1}}{(n+1)^n} \right]}{n(n+1)}$$

Hint: Use the properties of logarithms to show that the series telescopes.

38. Evaluate

$$\sum_{n=1}^{\infty} \frac{n}{(n+1)!}$$

Hint: Express as a telescoping series.

39. Find $\sum_{k=0}^{\infty} (2a_k + 2^{-k})$ given that $\sum_{k=0}^{\infty} a_k = 3.57$.

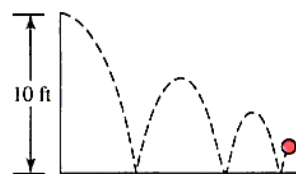
40. Find $\sum_{k=0}^{\infty} \frac{b_k - 3^{-k}}{2}$ given that $\sum_{k=0}^{\infty} b_k = 0.54$.

41. Evaluate $\sum_{k=0}^{\infty} \left(\frac{1}{2^k} + \frac{1}{3^k} \right)^2$.

42. Evaluate $\sum_{k=0}^{\infty} \left[\left(\frac{2}{3} \right)^k + \left(\frac{3}{4} \right)^k \right]^2$.

Modeling Problems: In Problems 43–53, set up an appropriate model to answer the given question.

43. How much should you invest today at an annual interest rate of 1.5% compounded continuously so that, starting next year, you can make annual withdrawals of \$2,000 forever?
44. A pendulum is released through an arc of length 20 cm from a vertical position and is allowed to swing free until it eventually comes to rest. Each subsequent swing of the bob of the pendulum is 90% as far as the preceding swing. How far will the bob travel before coming to rest?
45. A flywheel rotates at 500 rpm and slows in such a way that during each minute it revolves at two-thirds the rate of the preceding minute. How many total revolutions does the flywheel make before coming to rest?
46. A ball is dropped from a height of 10 ft. Each time the ball bounces, it rises 0.6 the distance it had previously fallen. What is the total distance traveled by the ball?



47. Suppose the ball in Problem 46 is dropped from a height of h feet and on each bounce rises 75% of the distance it had previously fallen. If it travels a total distance of 21 ft, what is h ?
48. Suppose that nationwide approximately 92% of all income is spent and 8% is saved. How much total spending will be generated by a \$50 billion tax cut if savings habits do not change?
49. Suppose that a piece of machinery costing \$10,000 depreciates 20% of its current value each year. That is, the first year $\$10,000(0.20) = \$2,000$ is depreciated. The second year's depreciation is $\$8,000(0.20) = \$1,600$, because the value for the second year is $\$10,000 - \$2,000 = \$8,000$. If the depreciation is calculated this way indefinitely, what is the total depreciation?

50. Winnie Winner wins \$100 in a pie-baking contest run by the Hi-Do Pie Co. The company gives Winnie the \$100. However, the tax collector wants 20% of the \$100. Winnie pays the tax. But then she realizes that she didn't really win a \$100 prize and tells her story to the Hi-Do Co. The friendly Hi-Do Co. gives Winnie the \$20 she paid in taxes. Unfortunately, the tax collector now wants 20% of the \$20. She pays the tax again and then goes back to the Hi-Do Co. with her story. Assume that this can go on indefinitely. How much money does the Hi-Do Co. have to give Winnie so that she will really win \$100? How much does she pay in taxes?
51. A patient is given an injection of 20 units of a certain drug every 24 hr. The drug is eliminated exponentially, so the part that remains in the patient's body after t days is $f(t) = e^{-t/2}$. If the treatment is to continue indefinitely, approximately how many units of the drug will eventually be in the patient's body just before an injection?
52. A certain drug is injected into a body. It is known that the amount of drug in the body at the end of a given hour is $\frac{1}{4}$ the amount that was present at the end of the previous hour. One unit is injected initially, and to keep the concentration up, an additional unit is injected at the end of each subsequent hour. What is the highest possible amount of the drug that can ever be present in the body?
53. Each January 1, the administration of a certain private college adds six new members to its board of trustees. If the fraction of trustees who remain active for at least t years is $f(t) = e^{-0.2t}$, approximately how many active trustees will the college have in the long run? Assume that there were six members on the board on January 1 of the present year and that the trustees are counted on December 31.
54. **Historical Quest**

A professor of mathematics at the University of Warsaw, Waclaw Sierpinski was the first person to give a lecture course devoted entirely to set theory. He became interested in set theory in 1907 when he came across a theorem that stated that points in the plane could be specified with a single coordinate. He wrote to a friend and asked how such a result could be possible, and he received a one word reply: "Cantor." Georg Cantor (1845-1918) wrote a series of eight papers which provided the basis for what we today call set theory. For this **Quest**, we investigate an interesting geometrical figure named after Sierpinski, namely the Sierpinski gasket, shown in Figure 8.11.



Waclaw Sierpinski
(1882-1969)

Karl Smith Library

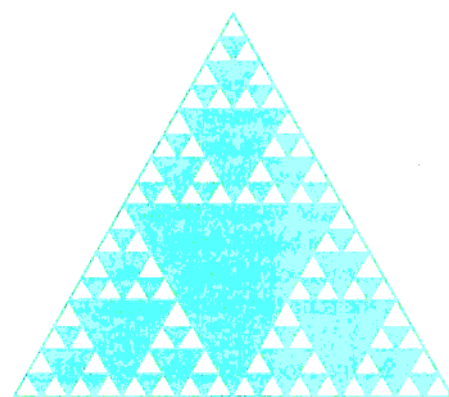


Figure 8.11 Interactive Sierpinski gasket

To form the gasket, begin by drawing an equilateral triangle, T , with sides of length 1. Remove the middle triangle formed by joining the midpoints of the sides of T , as shown in Figure 8.12a. This leaves three equal triangular regions, and the next step is to remove the middle triangle from each of these, as shown in Figure 8.12b. If you continue this process indefinitely, what remains is the Sierpinski gasket. Show that the Sierpinski gasket has area 0. *Hint*: What is the sum of the areas of the regions removed?

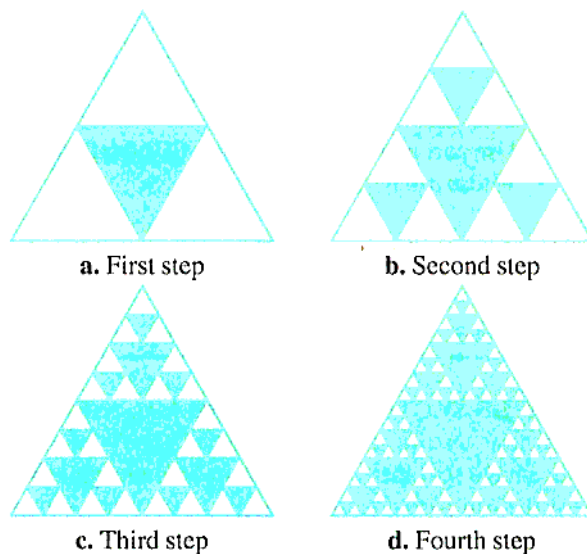


Figure 8.12 Construction of a Sierpinski gasket

55. **Historical Quest**

John von Neumann was one of the original six mathematics professors at the Institute for Advanced Study at Princeton; the other five were also world-class mathematicians or Nobel laureates: J. W. Alexander, Albert Einstein, Marston Morse, Oswald Veblen, and Hermann Weyl.



John von Neumann
(1903-1957)

Karl Smith Library

Von Neumann, a pioneer in computer science, received two Presidential Awards, the Medal for Merit and the Medal for Freedom. The fact that he had a brilliant mind is an understatement, as illustrated in the following anecdote, stated in the form of a quest problem.

Two trains, each traveling at 10 mph approach each other on the same straight track. When the trains are 5 miles apart, a fly begins flying back and forth from train to train at the rate of 16 mi/h. Between the time the fly starts and the time the trains collide, how far does the fly fly?

- Find the fly's total distance traveled by determining how long the fly flies.
- When von Neumann gave the correct answer almost immediately, the poser of the problem was convinced he had used the "trick" suggested in part a. "What trick?" asked von Neumann. "I summed the series." Follow von Neumann's lead by summing an infinite series whose terms are the distances traveled by the fly as it flies between the trains.

Level 3

56. a. Let $\sum_{k=1}^n a_k = S_n$ be the n th partial sum of an infinite series. Show that

$$S_n - S_{n-1} = a_n$$

- b. Use part a to find the n th term of the infinite series whose n th partial sum is

$$S_n = \frac{n}{2n+3}$$

57. Show that if $\sum a_k$ converges and $\sum b_k$ diverges, then $\sum (a_k - b_k)$ diverges.

Hint: Note that $b_k = a_k - (a_k - b_k)$.

58. Show that $\sum_{k=0}^{\infty} ar^k$ diverges if $|r| \geq 1$.

59. Evaluate

$$\sum_{k=1}^{\infty} \frac{12^k}{(4^k - 3^k)(4^{k+1} - 3^{k+1})}$$

Hint: Find A and B so that

$$\frac{3^k A}{4^k - 3^k} + \frac{3^k B}{4^{k+1} - 3^{k+1}} = \frac{12^k}{(4^k - 3^k)(4^{k+1} - 3^{k+1})}$$

60. Square $ABCD$ has sides of length 1. Square $EFGH$ is formed by connecting the midpoints of the sides of the first square, as shown in Figure 8.13.

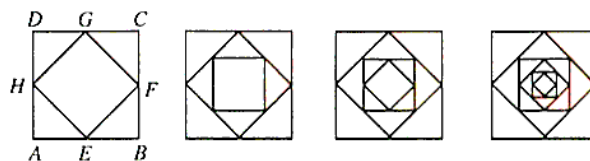


Figure 8.13 Find the area of the shaded regions

Assume that the pattern of shaded regions in the square is continued indefinitely. What is the total area of the shaded regions?

8.3 THE INTEGRAL TEST; P -SERIES

IN THIS SECTION: Divergence test, series of nonnegative numbers; the integral test, p -series

We now turn to developing some "tests" for determining the convergence of a given series. These tests have the advantage that they do not require knowledge of an explicit formula (such as the one we found for a geometric series) for the n th partial sum S_n and the disadvantage that they usually cannot be used to find the actual sum of a convergent series.

The convergence or divergence of an infinite series is determined by the behavior of its n th partial sum, S_n , as $n \rightarrow \infty$. In Section 8.2 we saw how algebraic methods can sometimes be used to find formulas for the n th partial sum of a series. Unfortunately, it is often difficult or even impossible to find a usable formula for the n th partial sum S_n of a series, and other techniques must be used to determine convergence or divergence.

Divergence Test

When investigating the infinite series $\sum a_k$, it is easy to confuse the sequence of general terms $\{a_k\}$ with the sequence of partial sums $\{S_n\}$, where

$$S_n = \sum_{k=1}^n a_k$$

Because the sequence $\{a_k\}$ is usually more accessible than $\{S_n\}$, it would be convenient if the convergence of $\{S_n\}$ could be settled by examining

$$\lim_{k \rightarrow \infty} a_k$$

Even though we do not have one simple, definitive test for convergence, the following theorem tells us that if certain conditions are not met, the series must diverge.

Theorem 8.9 The divergence test

If $\lim_{k \rightarrow \infty} a_k \neq 0$, then the series $\sum a_k$ must diverge.

What this says The divergence test can only tell us that $\sum a_k$ diverges if $\lim_{k \rightarrow \infty} a_k \neq 0$, **but it cannot be used to show convergence of a series.** In Example 2 we show that the series $\sum \frac{1}{k}$ diverges even though $\lim_{k \rightarrow \infty} \frac{1}{k} = 0$.

Proof: Suppose the sequence of partial sums $\{S_n\}$ converges with sum L , so that $\lim_{n \rightarrow \infty} S_n = L$. Then we also have

$$\lim_{n \rightarrow \infty} S_{n-1} = L$$

Because $S_k - S_{k-1} = a_k$, it follows that

$$\lim_{k \rightarrow \infty} a_k = \lim_{k \rightarrow \infty} (S_k - S_{k-1}) = \lim_{k \rightarrow \infty} S_k - \lim_{k \rightarrow \infty} S_{k-1} = L - L = 0$$

If the series $\sum a_k$ converges, then $\lim_{k \rightarrow \infty} a_k = 0$. Thus, if $\lim_{k \rightarrow \infty} a_k \neq 0$, the series $\sum a_k$ must diverge. ♦

Theorem 8.9 gives a necessary but not a sufficient condition for convergence. For example, clouds are necessary for rain; clouds are not sufficient for rain, just as $\lim_{k \rightarrow \infty} a_k = 0$ is necessary for $\sum a_k$ to converge, but is not sufficient to make it converge. Note also that this is therefore called a DIVERGENCE test, not a CONVERGENCE test.

Example 1 Divergence test to show divergence

Show that the series $\sum_{k=0}^{\infty} \frac{k-300}{4k+750}$ diverges.

Solution Taking the limit of the k th term as $k \rightarrow \infty$, we find

$$\lim_{k \rightarrow \infty} \frac{k-300}{4k+750} = \frac{1}{4}$$

Because this limit is not 0, the divergence test tells us that the series must diverge. ■

Series of Nonnegative Numbers; The Integral Test

Series whose terms are all nonnegative numbers play an important role in the general theory of infinite series and in applications. Our next goal is to develop methods for investigating the convergence of nonnegative-term series, and we begin by establishing the following general principle.

Theorem 8.10 Convergence criterion for series with nonnegative terms

A series $\sum a_k$ with $a_k \geq 0$ for all k converges if its sequence of partial sums is bounded from above and diverges otherwise.

Proof: Suppose $\sum a_k$ is a series of nonnegative terms, and let S_n denote its n th partial sum; that is

$$S_n = \sum_{k=1}^n a_k = a_1 + a_2 + \cdots + a_n$$

Because $S_{n+1} = S_n + a_{n+1}$ and because $a_k \geq 0$ for all k , it follows that

$$S_{n+1} \geq S_n$$

for all n , so that $\{S_n\}$ is an increasing sequence. According to the BMCT, the increasing sequence $\{S_n\}$ converges if it is bounded from above and diverges otherwise. Hence, because the series $\sum a_k$ represents the limit of the sequence $\{S_n\}$, the series $\sum a_k$ converges if the sequence $\{S_n\}$ is bounded from above and diverges if it is unbounded. ♦

This convergence criterion is often difficult to apply because it is not easy to determine whether the sequence of partial sums $\{S_n\}$ is bounded from above. Our next goal is to examine various “tests for convergence.” These are procedures that allow us to determine indirectly whether a given series converges or diverges without actually having to compute the limit of the partial sums. We begin with a convergence test that relates the convergence of a series to that of an improper integral.

Theorem 8.11 The integral test

If $a_k = f(k)$ for $k = 1, 2, \dots$, where f is a positive, continuous, and decreasing function of x for $x \geq 1$, then

$$\sum_{k=1}^{\infty} a_k \quad \text{and} \quad \int_1^{\infty} f(x) dx$$

either both converge or both diverge.

Proof: We will use a geometric argument to show that the sequence of partial sums $\{S_n\}$ of the series $\sum_{k=1}^{\infty} a_k$ is bounded from above if the improper integral $\int_1^{\infty} f(x) dx$ converges and that it is unbounded if the integral diverges. Accordingly, Figure 8.14a and Figure 8.14b show the graph of a continuous decreasing function f that satisfies $f(k) = a_k$ for $k = 1, 2, 3, \dots$. Rectangles have been constructed at unit intervals in both figures, but in Figure 8.14a, k th rectangle has height $f(k+1) = a_{k+1}$, whereas in Figure 8.14b, the comparable rectangle has height $f(k) = a_k$.

When applying the integral test, the function f need not be decreasing for all $x \geq 1$, only for all $x \geq N$ for some number N . That is, f must be decreasing “in the long run.”

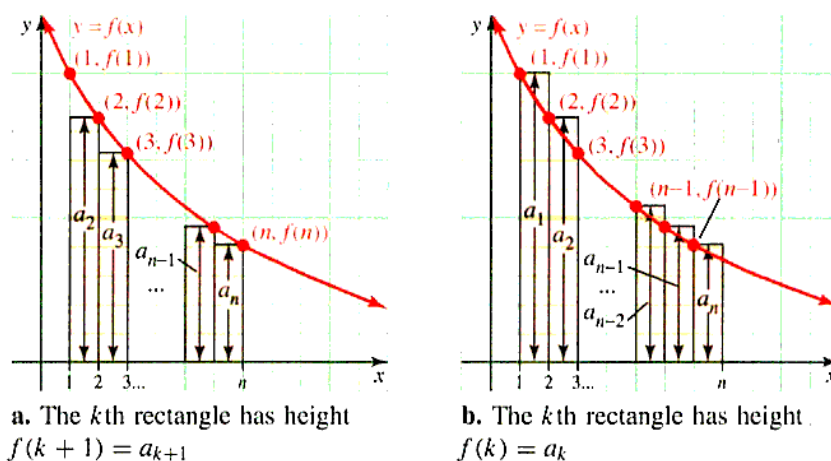


Figure 8.14 Interactive Integral test

Notice that in Figure 8.14a, the rectangles are inscribed so that

AREA OF THE FIRST $n - 1$ RECTANGLES < AREA UNDER $y = f(x)$ OVER $[1, n]$

$$\begin{aligned} a_2 + a_3 + \cdots + a_n &< \int_1^n f(x) dx \\ a_1 + a_2 + a_3 + \cdots + a_n &< a_1 + \int_1^n f(x) dx \quad \text{Add } a_1 \text{ to both sides.} \end{aligned}$$

Similarly, in Figure 8.14b, the rectangles are circumscribed so that

AREA UNDER $y = f(x)$ OVER $[1, n + 1] <$ AREA OF THE FIRST n RECTANGLES

$$\int_1^{n+1} f(x) dx < a_1 + a_2 + a_3 + \cdots + a_n$$

Let $S_n = a_1 + a_2 + a_3 + \cdots + a_n$ be the n th partial sum of the series so that

$$\int_1^{n+1} f(x) dx < S_n < a_1 + \int_1^n f(x) dx$$

Now, suppose the improper integral $\int_1^\infty f(x) dx$ converges and has the value I ; that is,

$$\int_1^\infty f(x) dx = I$$

Then

$$S_n < a_1 + \int_1^n f(x) dx < a_1 + \int_1^\infty f(x) dx = a_1 + I$$

It follows that the sequence of partial sums is bounded from above (by $a_1 + I$), and the convergence criterion for nonnegative term series tells us that the series $\sum_{k=1}^\infty a_k$ must converge.

On the other hand, if the improper integral $\int_1^\infty f(x) dx$ diverges, then

$$\lim_{n \rightarrow \infty} \int_1^{n+1} f(x) dx = \infty$$

It follows that $\lim_{n \rightarrow \infty} S_n = \infty$ also because

$$\int_1^{n+1} f(x) dx < S_n$$

This means that the series must diverge. Thus, the series and the improper integral either both converge or both diverge, as claimed. ♦

A series that arises in connection with overtones produced by a vibrating string is

$$\sum_{k=1}^\infty \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$$

⊙ This is called the **harmonic series**, and in the next example, we use the integral test to show that this series diverges. Another proof that does not use the integral test can be found in the supplementary problems at the end of this chapter.

Example 2 Harmonic series diverges

Test the series $\sum_{k=1}^{\infty} \frac{1}{k}$ for convergence.

Solution Because $f(x) = \frac{1}{x}$ is positive, continuous, and decreasing for $x \geq 1$, the conditions of the integral test are satisfied.

$$\int_1^{\infty} \frac{1}{x} dx = \lim_{b \rightarrow \infty} \int_1^b \frac{1}{x} dx = \lim_{b \rightarrow \infty} [\ln b - \ln 1] = \infty$$

The integral diverges, so the harmonic series diverges.

Example 3 Integral test for convergence

Test the series $\sum_{k=1}^{\infty} \frac{k}{e^{k/5}}$ for convergence.

Solution The function $f(x) = \frac{x}{e^{x/5}} = xe^{-x/5}$ is positive and continuous for all $x > 0$. We find that

$$f'(x) = x \left(-\frac{1}{5} e^{-x/5} \right) + e^{-x/5} = \left(1 - \frac{x}{5} \right) e^{-x/5}$$

The critical number is found when $f'(x) = 0$, so we solve

$$\begin{aligned} \left(1 - \frac{x}{5} \right) e^{-x/5} &= 0 \\ 1 - \frac{x}{5} &= 0 && \text{Since } e^{-x/5} \neq 0 \\ x &= 5 \end{aligned}$$

We see that $f'(x) < 0$ for $x > 5$, so it follows that f is decreasing for $x > 5$. Thus, the conditions for the integral test have been established, and the given series and the improper integral either both converge or both diverge. Computing the improper integral, we find

$$\begin{aligned} \int_5^{\infty} xe^{-x/5} dx &= \lim_{b \rightarrow \infty} \int_5^b xe^{-x/5} dx \\ &= \lim_{b \rightarrow \infty} \left[-5xe^{-x/5} \Big|_5^b - \int_5^b (-5e^{-x/5}) dx \right] \\ &= \lim_{b \rightarrow \infty} \left[-5xe^{-x/5} - 25e^{-x/5} \right]_5^b \\ &= \lim_{b \rightarrow \infty} \left[-5be^{-b/5} - 25e^{-b/5} + 25e^{-1} + 25e^{-1} \right] \\ &= -5 \lim_{b \rightarrow \infty} \frac{b+5}{e^{b/5}} + \lim_{b \rightarrow \infty} 50e^{-1} \\ &= -5 \lim_{b \rightarrow \infty} \frac{1}{\left(\frac{1}{5}\right) e^{b/5}} + 50e^{-1} \\ &= 50e^{-1} \end{aligned}$$

Let $u = x$ $dv = e^{-x/5} dx$
 $du = dx$ $v = -5e^{-x/5}$

Note that the series converges because the integral converges to a finite number $50e^{-1}$, but this does not mean that the series



$$\sum_{k=1}^{\infty} \frac{k}{e^{k/5}}$$

converges to the same number $50e^{-1}$.

l'Hôpital's rule

$$\lim_{b \rightarrow \infty} \frac{1}{e^{b/5}} = 0$$

Thus, the improper integral converges, which in turn assures the convergence of the given series. ■

p -Series

The harmonic series $\sum \frac{1}{k}$ is a special case of a more general series form called a p -series.

p -SERIES A series of the form

$$\sum_{k=1}^{\infty} \frac{1}{k^p} = \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \cdots$$

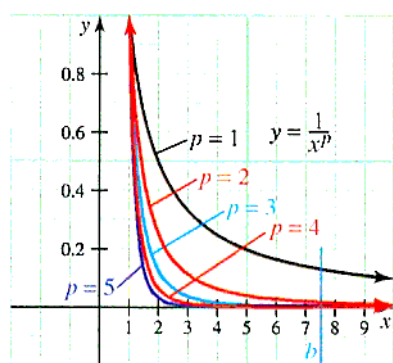
where p is a positive constant is called a **p -series**. The harmonic series (Example 2) is the case where $p = 1$.

The following theorem shows how the convergence of a p -series depends on the value of p .

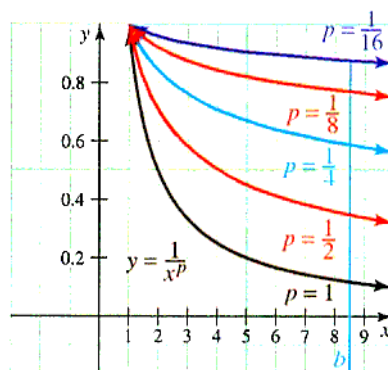
Theorem 8.12 The p -series test

The p -series $\sum_{k=1}^{\infty} \frac{1}{k^p}$ converges if $p > 1$ and diverges if $p \leq 1$.

What this says It should be clear that all functions of the form $y = \frac{1}{x^p}$ with $p > 0$ are decreasing, but why should the series $\sum \frac{1}{k^p}$ converge only for $p > 1$? The answer lies in the *rate* of decrease for $y = \frac{1}{x^p}$. If $p > 1$, the curve $y = \frac{1}{x^p}$ falls fast enough to guarantee that the area under the curve for $p > 1$ is finite (see Figure 8.15a, while if $p \leq 1$, the curve $\frac{1}{x^p}$ falls more gradually, and the area under the curve is infinite (Figure 8.15b).



a. Areas are bounded



b. Areas are not bounded

Figure 8.15 Graphs for the p -series test

Proof: We leave it for the reader to verify that $f(x) = \frac{1}{x^p}$ is continuous, positive, and decreasing for $x \geq 1$ and $p > 0$. We know the harmonic series ($p = 1$) diverges. For $p > 0$, $p \neq 1$, we have

$$\int_1^{\infty} \frac{dx}{x^p} = \lim_{b \rightarrow \infty} \int_1^b x^{-p} dx = \lim_{b \rightarrow \infty} \frac{b^{1-p} - 1}{1-p} = \begin{cases} \frac{1}{p-1} & \text{if } p > 1 \\ \infty & \text{if } 0 < p < 1 \end{cases}$$

That is, this improper integral converges if $p > 1$ and diverges if $0 < p < 1$. For $p = 0$, the series becomes

$$\sum_{k=1}^{\infty} \frac{1}{k^0} = \frac{1}{1} + \frac{1}{1} + \frac{1}{1} + \cdots$$

and if $p < 0$, we have $\lim_{k \rightarrow \infty} \frac{1}{k^p} = \infty$, so the series diverges by the divergence test (Theorem 8.9). Thus, a p -series converges only when $p > 1$. ♦

Example 4 p -series test

Test each of the following series for convergence.

a. $\sum_{k=1}^{\infty} \frac{1}{\sqrt{k^3}}$ b. $\sum_{k=1}^{\infty} \left(\frac{1}{e^k} - \frac{1}{\sqrt{k}} \right)$

Solution

a. Here $\sqrt{k^3} = k^{3/2}$, so $p = \frac{3}{2} > 1$, and the series converges.

b. $\sum_{k=1}^{\infty} \left(\frac{1}{e^k} - \frac{1}{\sqrt{k}} \right)$; We note that $\sum_{k=1}^{\infty} \frac{1}{e^k}$ converges, because it is geometric with $|r| = \frac{1}{e} < 1$ and $\sum_{k=1}^{\infty} \frac{1}{\sqrt{k}}$ diverges, because it is a p -series with $p = \frac{1}{2} < 1$. Because one series in the difference converges and the other diverges, the given series must diverge (Theorem 8.7). ■

PROBLEM SET 8.3

Level 1

- What does this say? What is a p -series?
- What does this say? What is the integral test?

For the p -series given in Problems 3–6, specify the value of p and tell whether the series converges or diverges.

- $\sum_{k=1}^{\infty} \frac{1}{k^3}$
- $\sum_{k=1}^{\infty} \frac{100}{\sqrt{k}}$
- $\sum_{k=1}^{\infty} \frac{1}{\sqrt[3]{k}}$
- $\sum_{k=1}^{\infty} \frac{1}{2k\sqrt{k}}$

Show that the function f determined by the n th term of each series given in Problems 7–12 satisfies the hypotheses of the integral test. Then use the integral test to determine whether the series converges or diverges.

- $\sum_{k=1}^{\infty} \frac{1}{(2+3k)^2}$
- $\sum_{k=1}^{\infty} (2+k)^{-3/2}$
- $\sum_{k=2}^{\infty} \frac{\ln k}{k}$
- $\sum_{k=2}^{\infty} \frac{1}{k(\ln k)^2}$
- $\sum_{k=1}^{\infty} \frac{(\tan^{-1} k)^2}{1+k^2}$
- $\sum_{k=1}^{\infty} ke^{-k^2}$

In Problems 13–18, either use the divergence test to show that the given series diverges or show that the divergence test does not apply.

- $\sum_{k=1}^{\infty} \frac{k}{k+1}$

- $\sum_{k=2}^{\infty} \frac{k}{\sqrt{k^2-1}}$

- $\sum_{k=0}^{\infty} \cos k\pi$

- $\sum_{k=1}^{\infty} \left(1 + \frac{1}{k} \right)^{-k}$

- $\sum_{k=1}^{\infty} e^{-1/k}$

- $\sum_{k=1}^{\infty} ke^{-k}$

Test the series in Problems 19–38 for convergence.

- $\sum_{k=1}^{\infty} \frac{\ln k}{k^2}$

- $\sum_{k=1}^{\infty} e^{-k}$

- $\sum_{k=1}^{\infty} \left(2 + \frac{3}{k} \right)^k$

- $\sum_{k=1}^{\infty} \left(1 + \frac{2}{k} \right)^k$

- $\sum_{k=1}^{\infty} \frac{1}{k^4}$

- $\sum_{k=1}^{\infty} \frac{5}{\sqrt{k}}$

- $\sum_{k=1}^{\infty} k^{-3/4}$

- $\sum_{k=1}^{\infty} k^{-4/3}$

- $\sum_{k=1}^{\infty} \frac{k - \sin k}{3k + 2 \sin k}$

- $\sum_{k=1}^{\infty} \frac{1}{(0.25)^k}$

- $\sum_{k=2}^{\infty} \frac{\ln \sqrt{k}}{\sqrt{k}}$

- $\sum_{k=2}^{\infty} \frac{1}{k\sqrt{\ln k}}$

- $\sum_{k=1}^{\infty} \frac{k}{e^k}$

- $\sum_{k=1}^{\infty} \frac{k^2}{e^k}$

33. $\sum_{k=1}^{\infty} \frac{\tan^{-1} k}{1+k^2}$

35. $\sum_{k=1}^{\infty} \tan^{-1} k$

37. $\sum_{k=1}^{\infty} k \sin \frac{1}{k}$

34. $\sum_{k=1}^{\infty} \cot^{-1} k$

36. $\sum_{k=1}^{\infty} \frac{1}{e^k + e^{-k}}$

38. $\sum_{k=4}^{\infty} \ln \left(\frac{k}{k-3} \right)$

53. $\sum_{k=3}^{\infty} \frac{1}{k^p \ln k}$

55. $\sum_{k=2}^{\infty} \frac{1}{k(\ln k)^p}$

54. $\sum_{k=2}^{\infty} \frac{\ln k}{k^p}$

56. $\sum_{k=3}^{\infty} \frac{1}{k \ln k [\ln(\ln k)]^p}$

Determine which of the series in Problems 39–50 converge. Write “inconclusive” if there is not yet enough information to settle convergence.

39. $\sum_{k=1}^{\infty} \frac{k-1}{k+1}$

41. $\sum_{k=1}^{\infty} \frac{k^2+1}{k^3}$

43. $\sum_{k=1}^{\infty} \frac{1}{k^2}$

45. $\sum_{k=1}^{\infty} \frac{k^{\sqrt{3}}+1}{k^{2.7321}}$

47. $\sum_{k=1}^{\infty} \left[\frac{1}{k} + \frac{k+1}{k+2} \right]$

49. $\sum_{k=1}^{\infty} \left[\frac{1}{2^k} - \frac{1}{k} \right]$

40. $\sum_{k=1}^{\infty} \frac{k^2+1}{k+5}$

42. $\sum_{k=1}^{\infty} \frac{2k^4+3}{k^5}$

44. $\sum_{k=1}^{\infty} \frac{1+k^2-k^5}{2+k^5}$

46. $\sum_{k=1}^{\infty} \frac{k^{\sqrt{5}}+1}{k^{2.236}}$

48. $\sum_{k=1}^{\infty} \left[\frac{1}{2k} + \left(\frac{3}{2} \right)^k \right]$

50. $\sum_{k=1}^{\infty} \left[\frac{1}{2^k} + \frac{2k+3}{3k+4} \right]$

Level 2

Find all values of p for which each series given in Problems 51–56 converges.

51. $\sum_{k=2}^{\infty} \frac{k}{(k^2-1)^p}$

52. $\sum_{k=1}^{\infty} \frac{k^2}{(k^2+4)^p}$

Level 3

57. Think Tank Problem If $\sum a_k$ converges, $\sum b_k$ diverges, and $\lim_{k \rightarrow \infty} b_k = 0$, then $\sum a_k b_k$ must diverge. Either prove this statement or provide a counterexample to show it is not generally true.

58. Think Tank Problem If $\lim_{k \rightarrow \infty} a_k = 0$, $\lim_{k \rightarrow \infty} b_k = 0$, and $\sum a_k$ and $\sum b_k$ both diverge, then $\sum a_k b_k$ must also diverge. Either prove this statement or provide a counterexample to show it is not generally true.

59. Let $f(x)$ be a positive, continuous, decreasing function for $x \geq 1$, and let $a_k = f(k)$ for $k = 1, 2, \dots$. If $\sum a_k$ converges, show that

$$\int_1^{\infty} f(x) dx \leq \sum_{k=1}^{\infty} a_k \leq a_1 + \int_1^{\infty} f(x) dx$$

60. Getting a feeling for a series by examining its graph is a good idea, but you should be careful because you cannot always obtain a graph. Decide whether

$$\sum \left(1 - \frac{100}{n} \right)^n$$

converges or diverges, and then see if you can verify your result graphically.

8.4 COMPARISON TESTS

IN THIS SECTION: Direct comparison test, limit comparison tests

Often a given series will have the same general appearance as another series whose convergence properties are known. In such case, it is convenient to use the properties of the known series to determine those of the given series. The goal of this section is to examine three comparison tests for making these determinations.

Direct Comparison Test

The first of the comparison tests is called the **direct comparison test**.

Theorem 8.13 Direct comparison test

Suppose $0 \leq a_k \leq c_k$ for all k .

If $\sum_{k=1}^{\infty} c_k$ converges, then $\sum_{k=1}^{\infty} a_k$ also converges

Let $0 \leq d_k \leq a_k$ for all k .

If $\sum_{k=1}^{\infty} d_k$ diverges, then $\sum_{k=1}^{\infty} a_k$ also diverges

What this says Let $\sum a_k$, $\sum c_k$ and $\sum d_k$ be series with positive terms. The series $\sum a_k$ *converges* if it is “smaller” than (dominated by) a known convergent series $\sum c_k$ and *diverges* if it is “larger” than (dominates) a known divergent series $\sum d_k$. That is, “smaller than convergent is convergent,” and “bigger than divergent is divergent.”

Proof: Suppose $\sum a_k$ is a given nonnegative term series that is **dominated** by a convergent series $\sum c_k$ with $0 \leq a_k \leq c_k$ for all k . Then, because the series $\sum c_k$ converges, its sequence of partial sums is bounded from above (say, by M), and we have

$$\sum_{k=1}^n a_k \leq \sum_{k=1}^n c_k < M \quad \text{for all } n$$

Thus, the sequence of partial sums of the dominated series $\sum a_k$ is also bounded from above by M , and it, too, must converge.

On the other hand, suppose the given series $\sum a_k$ **dominates** a divergent series $\sum d_k$ with $0 \leq d_k \leq a_k$. Then because the sequence of partial sums of $\sum d_k$ is unbounded, the same must be true for the partial sums of the dominating series $\sum a_k$, and $\sum a_k$ must also diverge. ♦

Remark:

The above theorem is also true if the dominance only holds for all $k \geq N$ for some positive N because whether a series converges or diverges depends on only what happens for large k . That is, $\sum_{k=1}^{\infty} a_k$ converges if and only if $\sum_{k=N}^{\infty} a_k$ converges for any integer N .

Example 1 Convergence using the direct comparison test

Test the series $\sum_{k=1}^{\infty} \frac{1}{3^k + 1}$ for convergence.

Solution For $k \geq 0$, we have $3^k + 1 > 3^k > 0$, and $0 < \frac{1}{3^k + 1} < \frac{1}{3^k}$. Thus, the given series is *dominated* by the convergent geometric series $\sum_{k=1}^{\infty} \frac{1}{3^k}$ (convergent because $r = \frac{1}{3}$). The direct comparison test tells us the given series must converge. ■

Example 2 Divergence using the direct comparison test

Test the series $\sum_{k=2}^{\infty} \frac{1}{\sqrt{k} - 1}$ for convergence.

Solution For $k \geq 2$, we have

$$\frac{1}{\sqrt{k} - 1} > \frac{1}{\sqrt{k}} > 0$$

so the given series *dominates* the divergent p -series $\sum_{k=2}^{\infty} \frac{1}{k^{1/2}}$ (divergent because $p = \frac{1}{2} < 1$). The direct comparison test tells us the given series must diverge. ■

Example 3 Applying the direct comparison test

Test the series $\sum_{k=1}^{\infty} \frac{1}{k!}$ for convergence.

Solution We will compare the given series with a geometric series. Specifically, note that if $k \geq 2$, then $1/k! > 0$ and

$$k! = k(k-1)(k-2) \cdots 3 \cdot 2 \cdot 1 \geq \underbrace{2 \cdot 2 \cdot 2 \cdots 2}_{k-1 \text{ terms}} \cdot 2 \cdot 1 = 2^{k-1}$$

Therefore, we have $0 < \frac{1}{k!} \leq \frac{1}{2^{k-1}}$, and because the given series is *dominated by* the convergent geometric series $\sum_{k=1}^{\infty} \frac{1}{2^{k-1}}$ (with $r = \frac{1}{2}$), it must also converge. ■

Limit Comparison Tests

It is not always easy or even possible to make a suitable direct comparison between two similar series. For example, we would expect the series $\sum \frac{1}{2^k - 5}$ to converge because it is so much like the convergent geometric series $\sum \frac{1}{2^k}$. To compare the series, we must first note that $\frac{1}{2^k - 5}$ is negative for $k = 1$ and $k = 2$ and positive for $k \geq 3$. Thus, if $k \geq 3$,

$$0 \leq \frac{1}{2^k} < \frac{1}{2^k - 5}$$

That is, $\sum \frac{1}{2^k - 5}$ *dominates* the convergent series $\sum \frac{1}{2^k}$, which means the comparison test cannot be used to determine the convergence of $\sum \frac{1}{2^k - 5}$ by comparing with $\sum \frac{1}{2^k}$. In such situations, the following test is often useful.

Theorem 8.14 Limit comparison test

Suppose $a_k > 0$ and $b_k > 0$ for all sufficiently large k and that

$$\lim_{k \rightarrow \infty} \frac{a_k}{b_k} = L$$

where L is finite and positive ($0 < L < \infty$). Then $\sum a_k$ and $\sum b_k$ either both converge or both diverge.

■ **What this says** We have the following procedure for testing the convergence of $\sum a_k$.

- Step 1.* Find a series $\sum b_k$ whose convergence properties are known and whose general term b_k is “essentially the same” as a_k .
- Step 2.* Verify that $\lim_{k \rightarrow \infty} \frac{a_k}{b_k}$ exists and is positive.
- Step 3.* Determine whether $\sum b_k$ converges or diverges. Then the limit comparison test tells us that $\sum a_k$ does the same.

Proof: The limit comparison test is proved by using the direct comparison test. Details of the proof are given in Appendix B.

Example 4 Convergence using the limit comparison test

Test the series $\sum_{k=1}^{\infty} \frac{1}{2^k - 5}$ for convergence.

Solution Because the given series has the same general appearance as the convergent geometric series $\sum 1/2^k$, compute the limit

$$\lim_{k \rightarrow \infty} \frac{\frac{1}{2^k - 5}}{\frac{1}{2^k}} = \lim_{k \rightarrow \infty} \frac{2^k}{2^k - 5} = 1$$

This limit is finite and positive, so the limit comparison test tells us that the given series will have the same convergence properties as $\sum 1/2^k$. Thus, the given series converges. 

Example 5 Divergence using the limit comparison test

Test the series $\sum_{k=1}^{\infty} \frac{3k + 2}{\sqrt{k}(3k - 5)}$ for convergence.

Solution For large values of k , the general term of the given series

$$a_k = \frac{3k + 2}{\sqrt{k}(3k - 5)}$$

seems to be similar to

$$b_k = \frac{3k}{\sqrt{k}(3k)} = \frac{1}{\sqrt{k}}$$

To apply the limit comparison test, we first compute the limit

$$\lim_{k \rightarrow \infty} \frac{a_k}{b_k} = \lim_{k \rightarrow \infty} \frac{\frac{3k+2}{\sqrt{k}(3k-5)}}{\frac{1}{\sqrt{k}}} = \lim_{k \rightarrow \infty} \frac{3k+2}{3k-5} = 1$$

Because the limit is finite and positive, it follows that the given series behaves like the series $\sum 1/\sqrt{k}$, which is a divergent p -series ($p = \frac{1}{2}$). We conclude that the given series diverges. 

Example 6 Applying the limit comparison test

Test the series $\sum_{k=1}^{\infty} \frac{7k + 100}{e^{k/5} - 70}$ for convergence.

Solution For large k ,

$$a_k = \frac{7k + 100}{e^{k/5} - 70}$$

seems to behave like $b_k = \frac{k}{e^{k/5}}$ and indeed, we find that

$$\lim_{k \rightarrow \infty} \frac{\frac{7k+100}{e^{k/5}-70}}{\frac{k}{e^{k/5}}} = \lim_{k \rightarrow \infty} \frac{e^{k/5}(7k+100)}{k(e^{k/5}-70)} = 7$$

In Example 3, Section 8.3, we showed that the series $\sum k/e^{k/5}$ converges, and therefore the limit comparison test tells us that the given series also converges. 

Occasionally, two series Σa_k and Σb_k appear to have similar convergence properties, but it turns out that

$$\lim_{k \rightarrow \infty} \frac{a_k}{b_k}$$

is either 0 or ∞ , and the limit comparison test therefore does not apply. In such cases, it is often useful to have the following generalization of the limit comparison test.

Theorem 8.15 The zero-infinity limit comparison test

Suppose $a_k > 0$ and $b_k > 0$ for all sufficiently large k .

If $\lim_{k \rightarrow \infty} \frac{a_k}{b_k} = 0$ and Σb_k converges, then the series Σa_k converges.
 If $\lim_{k \rightarrow \infty} \frac{a_k}{b_k} = \infty$ and Σb_k diverges, then the series Σa_k diverges.

Proof: The first part of the proof is outlined in Problem 59; the second part follows similarly. ♦

Example 7 Convergence of the q -log series

Show that the series $\sum_{k=1}^{\infty} \frac{\ln k}{k^q}$ converges if $q > 1$ and diverges if $q \leq 1$. We call this the q -log series.

Solution We will carry out this proof in three parts:

Case I: ($q > 1$)

Let p be a number that satisfies $1 < p < q$, and let

$$a_k = \frac{\ln k}{k^q} \quad \text{and} \quad b_k = \frac{1}{k^p}$$

Then

$$\begin{aligned} \lim_{k \rightarrow \infty} \frac{a_k}{b_k} &= \lim_{k \rightarrow \infty} \frac{\frac{\ln k}{k^q}}{\frac{1}{k^p}} \\ &= \lim_{k \rightarrow \infty} \frac{\ln k}{k^{q-p}} && \text{This is the indeterminate form } \infty/\infty \text{ because } q - p > 0. \\ &= \lim_{k \rightarrow \infty} \frac{\frac{1}{k}}{(q-p)k^{q-p-1}} && \text{l'Hôpital's rule} \\ &= \lim_{k \rightarrow \infty} \frac{1}{(q-p)k^{q-p}} && \text{Because } q - p > 0 \\ &= 0 \end{aligned}$$

Because $\Sigma 1/k^p$ converges (p -series with $p > 1$), the series converges by the zero-infinity limit comparison test.

Case II: ($q < 1$)

Now, let p satisfy $q < p < 1$. Then with a_k and b_k defined as in case I, we have

$$\lim_{k \rightarrow \infty} \frac{a_k}{b_k} = \lim_{k \rightarrow \infty} \frac{\ln k}{k^{q-p}} = \lim_{k \rightarrow \infty} [(\ln k)k^{p-q}] = \infty$$

Because $\sum_{k=1}^{\infty} b_k = \sum_{k=1}^{\infty} \frac{1}{k^p}$ diverges (we know $p < 1$), it follows from the zero-infinity comparison test that $\sum_{k=1}^{\infty} \frac{\ln k}{k^q}$ diverges when $q < 1$.

Case III: ($q = 1$)

Here, the series is $\sum_{k=1}^{\infty} \frac{\ln k}{k}$, which diverges by the integral test (where we let $u = \ln x$; $du = \frac{1}{x} dx$):

$$\int_1^{\infty} \frac{\ln x}{x} dx = \lim_{b \rightarrow \infty} \left. \frac{(\ln x)^2}{2} \right|_1^b = \infty$$

PROBLEM SET 8.4

Level 1

The most common series used for comparison are given in Problems 1 and 2. Tell when each converges and when it diverges.

1. geometric series: $\sum_{k=0}^{\infty} r^k$ 2. p -series: $\sum_{k=1}^{\infty} \frac{1}{k^p}$

Each series in Problems 3–12 can be compared to the geometric series or p -series given in Problems 1 and 2. State which, and then determine whether it converges or diverges.

3. $\sum_{k=1}^{\infty} \cos^k \left(\frac{\pi}{6} \right)$
4. $\sum_{k=0}^{\infty} (0.5)^k$
5. $\sum_{k=0}^{\infty} (1.5)^k$
6. $\sum_{k=0}^{\infty} 2^{k/2}$
7. $\sum_{k=1}^{\infty} \frac{1}{k}$
8. $\sum_{k=1}^{\infty} \frac{1}{k^{0.5}}$
9. $\sum_{k=1}^{\infty} \frac{1}{k^{3/2}}$
10. $\sum_{k=1}^{\infty} \sqrt{\frac{2}{k}}$
11. $\sum_{k=0}^{\infty} 1^k$
12. $\sum_{k=1}^{\infty} e^k$

Test the series in Problems 13–42 for convergence.

13. $\sum_{k=1}^{\infty} \frac{1}{k^2 + k}$
14. $\sum_{k=1}^{\infty} \frac{1}{k^2 + 3k + 2}$
15. $\sum_{k=1}^{\infty} \frac{1}{\sqrt{k}}$
16. $\sum_{k=1}^{\infty} \frac{1}{k\sqrt{k}}$
17. $\sum_{k=1}^{\infty} \frac{1}{\sqrt{2k+3}}$
18. $\sum_{k=1}^{\infty} \frac{1}{\sqrt{k(k+1)}}$

19. $\sum_{k=1}^{\infty} \frac{1}{\sqrt{k^3 + 2}}$
20. $\sum_{k=1}^{\infty} \frac{1}{\sqrt{k^2 + 1}}$
21. $\sum_{k=1}^{\infty} \frac{2k^2}{k^4 - 4}$
22. $\sum_{k=1}^{\infty} \frac{k+1}{k^2 + 1}$
23. $\sum_{k=1}^{\infty} \frac{(k+2)(k+3)}{k^{7/2}}$
24. $\sum_{k=1}^{\infty} \frac{(k+1)^3}{k^{9/2}}$
25. $\sum_{k=1}^{\infty} \frac{2k+3}{k^2 + 3k + 2}$
26. $\sum_{k=1}^{\infty} \frac{3k^2 + 2}{k^2 + 3k + 2}$
27. $\sum_{k=1}^{\infty} \frac{k}{(k+2)2^k}$
28. $\sum_{k=1}^{\infty} \frac{5}{4^k + 3}$
29. $\sum_{k=1}^{\infty} \frac{1}{k(k+2)}$
30. $\sum_{k=1}^{\infty} \frac{1}{(k+2)(k+3)}$
31. $\sum_{k=1}^{\infty} \frac{1}{\sqrt{k} 2^k}$
32. $\sum_{k=1}^{\infty} \frac{1,000}{\sqrt{k} 3^k}$
33. $\sum_{k=2}^{\infty} \frac{1}{\sqrt{k} \ln k}$
34. $\sum_{k=1}^{\infty} \frac{\ln k}{\sqrt{2k+3}}$
35. $\sum_{k=1}^{\infty} \frac{2k^3 + k + 1}{k^3 + k^2 + 1}$
36. $\sum_{k=1}^{\infty} \frac{6k^3 - k - 4}{k^3 - k^2 - 3}$
37. $\sum_{k=1}^{\infty} \frac{|\sin(k!)|}{k^2}$
38. $\sum_{k=1}^{\infty} \sin \frac{1}{k}$
39. $\sum_{k=1}^{\infty} \frac{6k^2 + 2k + 1}{k^{1.1}(4k^2 + k + 4)}$
40. $\sum_{k=1}^{\infty} \frac{6k^2 + 2k + 1}{k^{0.9}(4k^2 + k + 4)}$
41. $\sum_{k=1}^{\infty} \frac{\sqrt[5]{k}}{\sqrt[4]{k^3 + 2} \sqrt[8]{k}}$
42. $\sum_{k=1}^{\infty} \frac{\sqrt{k}}{\sqrt[3]{k^3 + 1} \sqrt[6]{k^5}}$

Level 2

Test the series given in Problems 43–50 for convergence.

43. $\sum_{k=1}^{\infty} \frac{1}{k^3 + 4}$

44. $\sum_{k=2}^{\infty} \frac{\ln k}{k-1}$

45. $\sum_{k=1}^{\infty} \frac{\ln(k+1)}{(k+1)^3}$

46. $\sum_{k=1}^{\infty} \frac{\ln k}{k^2}$

47. $\sum_{k=2}^{\infty} \frac{1}{(k+3)(\ln k)^{1.1}}$

48. $\sum_{k=2}^{\infty} \frac{1}{(k+3)(\ln k)^{0.9}}$

49. $\sum_{k=1}^{\infty} k^{(1-k)/k}$

50. $\sum_{k=1}^{\infty} k^{(1+k)/k}$

51. Show that the series

$$\sum_{k=1}^{\infty} \frac{k^2}{(k+3)!} = \frac{1}{4!} + \frac{4}{5!} + \frac{9}{6!} + \cdots$$

converges by using the limit comparison test.

52. Show that the series

$$1 + \frac{1}{1 \cdot 3} + \frac{1}{1 \cdot 3 \cdot 5} + \frac{1}{1 \cdot 3 \cdot 5 \cdot 7} + \cdots + \frac{2^k k!}{(2k+1)!} + \cdots$$

converges. *Hint:* Compare with the convergent series $\sum 1/k!$.

53. Use a comparison test to show that

$$\sum_{k=2}^{\infty} \frac{1}{(\ln k)^{\ln k}}$$

converges. *Hint:* Use the fact that $\ln k > e^2$ for sufficiently large k .

Level 3

54. a. Let $\{a_n\}$ be a positive sequence such that

$$\lim_{k \rightarrow \infty} k^p a_k$$

exists. Use the limit comparison test to show that $\sum a_k$ converges if $p > 1$.

b. Show that $\sum e^{-k^2}$ converges. *Hint:* Show that $\lim_{k \rightarrow \infty} k^2 e^{-k^2}$ exists and use part a.

55. Let $\sum a_k$ be a series of positive terms and let $\{b_k\}$ be a sequence of positive numbers that converge to a positive number. Show that $\sum a_k$ converges if and only if $\sum a_k b_k$ converges. *Hint:* Use the limit comparison test.

56. Suppose $0 \leq a_k \leq A$ for some number A and $b_k \geq k^2$. Show that $\sum a_k/b_k$ converges.

57. Suppose $a_k > 0$ for all k and that $\sum a_k$ converges. Show that $\sum 1/a_k$ diverges.

58. **Think Tank Problem** Suppose that $0 < a_k < 1$ for every k , and that $\sum a_k$ converges. Use a comparison test to show that $\sum a_k^2$ converges. Find a counterexample where $\sum a_k^2$ converges, but $\sum a_k$ diverges.

59. Show that if $a_k > 0$, $b_k > 0$, and

$$\lim_{k \rightarrow \infty} \frac{a_k}{b_k} = 0$$

then $\sum a_k$ converges whenever $\sum b_k$ converges. *Hint:* Show that there is an integer N such that $a_k < b_k$ if $k > N$. Then use a comparison test.

60. Show that

$$\sum_{k=2}^{\infty} \frac{1}{k(\ln k)^q}$$

converges if and only if $q > 1$.

8.5 THE RATIO TEST AND THE ROOT TEST

IN THIS SECTION: Ratio test, root test

In this section, we develop two additional tests, the *ratio test* and the *root test*.

Ratio Test

Intuitively, a series of positive terms $\sum a_k$ converges if and only if the sequence $\{a_k\}$ decreases *rapidly enough* toward 0. One way to measure the rate at which the sequence $\{a_k\}$ is decreasing is to examine the ratio a_{k+1}/a_k as k grows large. This approach leads to the following result.

Theorem 8.16 The ratio test

Given the series $\sum a_k$ with $a_k > 0$, suppose that

$$\lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} = L$$

The **ratio test** states the following:

If $L < 1$, then $\sum a_k$ converges.

If $L > 1$, or if L is infinite, then $\sum a_k$ diverges.

If $L = 1$, then the test is inconclusive.

Proof: In a sense, the ratio test is a limit comparison test in which $\sum a_k$ is compared to itself. We will use the direct comparison test to show that $\sum a_k$ converges if $L < 1$. Choose R such that $0 < L < R < 1$. Then there exists some $N > 0$ such that

$$\frac{a_{k+1}}{a_k} < R \text{ for all } k \geq N \quad \text{The ratios } \frac{a_{k+1}}{a_k} \text{ are eventually very close to } L.$$

Therefore,

$$\begin{aligned} a_{N+1} &< a_N R \\ a_{N+2} &< a_{N+1} R < a_N R^2 \\ a_{N+3} &< a_{N+2} R < a_{N+1} R^2 < a_N R^3 \\ &\vdots \end{aligned}$$

The geometric series

$$\sum_{k=1}^{\infty} a_N R^k = a_N R + a_N R^2 + \cdots + a_N R^n + \cdots$$

converges because $0 < R < 1$, which means the “smaller” series

$$\sum_{k=1}^{\infty} a_{N+k} = a_{N+1} + a_{N+2} + \cdots + a_{N+n} + \cdots$$

also converges by the direct comparison test. Thus, $\sum a_k$ converges, because we can discard a finite number of terms ($k \leq N$).

The proof of the second part is similar, except now we choose R such that

$$\lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} = L > R > 1$$

and show there exists some $M > 0$ so that $a_{M+k} > a_M R^k$.

To prove that the ratio test is inconclusive if $L = 1$, it is enough to note that the harmonic series

harmonic series diverges

$$\sum_{k=1}^{\infty} \frac{1}{k} \text{ diverges with } \lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} = \lim_{k \rightarrow \infty} \frac{\frac{1}{k+1}}{\frac{1}{k}} = 1$$



while the p -series

p -series converges for $p > 1$

$$\sum_{k=1}^{\infty} \frac{1}{k^2} \text{ converges with } \lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} = \lim_{k \rightarrow \infty} \frac{\frac{1}{(k+1)^2}}{\frac{1}{k^2}} = 1$$

Example 1 Convergence using the ratio test

Test the series $\sum_{k=1}^{\infty} \frac{2^k}{k!}$ for convergence.

You will find the ratio test most useful with series involving factorials or exponentials.

Solution Let $a_k = \frac{2^k}{k!}$ and note that

$$\begin{aligned} L &= \lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} \\ &= \lim_{k \rightarrow \infty} \frac{\frac{2^{k+1}}{(k+1)!}}{\frac{2^k}{k!}} \\ &= \lim_{k \rightarrow \infty} \frac{k! 2^{k+1}}{(k+1)! 2^k} \\ &= \lim_{k \rightarrow \infty} \frac{2}{k+1} \\ &= 0 \end{aligned}$$

Thus $L < 1$, and the ratio test tells us that the given series converges.

Example 2 Divergence using the ratio test

Test the series $\sum_{k=1}^{\infty} \frac{k^k}{k!}$ for convergence.

Solution Let $a_k = \frac{k^k}{k!}$ and note that

$$\begin{aligned} L &= \lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} \\ &= \lim_{k \rightarrow \infty} \frac{\frac{(k+1)^{k+1}}{(k+1)!}}{\frac{k^k}{k!}} \\ &= \lim_{k \rightarrow \infty} \frac{k!(k+1)^{k+1}}{k^k(k+1)!} \\ &= \lim_{k \rightarrow \infty} \frac{(k+1)^k}{k^k} \\ &= \lim_{k \rightarrow \infty} \left(1 + \frac{1}{k}\right)^k \\ &= e \\ &> 1 \end{aligned}$$

Because $L > 1$, the given series diverges by the ratio test.

Window

TECHNOLOGY NOTE

It is instructive to compare and contrast two different ways you can use technology to help you with problems such as Example 2. First, as you have seen, you can use a calculator or a computer to calculate the partial sums directly. This table is shown below at the left.

A second procedure is to use the ratio test for Example 2. In this case, we are interested in

$$\lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} = \lim_{k \rightarrow \infty} \frac{\frac{(k+1)^{k+1}}{(k+1)!}}{\frac{k^k}{k!}} = \lim_{k \rightarrow \infty} \left(1 + \frac{1}{k}\right)^k$$

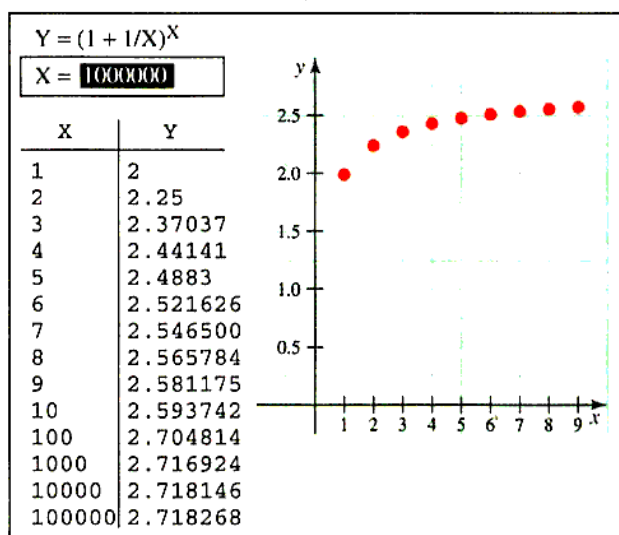
We can look at terms of the sequence $b_k = \left(1 + \frac{1}{k}\right)^k$ to see whether this limit is greater than, less than, or equal to 1. This is shown below on the right.

By table:

N	$\sum_{k=1}^N \frac{k^k}{k!}$
1	1
2	3
3	7.5
4	18.16667
5	44.20833
6	109.0083
7	272.4097
8	688.5113
9	1756.138
10	4511.87
11	11659.53
12	30273.46
13	78912.3
14	206375.3
15	541239.9

By graph:

$$\lim_{k \rightarrow \infty} \left(1 + \frac{1}{k}\right)^k$$



Interactive The partial sums seem to diverge.

Interactive The ratio test gives a limit greater than 1.

Finally, you might test this using a calculator to verify that it seems to diverge. A word of caution: in some cases, plugging in certain large values for k on various types of calculators will give an erroneous answer (1) instead of a number that approaches Euler's constant e . This is due to the so-called "machine limit" and is a direct consequence of round-off error accumulations. A more detailed discussion about this topic can be found in Appendix C, section "Calculator Experiments."

Example 3 Ratio test fails

Test the series $\sum_{k=2}^{\infty} \frac{1}{2k-3}$ for convergence.

Solution Let $a_k = \frac{1}{2k-3}$ and find

$$L = \lim_{k \rightarrow \infty} \frac{\frac{1}{2(k+1)-3}}{\frac{1}{2k-3}} = \lim_{k \rightarrow \infty} \frac{2k-3}{2k-1} = 1$$

The ratio test is inconclusive. We can use the integral test or the limit comparison test to determine convergence. Note that

$$\sum_{k=2}^{\infty} \frac{1}{2k-3} \text{ is similar to the known divergent series } \sum_{k=2}^{\infty} \frac{1}{2k}$$

so we conclude the given series is divergent.

Example 4 Convergence of a series of power functions

Find all numbers $x > 0$ for which the series

$$\sum_{k=1}^{\infty} k^3 x^k = x + 2^3 x^2 + 3^3 x^3 + \dots$$

converges. This is called a *power series*, and will be discussed in more detail in Sections 8.7 and 8.8.

Solution Viewing x as fixed (that is, as a constant), we apply the ratio test:

$$L = \lim_{k \rightarrow \infty} \frac{(k+1)^3 x^{k+1}}{k^3 x^k} = \lim_{k \rightarrow \infty} \left(\frac{k+1}{k} \right)^3 x = x$$

Thus, the series converges if $L = x < 1$ and diverges if $x > 1$. When $x = 1$, the series becomes $\sum_{k=1}^{\infty} k^3$, which diverges by the divergence test. ■

Root Test

Of all the tests we have developed, the divergence test is perhaps the easiest to apply, because it involves simply computing $\lim_{k \rightarrow \infty} a_k$ and observing whether that limit is zero. Unfortunately, most “interesting” series have $\lim_{k \rightarrow \infty} a_k = 0$, so the divergence test cannot be used to determine whether they converge or diverge. However, the following result shows that it may be possible to say more about the convergence of $\sum a_k$ by examining what happens to $\sqrt[k]{a_k}$ as $k \rightarrow \infty$. This test will prove particularly useful with a series involving a k th power.

Theorem 8.17 The root test

Given the series $\sum a_k$ with $a_k > 0$, suppose that

$$\lim_{k \rightarrow \infty} \sqrt[k]{a_k} = L$$

The **root test** states the following:

- If $L < 1$, then $\sum a_k$ converges.
- If $L > 1$, or if L is infinite, then $\sum a_k$ diverges.
- If $L = 1$, then the test is inconclusive.

Proof: The proof of this theorem is similar to that of the ratio test and is outlined in Problem 59. ◆

Example 5 Convergence with the root test

Test the series $\sum_{k=2}^{\infty} \frac{1}{(\ln k)^k}$ for convergence.

Solution Let $a_k = \frac{1}{(\ln k)^k}$ and note that

$$L = \lim_{k \rightarrow \infty} \sqrt[k]{a_k} = \lim_{k \rightarrow \infty} \sqrt[k]{(\ln k)^{-k}} = \lim_{k \rightarrow \infty} \frac{1}{\ln k} = 0$$

Because $L < 1$, the root test tells us that the given series converges.

Example 6 Divergence with the root test

Test the series $\sum_{k=1}^{\infty} \left(1 + \frac{1}{k}\right)^{k^2}$ for convergence.

$$\begin{aligned} \text{Solution} \quad L &= \lim_{k \rightarrow \infty} \left[\left(1 + \frac{1}{k}\right)^{k^2} \right]^{1/k} && \text{Root test} \\ &= \lim_{k \rightarrow \infty} \left(1 + \frac{1}{k}\right)^k \\ &= e \\ &> 1 \end{aligned}$$

Since $L > 1$, the series diverges by the root test.

We have seen that the ratio test is especially useful for series $\sum a_k$ for which the general term a_k involves factorials or powers, and the root test applies naturally if a_k involves a power of k . However, both the ratio test and the root test fail with certain very simple series such as the p -series $\sum \frac{1}{k^p}$ since

$$\lim_{k \rightarrow \infty} \frac{\frac{1}{(k+1)^p}}{\frac{1}{k^p}} = \lim_{k \rightarrow \infty} \left(\frac{k}{k+1} \right)^p = 1^p = 1 \quad \text{Inconclusive ratio test}$$

$$\lim_{k \rightarrow \infty} \sqrt[k]{\frac{1}{k^p}} = \left[\lim_{k \rightarrow \infty} k^{1/k} \right]^{-p} = 1^{-p} = 1 \quad \text{Inclusive root test where l'Hôpital's rule is used to evaluate the limit.}$$

Example 7 Testing for convergence

Test the following series for convergence:

$$\sum_{k=0}^{\infty} \frac{k!}{1 \cdot 4 \cdot 7 \cdots (3k+1)} = 1 + \frac{1!}{1 \cdot 4} + \frac{2!}{1 \cdot 4 \cdot 7} + \frac{3!}{1 \cdot 4 \cdot 7 \cdot 10} + \cdots$$

Solution Let $a_k = \frac{k!}{1 \cdot 4 \cdot 7 \cdots (3k+1)}$. Because a_k involves $k!$, we try the ratio test.

$$\begin{aligned} \frac{a_{k+1}}{a_k} &= \frac{\frac{(k+1)!}{1 \cdot 4 \cdot 7 \cdots [3(k+1)+1]}}{\frac{k!}{1 \cdot 4 \cdot 7 \cdots (3k+1)}} \\ &= \frac{(k+1)! [1 \cdot 4 \cdot 7 \cdots (3k+1)]}{k! [1 \cdot 4 \cdot 7 \cdots (3k+4)]} \\ &= \frac{k+1}{3k+4} \end{aligned}$$

Thus,

$$L = \lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} = \lim_{k \rightarrow \infty} \frac{k+1}{3k+4} = \frac{1}{3}$$

Because $L < 1$, the given series converges by the ratio test.

Recall that $n!$ denotes the product of all positive integers up to n . So $(3k)!$ is not the same as $3k!$ and $(3k+1)!$ is not the same as $1 \cdot 4 \cdot 7 \cdots (3k+1)$.

Example 8 Testing for convergence

Test the series $\sum_{k=1}^{\infty} \frac{k!}{k^k}$ for convergence.

Solution The general term $a_k = \frac{k!}{k^k}$ involves a k th power, which suggests using the root test, but the presence of the factorial $k!$ makes using the ratio test much more reasonable.

$$\begin{aligned} L &= \lim_{k \rightarrow \infty} \frac{\frac{(k+1)!}{(k+1)^{k+1}}}{\frac{k!}{k^k}} \\ &= \lim_{k \rightarrow \infty} \frac{(k+1)!k^k}{k!(k+1)^{k+1}} \\ &= \lim_{k \rightarrow \infty} \frac{(k+1)k^k}{(k+1)^{k+1}} \\ &= \lim_{k \rightarrow \infty} \left(\frac{k}{k+1} \right)^k \\ &= \lim_{k \rightarrow \infty} \frac{1}{\left(1 + \frac{1}{k}\right)^k} \\ &= \frac{1}{e} \\ &< 1 \end{aligned}$$

Since $L < 1$, the series converges by the ratio test.

PROBLEM SET 8.5

Level 1

- What does this say? Describe the ratio test.
- What does this say? Describe the root test.

Use either the ratio test or the root test to determine the convergence of the series given in Problems 3–26.

3. $\sum_{k=1}^{\infty} \frac{1}{k!}$

4. $\sum_{k=1}^{\infty} \frac{k!}{2^k}$

5. $\sum_{k=1}^{\infty} \frac{k!}{2^{3k}}$

6. $\sum_{k=1}^{\infty} \frac{3^k}{k!}$

7. $\sum_{k=1}^{\infty} \frac{k}{2^k}$

8. $\sum_{k=1}^{\infty} \frac{2^k}{k^2}$

9. $\sum_{k=1}^{\infty} \frac{k^{100}}{e^k}$

11. $\sum_{k=1}^{\infty} k \left(\frac{4}{3} \right)^k$

13. $\sum_{k=1}^{\infty} \left(\frac{2}{k} \right)^k$

15. $\sum_{k=1}^{\infty} \frac{k^5}{10^k}$

17. $\sum_{k=1}^{\infty} \left(\frac{k}{3k+1} \right)^k$

19. $\sum_{k=1}^{\infty} \frac{k!}{(k+2)^4}$

10. $\sum_{k=1}^{\infty} k e^{-k}$

12. $\sum_{k=1}^{\infty} k \left(\frac{3}{4} \right)^k$

14. $\sum_{k=1}^{\infty} \frac{k^{10} 2^k}{k!}$

16. $\sum_{k=1}^{\infty} \frac{3^k}{k^2}$

18. $\sum_{k=1}^{\infty} \frac{3k+1}{2^k}$

20. $\sum_{k=1}^{\infty} \frac{k^5 + 100}{k!}$

21. $\sum_{k=1}^{\infty} \frac{(k!)^2}{(2k)!}$

22. $\sum_{k=1}^{\infty} \frac{(k!)^2}{[(2k)!]^2}$

23. $\sum_{k=1}^{\infty} k^2 2^{-k}$

24. $\sum_{k=1}^{\infty} k^4 3^{-k}$

25. $\sum_{k=1}^{\infty} \left(\frac{k-2}{k}\right)^{k^2}$

26. $\sum_{k=1}^{\infty} \left(\frac{k}{2k+1}\right)^k$

Test the series in Problems 27–46 for convergence. Justify your answers (that is, state explicitly which test you are using).

27. $\sum_{k=1}^{\infty} \frac{1,000}{k}$

28. $\sum_{k=1}^{\infty} \frac{5,000}{k\sqrt{k}}$

29. $\sum_{k=1}^{\infty} \frac{5k+2}{k2^k}$

30. $\sum_{k=1}^{\infty} \frac{(k!)^2}{k^k}$

31. $\sum_{k=1}^{\infty} \frac{\sqrt{k}!}{2^k}$

32. $\sum_{k=1}^{\infty} \frac{3k+5}{k3^k}$

33. $\sum_{k=1}^{\infty} \frac{2^k k!}{k^k}$

34. $\sum_{k=1}^{\infty} \frac{2^{2k} k!}{k^k}$

35. $\sum_{k=1}^{\infty} \frac{\sqrt{k+1}}{k^{k+0.5}}$

36. $\sum_{k=1}^{\infty} \frac{1}{k^k}$

37. $\sum_{k=1}^{\infty} \frac{k!}{(k+1)!}$

38. $\sum_{k=1}^{\infty} \frac{2^{1,000k}}{k^{k/2}}$

39. $\sum_{k=1}^{\infty} \frac{k}{4k^3 - 5}$

40. $\sum_{k=1}^{\infty} \frac{k^2 + 1}{(k^2 + 2)k^2}$

41. $\sum_{k=1}^{\infty} \left(1 + \frac{1}{k}\right)^{-k^2}$

42. $\sum_{k=1}^{\infty} \left(\frac{k+2}{k}\right)^{-k^2}$

43. $\sum_{k=1}^{\infty} \left|\frac{\cos k}{2^k}\right|$

44. $\sum_{k=1}^{\infty} \left|\frac{\sin k}{3^k}\right|$

45. $\sum_{k=2}^{\infty} \left(\frac{\ln k}{k}\right)^k$

46. $\sum_{k=2}^{\infty} \frac{1}{(\ln k)^k}$

Level 2

In Problems 47–54, assume $x \geq 0$ and find all x for which the given series converges.

47. $\sum_{k=1}^{\infty} k^2 x^k$

48. $\sum_{k=1}^{\infty} kx^k$

49. $\sum_{k=1}^{\infty} \frac{(x+0.5)^k}{k\sqrt{k}}$

50. $\sum_{k=1}^{\infty} \frac{(3x+0.4)^k}{k^2}$

51. $\sum_{k=1}^{\infty} \frac{x^k}{k!}$

52. $\sum_{k=1}^{\infty} \frac{x^{2k}}{k}$

53. $\sum_{k=1}^{\infty} (ax)^k, a > 0$

54. $\sum_{k=1}^{\infty} kx^{2k}$

55. Use the root test to show that $\sum k^p e^{-k}$ converges for any fixed positive real number p . What does this imply about the following improper integral?

$$\int_1^{\infty} x^p e^{-x} dx$$

56. Consider the series

$$\sum_{k=1}^{\infty} 2^{-k+(-1)^k}$$

What can you conclude about the convergence of this series if you use each of the following?

- the ratio test
- the root test

Level 3

57. Let $\{a_k\}$ be a sequence of positive numbers and suppose that

$$\lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} = L$$

where $0 < L < 1$.

a. Show that $\lim_{k \rightarrow \infty} a_k = 0$.

Hint: Note that $\sum a_k$ converges.

b. Show that

$$\lim_{k \rightarrow \infty} \frac{x^k}{k!} = 0$$

for any $x > 0$.

58. Consider the series

$$1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{4} + \frac{1}{4} + \frac{1}{8} + \frac{1}{8} + \frac{1}{16} + \cdots$$

- Show that the ratio test fails.
- What is the result of applying the root test?

59. Prove the root test by completing the steps in the following outline:

a. Let

$$\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = R$$

If $R < 1$, choose x so $R < x < 1$. Explain why there is an N such that

$$0 \leq \sqrt[n]{a_n} \leq x$$

for all $n \geq N$. Then use the direct comparison test to show that $\sum a_n$ converges.

- b. If $R > 1$, explain why $a_n > 1$ for all but a finite number of n . Use the divergence test to show $\sum a_n$ diverges.
- c. Find two series $\sum a_n$ and $\sum b_n$ with $R = 1$, so that $\sum a_n$ converges but $\sum b_n$ diverges.

60. EXPLORATION PROBLEM Recall from Problem 50, Section 8.1, that the Fibonacci sequence $\{a_n\}$ is

$$1, 1, 2, 3, 5, 8, 13, 21, \dots$$

where $a_1 = 1$, $a_2 = 1$, and the rest of the terms are defined recursively by the formula

$$a_{n+1} = a_n + a_{n-1}$$

for $n = 2, 3, \dots$. Determine whether the series of reciprocal Fibonacci numbers

$$\sum_{k=1}^{\infty} \frac{1}{a_k} = 1 + 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{5} + \dots$$

converges or diverges.

8.6 ALTERNATING SERIES; ABSOLUTE AND CONDITIONAL CONVERGENCE

IN THIS SECTION: *Alternating series test, error estimates for alternating series, absolute and conditional convergence, summary of convergence tests, rearrangement of terms in an absolutely convergent series*

We now turn our attention from nonnegative term series to series with both positive and negative terms.

There are two classes of series for which the successive terms alternate in sign, and each of these series is appropriately called an **alternating series**:

$$\text{odd-indexed terms are negative: } \sum_{k=1}^{\infty} (-1)^k a_k = -a_1 + a_2 - a_3 + \dots$$

$$\text{even-indexed terms are negative: } \sum_{k=1}^{\infty} (-1)^{k+1} a_k = a_1 - a_2 + a_3 - \dots$$

where in both cases $a_k > 0$.

Alternating Series Test

In general, just knowing that $\lim_{k \rightarrow \infty} a_k = 0$ tells us very little about the convergence properties of the series $\sum a_k$, but it turns out that an alternating series must converge if the absolute value of its terms decreases monotonically toward zero. This result, first established by Leibniz in the 17th century, may be stated as shown in the following theorem.

Theorem 8.18 Alternating series test

An alternating series

$$\sum_{k=1}^{\infty} (-1)^k a_k \quad \text{or} \quad \sum_{k=1}^{\infty} (-1)^{k+1} a_k$$

where $a_k > 0$ for all k converges if both of the following two conditions are satisfied:

1. $\lim_{k \rightarrow \infty} a_k = 0$
2. $\{a_k\}$ is a decreasing sequence; that is, $a_{k+1} \leq a_k$ for all k .

Proof: We will show that when an alternating series of the form

$$\sum_{k=1}^{\infty} (-1)^{k+1} a_k$$

The alternating series test only applies to series whose terms strictly alternate in sign, not to series with other patterns of + and - signs. See Example 6.

satisfies the two required properties, it converges. The steps for the other type of alternating series are similar. For this proof we are given that

$$\lim_{k \rightarrow \infty} a_k = 0 \quad \text{and} \quad a_{k+1} \leq a_k \text{ for all } k$$

We need to prove that the sequence of partial sums $\{S_n\}$ converges, where

$$S_n = \sum_{k=1}^n (-1)^{k+1} a_k = a_1 - a_2 + a_3 - a_4 + \cdots + (-1)^{n+1} a_n$$

Our strategy will be to show first that the sequence of *even-indexed* partial sums $\{S_{2n}\}$ is *increasing* and converges to a certain limit L . Then we will show that the sequence of *odd-indexed* partial sums $\{S_{2n-1}\}$ also converges to L .

To understand why we would think to break up S_n into even- and odd-indexed partial sums, consider Figure 8.16. Start at the origin.

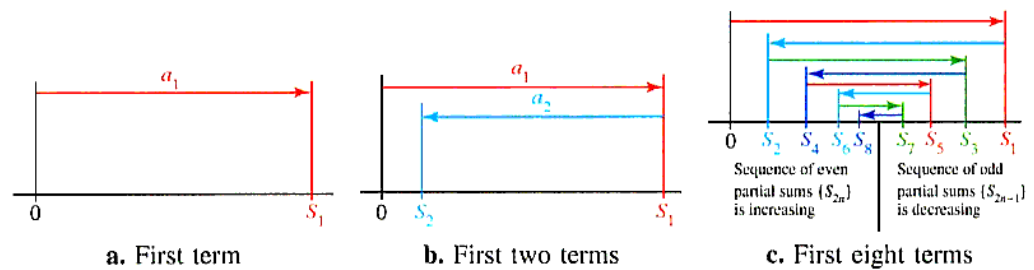


Figure 8.16 Alternating series test

Because $a_1 > 0$, S_1 will be somewhere to the right of 0 as shown in Figure 8.16a. We know that $a_2 \leq a_1$ (decreasing sequence) so S_2 is to the left of S_1 but to the right of 0, as shown in Figure 8.16b. As you continue this process you can see that the partial sums oscillate back and forth. Because $a_k \rightarrow 0$, the successive steps are getting smaller and smaller, as shown in Figure 8.16c.

Note that the even partial sums of the alternating series satisfy

$$\begin{aligned} S_2 &= a_1 - a_2 \\ S_4 &= (a_1 - a_2) + (a_3 - a_4) \\ &\vdots \\ S_{2n} &= (a_1 - a_2) + (a_3 - a_4) + \cdots + (a_{2n-1} - a_{2n}) \end{aligned}$$

Because $\{a_k\}$ is a decreasing sequence, each of the quantities in parentheses ($a_{2j-1} - a_{2j}$) is nonnegative, and it follows that $\{S_{2n}\}$ is an increasing sequence:

$$S_2 \leq S_4 \leq S_6 \leq \cdots$$

Moreover, because

$$\begin{aligned} S_2 &= a_1 - a_2 \leq a_1 \\ S_4 &= a_1 - (a_2 - a_3) - a_4 \leq a_1 \\ &\vdots \\ S_{2n} &= a_1 - (a_2 - a_3) - (a_4 - a_5) - \cdots - (a_{2n-1} - a_{2n}) - a_{2n} \leq a_1 \end{aligned}$$

the sequence of even partial sums $\{S_{2n}\}$ is bounded above by a_2 , and because it is also an increasing sequence, the BMCT tells us that this sequence must converge, say, to L ; that is,

$$\lim_{n \rightarrow \infty} S_{2n} = L$$

Next, consider $\{S_{2n-1}\}$, the sequence of odd partial sums. Because $S_{2n} - S_{2n-1} = a_{2n}$ and $\lim_{n \rightarrow \infty} a_n = 0$, it follows that $S_{2n-1} = S_{2n} - a_{2n}$ and

$$\lim_{n \rightarrow \infty} S_{2n-1} = \lim_{n \rightarrow \infty} S_{2n} - \lim_{n \rightarrow \infty} a_{2n} = L - 0 = L$$

Since the sequence of even partial sums and the sequence of odd partial sums both converge to L , it follows that $\{S_n\}$ converges to L , and the alternating series must converge (in fact, to L). $\blacklozenge$

NOTE: When you are testing the alternating series $\sum (-1)^{k+1} a_k$ for convergence, it is wise to begin by computing $\lim_{k \rightarrow \infty} a_k$. If $\lim_{k \rightarrow \infty} a_k \neq 0$, you know immediately that the series diverges (by the divergence test), and no further computation is necessary. However, if this limit is 0, you can show that the alternating series converges by verifying that $\{a_k\}$ is a decreasing sequence.

Example 1 Convergence using the alternating series test

Test the series $\sum_{k=1}^{\infty} \frac{(-1)^k}{k}$ for convergence. This series is called the **alternating harmonic series**.

Solution The series can be expressed as $\sum (-1)^k a_k$, where $a_k = \frac{1}{k}$. We have $\lim_{k \rightarrow \infty} \frac{1}{k} = 0$ and because

$$\frac{1}{k+1} < \frac{1}{k}$$

for all $k > 0$, the sequence $\{a_k\}$ is decreasing. Thus, the alternating series test tells us that the given series must converge. $\blacksquare$

Remember that the a_k used in the alternating series test does NOT include the $(-1)^{k+1}$.

Example 2 Using l'Hôpital's rule with the alternating series test

Test the series $\sum_{k=1}^{\infty} \frac{(-1)^{k+1} \ln k}{k}$ for convergence.

Solution Express the series in the form $\sum (-1)^{k+1} a_k$, where $a_k = \frac{\ln k}{k}$ (note that $a_k > 0$ for $k > 1$). The graph of $y = (\ln x)/x$ is shown in Figure 8.17.

By applying l'Hôpital's rule, we find that

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x} = \lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

It remains to show that the sequence $\{a_k\}$ is decreasing. We can do this by setting $f(x) = (\ln x)/x$ and noting that the derivative

$$f'(x) = \frac{1 - \ln x}{x^2}$$

satisfies $f'(x) < 0$ for all $x > e$. Thus, the sequence $\{a_k\}$ is (eventually) decreasing for all $k > 3 > e$. The conditions of the alternating series test are satisfied, and the given alternating series must converge. $\blacksquare$

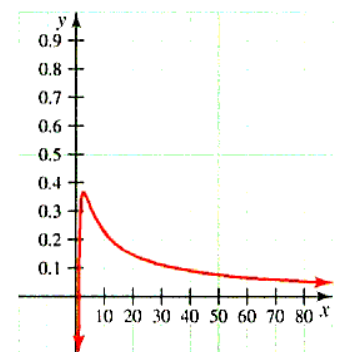


Figure 8.17 Graph of $y = \frac{\ln x}{x}$

Example 3 Divergence of an alternating series

Test the series $\sum_{k=1}^{\infty} (-1)^{k+1} e^{1/k}$ for convergence.

Solution Since

$$\lim_{k \rightarrow \infty} e^{1/k} = \exp\left(\lim_{k \rightarrow \infty} \frac{1}{k}\right) = e^0 = 1 \neq 0$$

the series diverges by the divergence test.

ALTERNATING p -SERIES The series $\sum_{k=2}^{\infty} \frac{(-1)^{k+1}}{k^p}$ is called the **alternating p -series**.

The following example shows that the alternating p -series converges for all $p > 0$.

Example 4 Alternating p -series

Prove that the alternating p -series $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^p}$ converges for $p > 0$.

Solution If $a^k = \frac{1}{k^p}$, then $\lim_{k \rightarrow \infty} a_k = 0$ for $p > 0$. To show that the sequence $\{a_k\}$ is decreasing, we note that

$$\frac{a_{k+1}}{a_k} = \frac{\frac{1}{(k+1)^p}}{\frac{1}{k^p}} = \frac{k^p}{(k+1)^p} < 1$$

so $a_k > a_{k+1}$. Thus, the alternating p -series converges for all $p > 0$.

Error Estimates for Alternating Series

For any convergent series with sum L , the n th partial sum is an approximation to L that we expect to improve as n increases. In general, it is difficult to know how large an n to pick in order to ensure that the approximation will have a desired degree of accuracy. However, if the series in question satisfies the conditions of the alternating series test, the following theorem provides a convenient criterion for making this determination.

Theorem 8.19 The error estimate for an alternating series

Suppose an alternating series

$$\sum_{k=1}^{\infty} (-1)^k a_k \quad \text{or} \quad \sum_{k=1}^{\infty} (-1)^{k+1} a_k$$

satisfies the conditions of the alternating series test; namely, $\lim_{k \rightarrow \infty} a_k = 0$ and $\{a_k\}$ is a decreasing sequence ($a_{k+1} \leq a_k$). If the series has sum S , then

$$|S - S_n| \leq a_{n+1}$$

where S_n is the n th partial sum of the series.

■ **What this says** If an alternating series satisfies the conditions of the alternating series test, you can approximate the sum of the series by using the n th partial sum $\{S_n\}$, and your error will have an absolute value no greater than the first term left off (namely, a_{n+1}).

Proof: We will prove the result for the second form of the alternating series and leave the first form for the reader. That is, let

$$S = \sum_{k=1}^{\infty} (-1)^{k+1} a_k \quad \text{and} \quad S_n = \sum_{k=1}^n (-1)^{k+1} a_k$$

Begin with $S - S_n$:

$$\begin{aligned} S - S_n &= \sum_{k=1}^{\infty} (-1)^{k+1} a_k - \sum_{k=1}^n (-1)^{k+1} a_k \\ &= \sum_{k=n+1}^{\infty} (-1)^{k+1} a_k \\ &= (-1)^{n+2} a_{n+1} + (-1)^{n+3} a_{n+3} + \cdots \\ &= (-1)^n (-1)^2 a_{n+1} + (-1)^n (-1)^3 a_{n+2} + (-1)^n (-1)^4 a_{n+3} + \cdots \\ &\quad \vdots \\ &= (-1)^n [a_{n+1} - a_{n+2} + a_{n+3} - a_{n+4} + \cdots] \\ &= (-1)^n [a_{n+1} - (a_{n+2} - a_{n+3}) - (a_{n+4} - a_{n+5}) - \cdots] \end{aligned}$$

Because the sequence $\{a_n\}$ is decreasing, we have $a_k - a_{k+1} \geq 0$ for all k , and it follows that

$$|S - S_n| = |a_{n+1} - (a_{n+2} - a_{n+3}) - (a_{n+4} - a_{n+5}) - \cdots| \leq a_{n+1}$$

since every quantity in parentheses is positive.

Example 5 Error estimate for an alternating series

Consider the convergent alternating series $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^4}$.

- Estimate the sum of the series by taking the sum of the first four terms. How accurate is this estimate?
- How many terms of the series are necessary to estimate its sum with three decimal-place accuracy? What is this estimate?

Solution

- Let $a_k = \frac{1}{k^4}$ and let S denote the actual sum of the series. The error estimate tells us that

$$|S - S_4| \leq a_5$$

where S_4 is the sum of the first four terms of the series. Using a calculator, we find

$$S_4 = \frac{1}{1^4} - \frac{1}{2^4} + \frac{1}{3^4} - \frac{1}{4^4} \approx 0.9459394$$

and $a_5 = \frac{1}{5^4} = 0.0016$. This first term omitted is positive. Thus, if we estimate S by $S_4 \approx 0.9459$, we incur an error of at most 0.0016, which means that

$$\begin{aligned} |S - S_4| &\leq 0.0016 \\ 0.9459394 &\leq S \leq 0.9459394 + 0.0016 \\ 0.9459394 &\leq S \leq 0.9475394 \end{aligned}$$

You may think that "three decimal place accuracy" means an error no greater than 0.001, but to insure against round-off error, we really need 0.0005.

- b. Because we want to approximate S by a partial sum S_n with three-decimal-place accuracy, we can allow an error of no more than 0.0005. According to Theorem 8.19, we want to find an index n such that $a_{n+1} \leq 0.0005$; that is,

$$\begin{aligned} \frac{1}{(n+1)^4} &\leq 0.0005 \\ \frac{1}{0.0005} &\leq (n+1)^4 \\ \sqrt[4]{2,000} &\leq n+1 \\ 6.687403 - 1 &\leq n \end{aligned}$$

Thus, $n \geq 5.687403$, so we need at least 6 terms to achieve the required accuracy. We find that

$$S_6 = \frac{1}{1^4} - \frac{1}{2^4} + \frac{1}{3^4} - \frac{1}{4^4} + \frac{1}{5^4} - \frac{1}{6^4} \approx 0.94677$$

The first term left off (namely $\frac{1}{7^4} \approx 0.00042$) is positive, so

$$\begin{aligned} 0.94677 &< S < 0.94677 + 0.00042 \\ 0.94677 &< S < 0.94719 \end{aligned}$$

Both bounds round to 0.947, so $S \approx 0.947$, with three decimal-place accuracy.

Absolute and Conditional Convergence

The convergence tests we have developed cannot be applied to a series that has mixed terms or does not strictly alternate. In such cases, it is often useful to apply the following result.

Theorem 8.20 The absolute convergence test

A series of real numbers $\sum a_k$ must converge if the related absolute value series $\sum |a_k|$ converges.

Proof: Assume $\sum |a_k|$ converges and let $b_k = a_k + |a_k|$ for all k . Note that

$$b_k = \begin{cases} 2|a_k| & \text{if } a_k > 0 \\ 0 & \text{if } a_k \leq 0 \end{cases}$$

Thus, we have $0 \leq b_k \leq 2|a_k|$ for all k . Because the series $\sum |a_k|$ converges and both $\sum b_k$ and $\sum |a_k|$ are series of nonnegative terms, the direct comparison test tells us that the dominated series $\sum b_k$ also converges. Finally, because

$$a_k = b_k - |a_k|$$

and both $\sum b_k$ and $\sum |a_k|$ converge, it follows that $\sum a_k$ must also converge.

Example 6 Convergence using the absolute convergence test

Test the series

$$1 + \frac{1}{4} + \frac{1}{9} - \frac{1}{16} - \frac{1}{25} + \frac{1}{36} + \frac{1}{49} + \frac{1}{64} - \frac{1}{81} - \frac{1}{100} + \cdots$$

for convergence. This is the series in which the absolute value of the general term is $\frac{1}{k^2}$ and the pattern of the signs is $+++- --+++- -- \cdots$.

Solution This is not a series of positive terms, nor is it strictly alternating. However, we find that the corresponding series of absolute values is the convergent p -series

$$1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots = \sum_{k=1}^{\infty} \frac{1}{k^2}$$

Because the absolute value series converges, the absolute convergence test assures us that the given series also converges. ■

Example 7 Convergence of a trigonometric series

Test the series $\sum_{k=1}^{\infty} \frac{\sin k}{2^k}$ for convergence.

Solution Because $\sin k$ takes on both positive and negative values, the series cannot be analyzed by methods that apply only to series of positive terms. Moreover, note that the series is not strictly alternating:

$$\sum_{k=1}^{\infty} \frac{\sin k}{2^k} = 0.421 + 0.227 + 0.018 - 0.047 - 0.030 - 0.004 + 0.005 + \cdots$$

The corresponding series of absolute values is

$$\sum_{k=1}^{\infty} \left| \frac{\sin k}{2^k} \right|$$

which is dominated by the convergent geometric series $\sum \frac{1}{2^k}$ because $|\sin k| \leq 1$ for all k ; that is,

$$0 \leq \left| \frac{\sin k}{2^k} \right| \leq \frac{1}{2^k} \quad \text{for all } k$$

Therefore, the absolute convergence test assures us that the given series converges. The graph of the related continuous function $y = (\sin x)/2^x$ is shown in Figure 8.18. ■

If $\sum a_k$ converges, then $\sum |a_k|$ may either converge or diverge. The two cases that can occur are given the following special names.

ABSOLUTELY AND CONDITIONALLY CONVERGENT The series $\sum a_k$ is **absolutely convergent** (or converges absolutely) if the related series $\sum |a_k|$ converges. The series $\sum a_k$ is **conditionally convergent** (or converges conditionally) if it converges but $\sum |a_k|$ diverges.

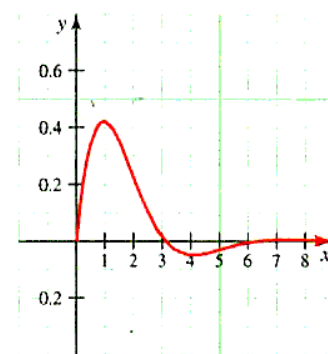


Figure 8.18 Graph of $y = \frac{\sin x}{2^x}$

Note that a convergent series is either absolutely convergent or conditionally convergent, but it cannot be both.

We remark that the series in Examples 6 and 7 are both absolutely convergent, but the alternating harmonic series

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}$$

is conditionally convergent because it converges (see Example 1), whereas the related series of absolute value terms $\sum 1/k$ (the harmonic series) diverges.

The ratio test and the root test for positive-term series can be generalized to apply to arbitrary series. The following is a statement and proof of the generalized ratio test.

Theorem 8.21 The generalized ratio test

For the series $\sum a_k$, suppose $a_k \neq 0$ for $k \geq 1$ and that

$$\lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| = L$$

where L is a real number or ∞ . Then:

If $L < 1$, the series $\sum a_k$ converges absolutely and hence converges.

If $L > 1$ or if L is infinite, the series $\sum a_k$ diverges.

If $L = 1$, the test is inconclusive.

Proof: Assume $L < 1$; then the positive series $\sum |a_k|$ converges by the ratio test, and hence $\sum a_k$ converges absolutely. Assume $L > 1$; then the sequence $\{|a_k|\}$ is eventually increasing, which means that $\{a_k\}$ cannot converge to 0 as $k \rightarrow \infty$, so $\sum a_k$ diverges by the divergence test. Assume $L = 1$. See Problem 58 for this part of the proof. ♦

Example 8 Convergence and divergence using the generalized ratio test

Find all values of x for which the series $\sum kx^k$ converges.

Solution Let $a_k = kx^k$.

$$\begin{aligned} L &= \lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| \\ &= \lim_{k \rightarrow \infty} \left| \frac{(k+1)x^{k+1}}{kx^k} \right| \\ &= \lim_{k \rightarrow \infty} \left(\frac{k+1}{k} \right) \left| \frac{x^{k+1}}{x^k} \right| \\ &= \lim_{k \rightarrow \infty} \left(1 + \frac{1}{k} \right) |x| \\ &= |x| \end{aligned}$$

Thus, by the generalized ratio test, since $L = |x|$ the series converges for $|x| < 1$ and diverges if $|x| > 1$. If $x = 1$, the series is $\sum k(1)^k$, which clearly diverges by the divergence test. Similarly, if $x = -1$, $\sum k(-1)^k$ diverges by the divergence test. Thus, the series converges for $|x| < 1$ and diverges for $|x| \geq 1$. ■

The statement and proof of a generalized root test is left to the problems (see Problem 56).

Summary of Convergence Tests

This concludes our basic study of convergence tests. There are no firm rules for deciding how to test the convergence of a given series Σa_k , but we can offer a few guidelines (Table 8.1).

Table 8.1 Guidelines for determining convergence of the series Σa_k

Series with known convergence properties	
Geometric series, $\sum_{k=1}^{\infty} ar^k$	diverges if $ r \geq 1$, and converges if $ r < 1$ with sum $S = \frac{a}{1-r}$
p -series, $\sum_{k=1}^{\infty} \frac{1}{k^p}$	converges if $p > 1$ and diverges if $p \leq 1$
Harmonic series, $\sum_{k=1}^{\infty} \frac{1}{k}$	special case of the p -series where $p = 1$; it diverges
q -log series, $\sum_{k=1}^{\infty} \frac{\ln k}{k^q}$	converges if $q > 1$ and diverges if $q \leq 1$
Divergence test	Compute $\lim_{k \rightarrow \infty} a_k$; if this limit is anything but 0, the series diverges.
Tests for series Σa_k with $a_k > 0$	
Limit comparison test	Check to see whether Σa_k is similar in appearance to a series Σb_k . If $\lim_{k \rightarrow \infty} a_k/b_k = L$, where L is finite and positive, then Σa_k and Σb_k either both converge or both diverge.
Ratio test	If a_k involves $k!$, k^p , or c^k , try the ratio test. Set $\lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} = L$. Converges if $L < 1$, diverges if $L > 1$, inconclusive if $L = 1$.
Root test	If it is easy to find $\lim_{k \rightarrow \infty} \sqrt[k]{a_k} = L$, try the root test. Converges if $L < 1$, diverges if $L > 1$, is inconclusive if $L = 1$.
Integral test	If f is continuous, positive, and decreasing, and if $a_k = f(k)$ for all k , then $\sum_{k=1}^{\infty} a_k$ and the improper integral $\int_1^{\infty} f(x) dx$ either both converge or both diverge. Consider using this test when f is easy to integrate or if a_k involves a logarithm, a trigonometric function, or an inverse trigonometric function.
Direct comparison test	If $0 \leq a_k \leq c_k$ and $\sum_{k=1}^{\infty} c_k$ converges, then $\sum_{k=1}^{\infty} a_k$ converges. If $0 \leq d_k \leq a_k$ and $\sum_{k=1}^{\infty} d_k$ diverges, then $\sum_{k=1}^{\infty} a_k$ diverges.
Zero-infinity limit comparison test	If $\lim_{k \rightarrow \infty} \frac{a_k}{b_k} = 0$ and Σb_k converges, then the series Σa_k converges. If $\lim_{k \rightarrow \infty} \frac{a_k}{b_k} = \infty$ and Σb_k diverges, then the series Σa_k diverges.
Tests for series Σa_k that do not require $a_k > 0$	
Test for absolute convergence	Take the absolute value of each term and proceed to test for convergence of positive series, as summarized above. 1. If $\Sigma a_k $ converges, then Σa_k converges. 2. If $\Sigma a_k $ diverges, then the series is <i>not absolutely convergent</i> , but may still converge <i>conditionally</i> .
Alternating series test (conditional convergence)	The alternating series $\Sigma (-1)^k a_k$ or $\Sigma (-1)^{k+1} a_k$ with $a_k > 0$ converges if $a_{k+1} \leq a_k$ for $k = 1, 2, \dots$ and $\lim_{k \rightarrow \infty} a_k = 0$.

Rearrangement of Terms in an Absolutely Convergent Series

You may be surprised to learn that if the terms in a conditionally convergent series are rearranged (that is, the order of the summands is changed), the new series may not converge or it may converge to a different sum from that of the original series!

For example, we know that the alternating harmonic series

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}$$

converges conditionally, and it can be shown (see Problem 51) that

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \frac{1}{8} + \frac{1}{9} - \cdots = \ln 2$$

If we rearrange this series by placing two successive subtracted terms after each added term, we obtain

$$\begin{aligned} 1 - \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{6} - \frac{1}{8} + \frac{1}{5} - \cdots &= \left(1 - \frac{1}{2}\right) - \frac{1}{4} + \left(\frac{1}{3} - \frac{1}{6}\right) - \frac{1}{8} + \cdots \\ &= \frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \cdots \\ &= \frac{1}{2} \left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots\right) \\ &= \frac{1}{2} \ln 2 \end{aligned}$$

In general, it can be shown that if $\sum a_k$ is conditionally convergent, there is a rearrangement of the terms of $\sum a_k$ so that the sum is equal to *any* specified finite number. In Problem 52, for example, you are asked to rearrange the terms of the series

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}$$

so that the sum is $\frac{3}{2} \ln 2$.

This information may be somewhat unsettling, because it is reasonable to expect a sum to be unaffected by the order in which the summands are taken. Absolutely convergent series behave more in the way we would expect. In fact, if the series $\sum a_k$ converges absolutely with sum S , then any rearrangement of the terms also converges absolutely to S . A detailed discussion of rearrangement of terms of a series requires the techniques of advanced calculus.

PROBLEM SET 8.6

Level 1

1. **What does this say?** Discuss absolute versus conditional convergence.
2. **What does this say?** Discuss the convergence of the alternating p -series.

Determine whether each series in Problems 3–30 converges absolutely, converges conditionally, or diverges.

$$3. \sum_{k=1}^{\infty} \frac{(-1)^{k+1}k}{k^2 + 1}$$

$$4. \sum_{k=1}^{\infty} \frac{(-1)^{k+1}k^2}{k^3 + 1}$$

$$5. \sum_{k=1}^{\infty} \frac{(-1)^{k+1}k}{2k + 1}$$

$$6. \sum_{k=1}^{\infty} \frac{(-1)^{k+1}k^2}{k^2 + 1}$$

$$7. \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^{3/2}}$$

$$9. \sum_{k=1}^{\infty} (-1)^{k+1} \frac{k^2}{e^k}$$

$$11. \sum_{k=1}^{\infty} (-1)^k \frac{(1 + k^2)}{k^3}$$

$$13. \sum_{k=2}^{\infty} (-1)^k \frac{k!}{\ln k}$$

$$15. \sum_{k=1}^{\infty} (-1)^{k+1} \frac{2^k}{k!}$$

$$8. \sum_{k=1}^{\infty} \frac{(-1)^{k+1}k}{2^k}$$

$$10. \sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{k}}$$

$$12. \sum_{k=1}^{\infty} \frac{(-1)^{k+1}k!}{k^k}$$

$$14. \sum_{k=1}^{\infty} \frac{(-1)^k (2k)!}{k^k}$$

$$16. \sum_{k=1}^{\infty} \frac{(-3)^k (k + 1)}{k!}$$

17. $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{2^{2k+1}}{k!}$ 18. $\sum_{k=2}^{\infty} \frac{(-1)^{k+1}}{\ln k}$
19. $\sum_{k=1}^{\infty} \frac{(-1)^{k+1} k}{(k+1)(k+2)}$ 20. $\sum_{k=2}^{\infty} \frac{(-1)^{k+1}}{(\ln k)^4}$
21. $\sum_{k=2}^{\infty} \frac{(-1)^{k+1}}{\ln(\ln k)}$ 22. $\sum_{k=2}^{\infty} \frac{(-1)^{k+1}}{k \ln k}$
23. $\sum_{k=1}^{\infty} \frac{(-1)^{k+1} \ln k}{k}$ 24. $\sum_{k=2}^{\infty} \frac{(-1)^{k+1} k}{\ln k}$
25. $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{\ln k}{k^2}$ 26. $\sum_{k=1}^{\infty} \frac{(-1)^{k+1} k}{(k+2)^2}$
27. $\sum_{k=1}^{\infty} (-1)^{k+1} \left(\frac{k}{k+1} \right)^k$ 28. $\sum_{k=2}^{\infty} (-1)^{k+1} \frac{\ln(\ln k)}{k \ln k}$
29. $\sum_{k=1}^{\infty} (-1)^{k+1} \left(\frac{1}{k} \right)^{1/k}$ 30. $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{k^5 5^{k+2}}{2^{3k}}$

Level 2

Given the series in Problems 31–36

- a. Estimate the sum of the series by taking the sum of the first four terms. How accurate is this estimate?
- b. How many terms of the series are necessary to estimate its sum with three-place accuracy? What is this estimate?

31. $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{2^{2k-2}}$ 32. $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k!}$
33. $\sum_{k=1}^{\infty} \frac{(-1)^k}{k^2}$ 34. $\sum_{k=1}^{\infty} \left(\frac{-1}{3} \right)^{k+1}$
35. $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^3}$ 36. $\sum_{k=1}^{\infty} \left(\frac{-1}{5} \right)^k$

Use the generalized ratio test in Problems 37–42 to find all numbers x for which the given series converges.

37. $\sum_{k=1}^{\infty} \frac{x^k}{k}$ 38. $\sum_{k=1}^{\infty} \frac{(2x)^k}{\sqrt{k}}$
39. $\sum_{k=1}^{\infty} \frac{2^k x^k}{k!}$ 40. $\sum_{k=1}^{\infty} \frac{(k+2)x^k}{k^2(k+3)}$
41. $\sum_{k=1}^{\infty} (-1)^{k+1} \left(\frac{x}{k} \right)^k$ 42. $\sum_{k=1}^{\infty} (-1)^k k^p x^k$,
for $p > 0$

43. Find an upper bound for the error if the convergent alternating series

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}$$

is approximated by the partial sum

$$S_5 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5}$$

44. Find an upper bound for the error if the convergent alternating series

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^2}$$

is approximated by the partial sum

$$S_5 = 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \frac{1}{5^2}$$

45. Find an upper bound for the error if the convergent alternating series

$$\sum_{k=2}^{\infty} \frac{(-1)^k}{\ln k}$$

is approximated by the partial sum

$$S_7 = \frac{1}{\ln 2} - \frac{1}{\ln 3} + \frac{1}{\ln 4} - \frac{1}{\ln 5} + \frac{1}{\ln 6} - \frac{1}{\ln 7}$$

46. Find an upper bound for the error if the convergent alternating series

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1} k}{2^k}$$

is approximated by the partial sum

$$S_6 = \frac{1}{2} - \frac{2}{2^2} + \frac{3}{2^3} - \frac{4}{2^4} + \frac{5}{2^5} - \frac{6}{2^6}$$

47. For what numbers p does the convergent alternating series

$$\sum_{k=2}^{\infty} \frac{(-1)^{k+1}}{k(\ln k)^p}$$

converge? For what numbers p does it converge absolutely?

48. Test the series $\sum_{k=1}^{\infty} \frac{\sin \sqrt[4]{2}}{k^2}$ for convergence.

49. Use series methods to evaluate $\lim_{k \rightarrow \infty} \frac{x^k}{k!}$ where x is a real number.

Level 3

50. Show that the sequence
- $\{a_n\}$
- converges, where

$$a_n = \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} - \ln n$$

51. Show that
- $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} = \ln 2$
- by completing the following steps.

- a. Let $S_m = \sum_{k=1}^m \frac{(-1)^{k+1}}{k}$ and let $H_m = \sum_{k=1}^m \frac{1}{k}$. Show that $S_{2m} = H_{2m} - H_m$.
- b. It can be shown (see Problem 50)

$$\lim_{m \rightarrow \infty} \left(1 + \frac{1}{2} + \cdots + \frac{1}{m} - \ln m \right)$$

exists. The value of this limit is a number $\gamma \approx 0.57722$, called **Euler's constant**. Use this fact, along with the relationship in part a, to show that

$$\lim_{m \rightarrow \infty} S_n = \ln 2.$$

Hint: Note that

$$\begin{aligned} S_{2m} &= H_{2m} - H_m \\ &= [H_{2m} - \ln(2m)] - [H_m - \ln m] \\ &\quad + \ln(2m) - \ln m \end{aligned}$$

52. Consider the following rearrangement of the conditionally convergent harmonic series

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}:$$

$$1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \frac{1}{9} + \frac{1}{11} - \frac{1}{6} + \cdots$$

- a. Let S_n and H_n denote the n th partial sum of the given rearranged series and the harmonic series $\sum_{k=1}^{\infty} \frac{1}{k}$, respectively. Show that if n is a multiple of 3, say $n = 3m$, then

$$S_{3m} = H_{4m} - \frac{1}{2}H_{2m} - \frac{1}{2}H_m$$

- b. Show that $\lim_{n \rightarrow \infty} S_n = \frac{3}{2} \ln 2$.

Hint: You may find it helpful to use

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{2} + \cdots + \frac{1}{n} - \ln n \right) = \gamma$$

given in Problem 51b.

- 53.
- Journal Problem**
- (
- School Science and Mathematics*
- by Michael Brozinsky)* Show that the series

$$\begin{aligned} &1 + \frac{1}{2} + \frac{1}{3} - \frac{1}{4} - \frac{1}{5} - \frac{1}{6} \\ &+ \frac{1}{7} + \frac{1}{8} + \frac{1}{9} - \cdots + + + \end{aligned}$$

converges and find its sum.

54. Suppose $\{a_k\}$ is a sequence with the property that $|a_n| < A^n$ for some positive number A and all positive integers n . Show that the series $\sum a_k x^k$ converges absolutely for $|x| \leq 1/A$.
55. **Think Tank Problem** What (if anything) is wrong with the following computation?

$$\begin{aligned} &1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \cdots \\ &= 1 + \left(\frac{1}{2} - 1\right) + \frac{1}{3} + \left(\frac{1}{4} - \frac{1}{2}\right) + \frac{1}{5} + \left(\frac{1}{6} - \frac{1}{3}\right) \\ &\quad + \cdots \\ &= \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots\right) - 1 - \frac{1}{2} - \frac{1}{3} - \frac{1}{4} - \cdots \\ &= \left(1 + \frac{1}{2} + \frac{1}{3} + \cdots\right) - \left(1 + \frac{1}{2} + \frac{1}{3} + \cdots\right) \\ &= 0 \end{aligned}$$

56. Prove the generalized root test, which may be stated as follows: Suppose that If $L < 1$, the series $\sum a_k$ converges absolutely. If $L > 1$ the series $\sum a_k$ diverges. If $L = 1$, the test is inconclusive.
57. Show that if

$$L = \lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| = 1$$

the series $\sum a_k$ can either converge or diverge.

Hint: Find a convergent series with $L = 1$ and a divergent series that satisfies the same condition.

58. Let $\sum (-1)^{k+1} a_k$ be an alternating series such that $\{a_k\}$ is decreasing and

$$\lim_{k \rightarrow \infty} a_k = 0$$

Show that the sequence of odd partial sums $\{S_{2n-1}\}$ is decreasing.

59. **Think Tank Problem** Let $\sum a_k$ be a series that converges. Either prove that $\sum a_k^2$ must also converge or find a counterexample.
60. **Think Tank Problem** Construct a divergent alternating series $\sum_{k=1}^{\infty} (-1)^{k+1} a_k$ such that $\lim_{k \rightarrow \infty} a_k = 0$, but $\{a_k\}$ is not decreasing.

*Volume 82, 1982, p. 175.

8.7 POWER SERIES

IN THIS SECTION: *Convergence of a power series, term-by-term differentiation and integration of power series*

A *power series* is a series whose terms are power functions of the form $a_k(x - c)^k$. We shall examine the properties of such series and shall find that under reasonable conditions, they can be differentiated and integrated term-by-term.

An infinite series of the form

$$\sum_{k=0}^{\infty} a_k(x - c)^k = a_0 + a_1(x - c) + a_2(x - c)^2 + \cdots$$

is called a **power series** in $(x - c)$. The numbers $a_0, a_1, a_2, \dots$ are the *coefficients* of the power series. If $c = 0$, the series has the form

$$\sum_{k=0}^{\infty} a_k x^k = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots$$

A power series in x may be considered as an extension of a polynomial in x .

Convergence of a Power Series

For what numbers x does a given power series converge? The following theorem answers this question for the case where $c = 0$.

Theorem 8.22 Convergence of a power series

For a power series $\sum_{k=0}^{\infty} a_k x^k$, exactly one of the following is true:

1. The series converges for all x .
2. The series converges only for $x = 0$.
3. The series converges absolutely for all x in an open interval $(-R, R)$ and diverges for $|x| > R$. It may either converge or diverge at the endpoints of the interval, $x = R$ and $x = -R$.

Proof: The proof is found in most advanced calculus textbooks.

The set of values for which a series converges is called the **convergence set** for the series. From this theorem we know it is always an interval, if one considers a point and the entire real line as intervals. The following three examples show each of these possibilities.

Example 1 When the convergence set is the entire x -axis

Show that the power series $\sum_{k=0}^{\infty} \frac{x^k}{k!}$ converges for all x .

Solution If $x = 0$, then the series is trivial and converges. For $x \neq 0$, we use the generalized ratio test to find

$$L = \lim_{k \rightarrow \infty} \left| \frac{\frac{x^{k+1}}{(k+1)!}}{\frac{x^k}{k!}} \right| = \lim_{k \rightarrow \infty} \left| \frac{x^{k+1} k!}{(k+1)! x^k} \right| = \lim_{k \rightarrow \infty} \frac{|x|}{k+1} = 0$$

Because $L = 0$ and thus $L < 1$, the series converges for all x .

Example 2 Convergence only at the point $x = 0$

Show that the power series $\sum_{k=1}^{\infty} k!x^k$ converges only when $x = 0$.

Solution We use the generalized ratio test to find

$$L = \lim_{k \rightarrow \infty} \left| \frac{(k+1)!x^{k+1}}{k!x^k} \right| = \lim_{k \rightarrow \infty} (k+1)|x|$$

For any x other than 0, we have $L = \infty$. Hence, the power series converges only when $x = 0$. ■

Example 3 Convergence set is a bounded interval

Find the convergence set for the power series

$$\sum_{k=1}^{\infty} \frac{x^k}{\sqrt{k}}$$

Solution Using the generalized ratio test, we find

$$L = \lim_{k \rightarrow \infty} \left| \frac{\frac{x^{k+1}}{\sqrt{k+1}}}{\frac{x^k}{\sqrt{k}}} \right| = \lim_{k \rightarrow \infty} \left| \frac{\sqrt{k}}{\sqrt{k+1}} \right| |x| = |x|$$

The power series converges absolutely if $|x| < 1$, and diverges if $|x| > 1$. We must also check the endpoints of the interval $|x| < 1$, namely $x = -1$ and $x = 1$:

$$x = -1: \sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{k}} \text{ converges (by the alternating series test)}$$

$$\frac{1}{\sqrt{k+1}} < \frac{1}{\sqrt{k}} \text{ and } \lim_{k \rightarrow \infty} \frac{1}{\sqrt{k}} = 0$$

$$x = 1: \sum_{k=1}^{\infty} \frac{(1)^k}{\sqrt{k}} \text{ diverges (} p\text{-series with } p = \frac{1}{2} < 1\text{)}.$$

Thus, the given power series converges for $-1 \leq x < 1$ and diverges otherwise.

According to Theorem 8.22, the set of numbers for which the power series $\sum a_k x^k$ converges is an interval centered at $x = 0$. We call this the **interval of convergence** of the power series. If this interval has length $2R$, then R is called the **radius of convergence** of the series, as shown in Figure 8.19

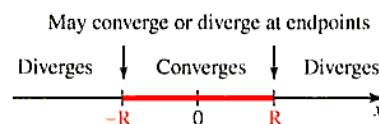


Figure 8.19 The interval of convergence of a power series

If the series converges only for $x = 0$, the series has radius of convergence $R = 0$, and if it converges for all x , we say that $R = \infty$.

In Example 1, we found that $\sum x^k/k!$ converges for all x , so $R = \infty$ and the interval of convergence is the entire real line. In Example 2, we found that the power series $\sum k!x^k$ has radius of convergence 0. In Example 3, we found that $\sum x^k/\sqrt{k}$ converges on $[-1, 1)$, so $R = 1$.

The procedure for determining the convergence set of a power series is further illustrated in the following examples.

Example 4 Finding the interval of convergence for a power series

Find the interval of convergence for the power series $\sum_{k=1}^{\infty} \frac{2^k x^k}{k}$. What is the radius of convergence?

Solution We apply the generalized ratio test to find

$$L = \lim_{k \rightarrow \infty} \left| \frac{\frac{2^{k+1} x^{k+1}}{k+1}}{\frac{2^k x^k}{k}} \right| = \lim_{k \rightarrow \infty} \left| \frac{2k}{k+1} \right| |x| = 2|x|$$

Thus, the series converges absolutely for $2|x| < 1$; that is for

$$-\frac{1}{2} < x < \frac{1}{2}$$

The radius of convergence is $R = \frac{1}{2}$. Checking the endpoints, we find:

Do not forget to check the endpoints.

$$x = \frac{1}{2}: \quad \sum_{k=1}^{\infty} \frac{2^k}{k} \left(\frac{1}{2}\right)^k = \sum_{k=1}^{\infty} \frac{1}{k} \text{ diverges (harmonic series)}$$

$$x = -\frac{1}{2}: \quad \sum_{k=1}^{\infty} \frac{2^k}{k} \left(-\frac{1}{2}\right)^k = \sum_{k=1}^{\infty} \frac{(-1)^k}{k} \text{ converges (alternating harmonic series)}$$

The interval of convergence is $-\frac{1}{2} \leq x < \frac{1}{2}$. ■

Example 5 Finding an interval of convergence

Find the radius of convergence and the interval of convergence for the power series $\sum_{k=1}^{\infty} \frac{2^k x^k}{k!}$.

Solution Applying the generalized ratio test, we find

$$L = \lim_{k \rightarrow \infty} \left| \frac{\frac{2^{k+1} x^{k+1}}{(k+1)!}}{\frac{2^k x^k}{k!}} \right| = \lim_{k \rightarrow \infty} \left| \frac{2}{k+1} \right| |x| = 0$$

Thus, the power series converges absolutely for x , and the radius of convergence is infinite. ■

Example 6 Radius of convergence using the root test

Find the radius of convergence of the power series $\sum_{k=1}^{\infty} \left(\frac{k+1}{k}\right)^{k^2} x^k$.

Solution Using the root test to examine absolute convergence, we find that

$$\begin{aligned} L &= \lim_{k \rightarrow \infty} \sqrt[k]{\left| \left(\frac{k+1}{k} \right)^{k^2} x^k \right|} \\ &= \lim_{k \rightarrow \infty} \left| \left(\frac{k+1}{k} \right)^{k^2} x^k \right|^{1/k} \\ &= \lim_{k \rightarrow \infty} \left(1 + \frac{1}{k} \right)^k |x| \\ &= e |x| \end{aligned}$$

Thus, the power series converges absolutely for $e|x| < 1$; that is, for $-e^{-1} < x < e^{-1}$. It follows that the radius of convergence is $R = e^{-1}$. We do not need to test the endpoints because we are not finding the interval of convergence. ■

In some applications, we will encounter power series of the form

$$\sum_{k=0}^{\infty} a_k (x - c)^k = a_0 + a_1(x - c) + a_2(x - c)^2 + \cdots$$

in which each term is a constant times a power of $x - c$ where c is a constant. The intervals of convergence are intervals of the form $-R < x - c < R$, including possibly one or both of the endpoints $x = c - R$ and $x = c + R$. The procedure for determining the interval of convergence of such power series is illustrated in the following example.

Example 7 Power series in $(x - c)$

Find the interval of convergence of the power series $\sum_{k=0}^{\infty} \frac{(x+1)^k}{3^k}$.

Solution Applying the generalized ratio test, we obtain

$$L = \lim_{k \rightarrow \infty} \left| \frac{\frac{(x+1)^{k+1}}{3^{k+1}}}{\frac{(x+1)^k}{3^k}} \right| = \lim_{k \rightarrow \infty} \left| \frac{3^k}{3^{k+1}} \right| |x+1| = \lim_{k \rightarrow \infty} \frac{1}{3} |x+1|$$

Thus, the power series converges absolutely for

$$\begin{aligned} \frac{1}{3} |x+1| &< 1 \\ |x+1| &< 3 \\ -3 &< x+1 < 3 \\ -4 &< x < 2 \end{aligned}$$

Checking the endpoints, we find

$$\begin{aligned} x = -4: \quad \sum_{k=0}^{\infty} \frac{(-4+1)^k}{3^k} &= \sum_{k=0}^{\infty} (-1)^k \text{ diverges (by oscillation)} \\ x = 2: \quad \sum_{k=0}^{\infty} \frac{(2+1)^k}{3^k} &= \sum_{k=0}^{\infty} 1^k \text{ diverges (by the divergence test)} \end{aligned}$$

The interval of convergence is $(-4, 2)$.

Term-by-Term Differentiation and Integration of Power Series

The power series $\sum_{k=0}^{\infty} a_k x^k$ can be thought of as a function defined on its interval of convergence. It is reasonable to ask if this function is differentiable and if it is integrable, and if so, what are its derivative and integral. This is useful because we can form new series using differentiation or integration of known series. If we regard a power series as an “infinite polynomial,” we would expect to be able to differentiate and integrate it term by term. The following theorem shows that this procedure is legitimate on the interval of absolute convergence.

Theorem 8.23 Term-by-term differentiation and integration of a power series

A power series $\sum_{k=0}^{\infty} a_k x^k$ with radius of convergence $R > 0$ can be differentiated or integrated term by term on its interval of absolute convergence $-R < x < R$. More specifically, if $f(x) = \sum_{k=0}^{\infty} a_k x^k$ for $|x| < R$, then for $|x| < R$ we have

$$f'(x) = \sum_{k=1}^{\infty} k a_k x^{k-1} = a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + \cdots$$

and

$$\int f(x) dx = \int \left(\sum_{k=0}^{\infty} a_k x^k \right) dx = \sum_{k=0}^{\infty} \left(\int a_k x^k dx \right) = \sum_{k=0}^{\infty} \frac{a_k}{k+1} x^{k+1} + C$$

Whether or not the differentiated or integrated series converges at the endpoints of the interval of convergence needs to be checked in each case. From the theorem we only know what happens inside the interval of convergence.

What this says In many ways, a function defined by a power series behaves like a polynomial. It is differentiable, hence continuous, inside its interval of absolute convergence, and its derivative and integral can be determined by differentiating and integrating term by term, respectively.

Proof: A proof can be found in most advanced calculus texts.

Example 8 Term-by-term differentiation of a power series

Let f be a function defined by the power series

$$f(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!} \quad \text{for all } x$$

Show that $f'(x) = f(x)$ for all x , and deduce that $f(x) = e^x$.

Solution The given power series converges for all x (by the generalized ratio test—see Example 1), and Theorem 8.23 tells us that it is differentiable for all x . Differentiating term by term, we find

$$\begin{aligned} f'(x) &= \frac{d}{dx} \left[1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots \right] \\ &= 0 + 1 + \frac{2x}{2!} + \frac{3x^2}{3!} + \frac{4x^3}{4!} + \cdots \end{aligned}$$

$$\begin{aligned}
 &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \\
 &= f(x)
 \end{aligned}$$

In Chapter 5, we found that the differential equation $f'(x) = f(x)$ has the general solution $f(x) = Ce^x$. Substituting $x = 0$ into the power series for $f(x)$, we obtain

$$f(0) = 1 + 0 + \frac{0^2}{2!} + \frac{0^3}{3!} + \cdots = 1$$

and by solving the equation $1 = Ce^0$ for C , we find $C = 1$; therefore, $f(x) = e^x$.

A power series can be differentiated term by term—not just once, but infinitely often in its interval of absolute convergence. The key to this fact lies in showing that if f satisfies

$$f(x) = \sum_{k=0}^{\infty} a_k x^k$$

and R is the radius of convergence of the power series on the right, then the derivative series

$$f'(x) = \sum_{k=1}^{\infty} k a_k x^{k-1}$$

also has the radius of convergence R . (You are asked to show this in Problem 56). Therefore, Theorem 8.23 can be applied to the derivative series to obtain the second derivative

$$f''(x) = \sum_{k=2}^{\infty} k(k-1)a_k x^{k-2}$$

for $|x| < R$. Continuing in this fashion, we can apply Theorem 8.23 to $f'''(x)$, $f^{(4)}(x)$, and all other higher derivatives of f .

For example, we know that the geometric series

$$\sum_{k=0}^{\infty} x^k \quad \text{converges absolutely to } f(x) = \frac{1}{1-x} \text{ for } |x| < 1$$

Thus, the term-by-term derivative

$$\frac{d}{dx} \left[\sum_{k=0}^{\infty} x^k \right] = \sum_{k=1}^{\infty} kx^{k-1} = 1 + 2x + 3x^2 + \cdots$$

converges to $f'(x) = \frac{1}{(1-x)^2}$ for $|x| < 1$, and the term-by-term *second derivative*

$$\frac{d^2}{dx^2} \left[\sum_{k=0}^{\infty} x^k \right] = \frac{d}{dx} \left[\sum_{k=1}^{\infty} kx^{k-1} \right] = \sum_{k=2}^{\infty} k(k-1)x^{k-2}$$

converges to $f''(x) = \frac{2}{(1-x)^3}$, and so on. These ideas are illustrated in our next example.

Example 9 Second derivative of a power series

Let f be the function defined by the power series

$$f(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!}$$

Do not forget, $0! = 1$.

for all x . Show that $f''(x) = -f(x)$ for all x .

Solution First, we use the ratio test to verify that the given power series converges absolutely for all x .

$$L = \lim_{k \rightarrow \infty} \left| \frac{\frac{(-1)^{k+1} x^{2(k+1)}}{[2(k+1)]!}}{\frac{(-1)^k x^{2k}}{(2k)!}} \right| = \lim_{k \rightarrow \infty} \frac{x^{2k+2}}{[2(k+1)]!} \cdot \frac{(2k)!}{x^{2k}} = \lim_{k \rightarrow \infty} \frac{x^2}{(2k+2)(2k+1)} = 0$$

Because $L < 1$, the series converges for all x . Next, differentiate the series term by term:

$$\begin{aligned} f'(x) &= \frac{d}{dx} \left[1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \right] \\ &= -\frac{2x}{2!} + \frac{4x^3}{4!} - \frac{6x^5}{6!} + \cdots \\ &= -\frac{x}{1!} + \frac{x^3}{3!} - \frac{x^5}{5!} + \cdots \end{aligned}$$

Finally, by differentiating term by term again, we obtain

$$\begin{aligned} f''(x) &= \frac{d}{dx} \left[-\frac{x}{1!} + \frac{x^3}{3!} - \frac{x^5}{5!} + \cdots \right] \\ &= -1 + \frac{3x^2}{3!} - \frac{5x^4}{5!} + \cdots \\ &= -\left[1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots \right] \\ &= -f(x) \end{aligned}$$

Example 10 Term-by-term integration of a power series

By integrating an appropriate geometric series term by term, show that

$$\sum_{k=0}^{\infty} \frac{x^{k+1}}{k+1} = -\ln(1-x) \quad \text{for } -1 < x < 1$$

Solution Integrating the geometric series $\sum_{k=0}^{\infty} u^k = \frac{1}{1-u}$ term-by-term in the interval $-1 < u < 1$, we obtain

$$\begin{aligned} \int_0^x \frac{1}{1-u} du &= \int_0^x \left[\sum_{k=0}^{\infty} u^k \right] du \\ &= \int_0^x [1 + u + u^2 + u^3 + \cdots] du \\ &= x + \frac{x^2}{2} + \frac{x^3}{3} + \cdots \\ &= \sum_{k=0}^{\infty} \frac{x^{k+1}}{k+1} \quad \text{for } -1 < x < 1 \end{aligned}$$

We also know

$$\int_0^x \frac{du}{1-u} = -\ln(1-x) \text{ for } -1 < x < 1$$

Thus,

$$-\ln(1-x) = \int_0^x \frac{du}{1-u} = \sum_{k=0}^{\infty} \frac{x^{k+1}}{k+1} \text{ for } -1 < x < 1$$

If we check endpoints, we find the series actually converges for $-1 < x < 1$.

Example 11 Term-by-term integration of a power series

Find a power series for $\tan^{-1} x$.

Solution We will use the fact that $\tan^{-1} x = \int_0^x \frac{dt}{1+t^2}$.

Consider the integrand as a geometric series:

$$\frac{1}{1+t^2} = 1 - t^2 + t^4 - t^6 + \cdots = \sum_{k=0}^{\infty} (-1)^k t^{2k}$$

Thus,

$$\begin{aligned} \tan^{-1} x &= \int_0^x \frac{1}{1+t^2} dt \\ &= \int_0^x [1 - t^2 + t^4 - t^6 + \cdots] dt \\ &= \sum_{k=0}^{\infty} \int_0^x (-1)^k t^{2k} dt \\ &= \sum_{k=0}^{\infty} (-1)^k \frac{t^{2k+1}}{2k+1} \Big|_0^x \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{2k+1} \text{ for } |x| < 1 \\ &= x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots \text{ for } |x| < 1 \end{aligned}$$

If we check the endpoints, we find the series converges for $|x| \leq 1$.

This series for $\tan^{-1} x$ is called **Gregory's series** after the British mathematician James Gregory (1638-1675), who developed it in 1671 (see Figure 8.20).

Because it converges at $x = 1$ and $x = -1$, it can be used to represent $\tan^{-1} x$ on $[-1, 1]$. For example, at $x = 1$

$$\begin{aligned} \frac{\pi}{4} &= \tan^{-1} 1 = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \cdots \\ \pi &= 4 \left[1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \cdots \right] \end{aligned}$$

This is the *Leibniz formula* for π . Unfortunately, as a means of estimating the actual value of π , it has a serious drawback—the convergence is very slow. Look at the partial sums for the 49th and 50th terms, and note that these are not very close to the limit (which is $\pi \approx 3.14159265359 \dots$).

No. of terms	Gregory's Series	
	term	partial sum
0	4	
1	-1.33333333333333	2.666666667
2	0.8	3.466666667
3	-0.57142857142857	2.895238095
4	0.44444444444444	3.33968254
5	-0.36363636363636	2.976046176
6	0.30769230769231	3.283738484
7	-0.26666666666667	3.017071817
8	0.23529411764706	3.252365935
9	-0.21052631578947	3.041839619
10	0.19047619047619	3.232315809
11	-0.17391304347826	3.058402766
12	0.16	3.218402766
13	-0.14814814814815	3.070254618
14	0.13793103448276	3.208185652
15	-0.12903225806452	3.079153394
16	0.12121212121212	3.200365515
17	-0.11428571428571	3.086079801
18	0.10810810810811	3.194187909
19	-0.1025641025641	3.091623807
20	0.09560975609756	3.189184782
21	-0.09302325581395	3.096161526
22	0.08888888888889	3.185050415
23	-0.08510638297872	3.099944032
24	0.081632653061224	3.181576685
25	-0.07843137254902	3.103145313
26	0.075471698113208	3.178617011
27	-0.07272727272727	3.105889738
28	0.070175438596491	3.176065177
29	-0.06779661016949	3.108268567
30	0.065573770491803	3.173842337
31	-0.06349206349206	3.110350274
32	0.061538461538462	3.171888735
33	-0.05970149253731	3.112187243
34	0.057971014492754	3.170158257
35	-0.05633802816901	3.113820229
36	0.054794520547945	3.16861475
37	-0.05333333333333	3.115281416
38	0.051948051948052	3.167229468
39	-0.05063291139241	3.116596557
40	0.0149382716049383	3.165979273
41	-0.04819277108434	3.117786502
42	0.047058823529412	3.164845325
43	-0.04597701149425	3.118868314
44	0.044943820224719	3.163812134
45	-0.04395604395604	3.11985609
46	0.043010752688172	3.162866843
47	-0.04210526315789	3.12076158
48	0.041237113402062	3.161998693
49	-0.04040404040404	3.121594653
50	0.03960396039604	3.161198613

Figure 8.20 Approximation of π using Gregory's series

In 1706, the mathematician John Machin (1680-1751) discovered another series that converges much faster. The identity

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

can be used to show that

$$\begin{aligned}\tan^{-1} 1 &= 4 \tan^{-1} \frac{1}{5} - \tan^{-1} \frac{1}{239} \\ \frac{\pi}{4} &= 4 \left[\frac{1}{5} - \frac{1}{3} \left(\frac{1}{5} \right)^3 + \frac{1}{5} \left(\frac{1}{5} \right)^5 - \cdots \right] \\ &\quad - \left[\frac{1}{239} - \frac{1}{3} \left(\frac{1}{239} \right)^3 + \frac{1}{5} \left(\frac{1}{239} \right)^5 - \cdots \right] \\ \pi &= 16 \left[\frac{1}{5} - \frac{1}{3} \left(\frac{1}{5} \right)^3 + \frac{1}{5} \left(\frac{1}{5} \right)^5 - \cdots \right] \\ &\quad - 4 \left[\frac{1}{239} - \frac{1}{3} \left(\frac{1}{239} \right)^3 + \frac{1}{5} \left(\frac{1}{239} \right)^5 - \cdots \right]\end{aligned}$$

By comparing the table in Figure 8.21 showing the partial sums for this series with the one in Figure 8.20, we see that Machin's formula for π converges much more rapidly than the Leibniz formula. Indeed, the sum of the first seven terms $M_7 = 3.141592654$ already coincides with π for the first eight decimal places.

Machin's Series		
No. of terms	term	partial sum
0	3.1832635983264	
1	-0.0426665690003	3.140597029
2	0.001023999998974	3.141621029
3	-2.9257142857E-05	3.141591772
4	9.10222222222E-07	3.141592682
5	-2.9789090909E-08	3.141592653
6	1.00824615385E-09	3.141592654
7	-3.4952533333E-11	3.141592654

Figure 8.21 Approximation of π using Machin's series

PROBLEM SET 8.7

Level 1

Find the convergence set for the power series given in Problems 1-28.

1. $\sum_{k=1}^{\infty} \frac{kx^k}{k+1}$

2. $\sum_{k=1}^{\infty} \frac{k^2 x^k}{k+1}$

3. $\sum_{k=1}^{\infty} \frac{k(k+1)x^k}{k+2}$

4. $\sum_{k=1}^{\infty} \sqrt{k-1} x^k$

5. $\sum_{k=1}^{\infty} k^2 3^k (x-3)^k$

6. $\sum_{k=1}^{\infty} \frac{k^2 (x-2)^k}{3^k}$

7. $\sum_{k=0}^{\infty} \frac{3^k (x+3)^k}{4^k}$

8. $\sum_{k=0}^{\infty} \frac{4^k (x+1)^k}{3^k}$

9. $\sum_{k=1}^{\infty} \frac{k!(x-1)^k}{5^k}$

10. $\sum_{k=1}^{\infty} \frac{(x-15)^k}{\ln(k+1)}$

11. $\sum_{k=1}^{\infty} \frac{k^2}{2^k} (x-1)^k$

12. $\sum_{k=1}^{\infty} \frac{2^k (x-3)^k}{k(k+1)}$

13. $\sum_{k=1}^{\infty} \frac{k(3x-4)^k}{(k+1)^2}$

15. $\sum_{k=1}^{\infty} \frac{kx^k}{7^k}$

17. $\sum_{k=1}^{\infty} \frac{(k!)^2 x^k}{k^k}$

19. $\sum_{k=2}^{\infty} \frac{(-1)^k x^k}{k(\ln k)^2}$

21. $\sum_{k=0}^{\infty} \frac{(2x)^{2k}}{k!}$

23. $\sum_{k=0}^{\infty} \frac{k!}{2^k} (3x)^{3k}$

25. $\sum_{k=0}^{\infty} \frac{2^k}{k!} (2x-1)^{2k}$

27. $\sum_{k=1}^{\infty} \frac{x^k}{k\sqrt{k}}$

14. $\sum_{k=0}^{\infty} \frac{(2x+3)^k}{4^k}$

16. $\sum_{k=0}^{\infty} \frac{(2k)! x^k}{(3k)!}$

18. $\sum_{k=1}^{\infty} \frac{(-1)^k kx^k}{\ln(k+2)}$

20. $\sum_{k=0}^{\infty} \frac{(3x)^k}{2^{k+1}}$

22. $\sum_{k=0}^{\infty} \frac{(x+2)^{2k}}{3^k}$

24. $\sum_{k=1}^{\infty} \frac{(3x)^{2k}}{\sqrt{k}}$

26. $\sum_{k=0}^{\infty} 2^k (3x)^{3k}$

28. $\sum_{k=1}^{\infty} \frac{(\ln k)x^k}{k}$

Level 2

Find the radius of convergence R in Problems 29–36.

$$29. \sum_{k=1}^{\infty} k^2(x+1)^{2k+1} \quad 30. \sum_{k=1}^{\infty} 2^{\sqrt{k}}(x-1)^k$$

$$31. \sum_{k=1}^{\infty} \frac{k!x^k}{k^k} \quad 32. \sum_{k=1}^{\infty} \frac{k^k x^k}{k!}$$

$$33. \sum_{k=1}^{\infty} \frac{(k!)^2 x^k}{(2k)!} \quad 34. \sum_{k=2}^{\infty} \frac{x^k}{(\ln k)^k}$$

$$35. \sum_{k=1}^{\infty} k(ax)^k \text{ for constant } a.$$

$$36. \sum_{k=1}^{\infty} (a^2 x)^k \text{ for constant } a.$$

In Problems 37–42, find the derivative $f'(x)$ by differentiating term by term.

$$37. f(x) = \sum_{k=0}^{\infty} \left(\frac{x}{2}\right)^k \quad 38. f(x) = \sum_{k=1}^{\infty} \frac{x^k}{k}$$

$$39. f(x) = \sum_{k=0}^{\infty} (k+2)x^k \quad 40. f(x) = \sum_{k=0}^{\infty} kx^k$$

$$41. f(x) = \sum_{k=0}^{\infty} x^k \quad 42. f(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k+1} x^k$$

In Problems 43–48, find $\int_0^x f(u) du$ by integrating term by term.

$$43. f(x) = \sum_{k=0}^{\infty} \left(\frac{x}{2}\right)^k \quad 44. f(x) = \sum_{k=1}^{\infty} \frac{x^k}{k}$$

$$45. f(x) = \sum_{k=0}^{\infty} (k+2)x^k \quad 46. f(x) = \sum_{k=0}^{\infty} kx^k$$

$$47. f(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!} \quad 48. f(x) = \sum_{k=0}^{\infty} (-1)^k x^k$$

49. Use term-by-term differentiation of a geometric series to find a power series for

$$f(x) = \frac{1}{(1-x)^3}$$

For what values of x does this power series converge?

50. Find the radius of convergence for the power series

$$\sum_{k=0}^{\infty} \frac{(qk)!}{(k!)^q} x^k$$

where q is a positive integer.

51. Find the radius of convergence for

$$\sum_{k=1}^{\infty} \frac{(k+3)!x^k}{k!(k+4)!}$$

52. Find the radius of convergence for

$$\sum_{k=1}^{\infty} \frac{1 \cdot 2 \cdot 3 \cdots k (-x)^{2k-1}}{1 \cdot 3 \cdot 5 \cdots (2k-1)}$$

53. Let f be the function defined by the power series

$$f(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}$$

for all x . Show that $f''(x) = -f(x)$ for all x .

54. Let f be the function defined by the power series

$$f(x) = \sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!}$$

for all x . Show that $f''(x) = f(x)$.

Level 3

55. **Think Tank Problem** Show that the series

$$S = \sum_{k=1}^{\infty} \frac{\sin(k!x)}{k^2}$$

converges for all x . Differentiate term by term to obtain the series

$$T = \sum_{k=1}^{\infty} \frac{k! \cos(k!x)}{k^2}$$

Show that this series diverges for all x . Why does this not violate Theorem 8.23?

56. Suppose $\{a_k\}$ is a sequence for which

$$\lim_{k \rightarrow \infty} \sqrt[k]{|a_k|} = \frac{1}{R}$$

Show that the power series

$$\sum_{k=1}^{\infty} k a_k x^{k-1}$$

has radius of convergence R .

57. Show that if $f(x) = \sum_{k=1}^{\infty} a_k x^k$ has radius of convergence $R > 0$, then the series

$$\sum_{k=1}^{\infty} a_k x^{kp}$$

where p is a positive integer, has radius of convergence $R^{1/p}$.

58. For what values of x does the series

$$\sum_{k=1}^{\infty} \frac{x}{k+x}$$

converge? *Note:* This is not a power series.

59. For what values of x does the series

$$\sum_{k=1}^{\infty} \frac{k}{x^k}$$

converge? *Note:* This is not a power series.

60. For what values of x does the series

$$\sum_{k=1}^{\infty} \frac{1}{kx^k}$$

converge? *Note:* This is not a power series.

8.8 TAYLOR AND MACLAURIN SERIES

IN THIS SECTION: *Taylor and Maclaurin polynomials, Taylor's theorem, Taylor and Maclaurin series, operations with Taylor and Maclaurin series*

In this section, we seek to represent a function as an infinite series in such a way that the value of the given function at $x = c$ and the value of the series differ by no more than some specified tolerance.

Taylor and Maclaurin Polynomials

Consider a function f that can be differentiated n times on some interval I . Our goal is to find a polynomial function that approximates f at a number c in its domain. For simplicity, we begin by considering an important special case where $c = 0$. For example, consider the function $f(x) = e^x$ at the point $x = 0$, as shown in Figure 8.22.

To approximate f by a polynomial function $M(x)$, we begin by making sure that both the polynomial function and f pass through the same point. That is, $M(0) = f(0)$. We say that the polynomial is **expanded about $c = 0$** or that it is **centered at 0**.

There are many polynomial functions that we could choose to approximate f at $x = 0$, and we proceed by making sure that both f and M have the same slope at $x = 0$. That is,

$$M'(0) = f'(0)$$

The graph in Figure 8.22 shows that we have $f(x) = e^x$, so $f'(x) = e^x$ and $f(0) = f'(0) = 1$.

To find M , we let

$$M_1(x) = a_0 + a_1x \quad M'_1(x) = a_1$$

and require $M_1(0) = 1$ and $M'_1(0) = 1$. Because $M_1(0) = a_0 = 1$ and $M'_1(0) = a_1 = 1$, we find that

$$M_1(x) = 1 + x$$

We see from Figure 8.22 that close to $x = c$ the approximation of f by M_1 is good, but as we move away from $(0, 1)$, M_1 no longer serves as a good approximation. To improve the approximation, we impose the requirement that the values of the second derivatives of M and f agree at $x = 0$. Using this criterion, we find

$$M_2(x) = a_0 + a_1x + a_2x^2$$

so that

$$M'_2(x) = a_1 + 2a_2x \quad \text{and} \quad M''_2(x) = 2a_2$$

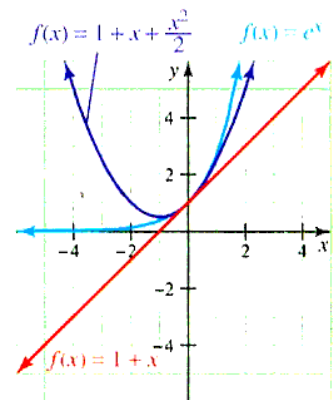


Figure 8.22 Interactive
Graph of $f(x) = e^x$ and
approximating polynomials

Because $f''(x) = e^x$ and $f''(0) = 1$, we want

$$2a_2 = 1 \quad \text{or} \quad a_2 = \frac{1}{2}$$

Thus,

$$M_2(x) = 1 + x + \frac{1}{2}x^2$$

To improve the approximation even further, we can require that the values of the approximating polynomials $M_3, M_4, \dots, M_n$ at $x = 0$ have derivatives that match those of f at $x = 0$. We find that

$$\begin{aligned} M_3(x) &= 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 \\ M_4(x) &= 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4 \\ M_5(x) &= 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4 + \frac{1}{120}x^5 \\ &\vdots \\ M_n(x) &= 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} \end{aligned}$$

This n th degree polynomial approximation of f at $x = 0$ is called a **n th degree Maclaurin polynomial** for f . If we repeat the steps for $x = c$ rather than for $x = 0$, we go through the same steps as above to get a polynomial $T_n(x)$ in powers of $(x - c)$ and we find

$$T_n(x) = e^c + \frac{(x - c)}{1!}e^c + \frac{(x - c)^2}{2!}e^c + \dots + \frac{(x - c)^n}{n!}e^c$$

which is called the **n th-degree Taylor polynomial** of the function $f(x) = e^x$ at $x = c$.

Taylor's Theorem

Instead of stopping at the n th term, we will now approximate a function f by an infinite series. A function f is said to be represented by the power series

$$\sum_{k=0}^{\infty} a_k(x - c)^k$$

on an interval I if $f(x) = \sum_{k=0}^{\infty} a_k(x - c)^k$ for all x in I . Power series representation of functions are extremely useful, but before we can deal effectively with such representations, we must answer two questions.

1. **Existence:** Under what conditions does a given function have a power series representation?
2. **Uniqueness:** When f can be represented by a power series, is there only one such series, and, if so, what is it?

The uniqueness issue is addressed in the following theorem.

Theorem 8.24 The uniqueness theorem for power series representation

Suppose an infinitely differentiable function f is known to have the power series representation

$$f(x) = \sum_{k=0}^{\infty} a_k(x - c)^k$$

for $-R < x - c < R$. Then there is exactly one such representation, and the coefficients a_k must satisfy

$$a_k = \frac{f^{(k)}(c)}{k!} \text{ for } k = 0, 1, 2, \dots$$

Proof: The uniqueness theorem may be established by differentiating the given power series term by term and evaluating successive derivatives at c . We start with

$$f(x) = \sum_{k=0}^{\infty} a_k (x - c)^k$$

and substitute $x = c$ to obtain

$$f(c) = a_0 + a_1(c - c) + a_2(c - c)^2 + \dots = a_0$$

Next, differentiate the original series, term by term, to get

$$f'(x) = a_1 + 2a_2(x - c) + 3a_3(x - c)^2 + \dots$$

and thus,

$$f'(c) = a_1 + 2a_2(c - c) + 3a_3(c - c)^2 + \dots = a_1$$

Differentiating once again and substituting $x = c$, we find

$$f''(c) = 2a_2 \text{ so that } a_2 = \frac{f''(c)}{2}$$

In general, the k th derivative of f at $x = c$ is given by

$$f^{(k)}(c) = k!a_k \text{ so that } a_k = \frac{f^{(k)}(c)}{k!}$$

The question of existence—that is, under what conditions a given function has a power series representation—is answered by considering what is called the **Taylor remainder function**:

$$R_n(x) = f(x) - T_n(x)$$

We see that f is represented by its Taylor series in I if and only if

$$\lim_{n \rightarrow \infty} R_n(x) = 0 \text{ for all } x \text{ in } I$$

This relationship among f , its Taylor polynomial $T_n(x)$, and the Taylor remainder function $R_n(x)$ is summarized in the following theorem.

Theorem 8.25 Taylor's theorem

If f and all its derivatives exist in an open interval I containing c , then for each x in I ,

$$f(x) = f(c) + \frac{f'(c)}{1!}(x - c) + \frac{f''(c)}{2!}(x - c)^2 + \dots + \frac{f^{(n)}(c)}{n!}(x - c)^n + R_n(x)$$

where the remainder function $R_n(x)$ is given by

$$R_n(x) = \frac{f^{(n+1)}(z_n)}{(n+1)!}(x - c)^{n+1}$$

for some z_n that depends on x and lies between c and x .

Proof: The proof of this theorem is given in Appendix B.

The formula for $R_n(x)$ is called the *Lagrange form* of the Taylor remainder function, after the French/Italian mathematician Joseph Lagrange (1736–1813). There are others forms for the remainder, principally one developed by Cauchy (see [Historical Quest](#), Problem 53 of Section 2.1), but we will consider only the Lagrange form in this text.

When applying Taylor's theorem, we do not expect to be able to find the exact value of z_n . However, we can often determine an upper bound for $|R_n(x)|$ in an open interval. We will illustrate this situation later in this section.

Taylor and Maclaurin Series

The uniqueness theorem tells us that if f has a power series representation at c , it must be the series

$$f(c) + \frac{f'(c)}{1!} + \frac{f''(c)}{2!}(x-c)^2 + \frac{f'''(c)}{3!}(x-c)^3 + \cdots$$

and Taylor's theorem says that $f(x)$ is represented by such a series precisely where the remainder term $R_n(x) \rightarrow 0$ as $n \rightarrow \infty$. It is convenient to have the following terminology.

The uniqueness theorem may be summarized by saying that the Taylor series of f at c is the only power series of the form $\sum a_k(x-c)^k$ that can possibly represent f on I . But it does not say that f is equal to the power series on I . All we really know is that if f has a power series representation at c , then that representation must be its Taylor series.

TAYLOR SERIES Suppose there is an open interval I containing c throughout which the function f and all its derivatives exist. Then the power series

$$f(c) + \frac{f'(c)}{1!} + \frac{f''(c)}{2!}(x-c)^2 + \frac{f'''(c)}{3!}(x-c)^3 + \cdots$$

is called the **Taylor series of f at c** .

MACLAURIN SERIES The special case where $c = 0$ is called the **Maclaurin series of f** :

$$f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots$$

Example 1 Maclaurin series

Find the Maclaurin series for $f(x) = \cos x$.

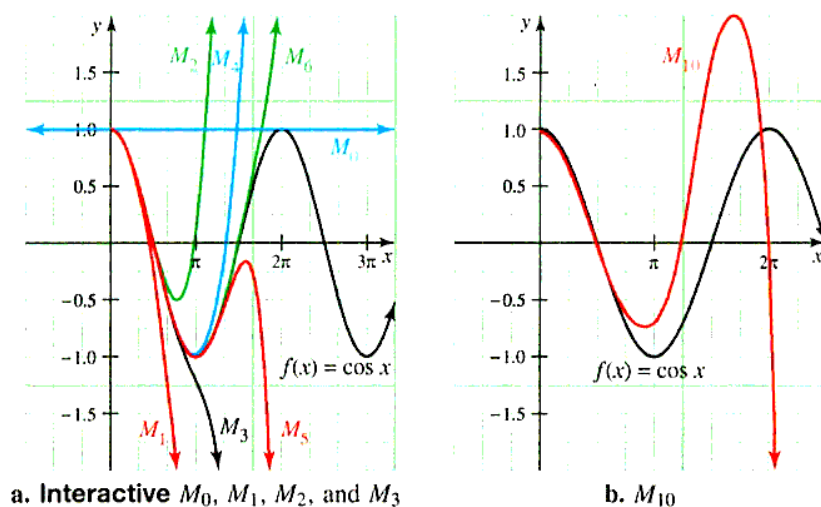
Solution First, note that f is infinitely differentiable at $x = 0$. We find

$$\begin{array}{ll} f(x) = \cos x & f(0) = 1 \\ f'(x) = -\sin x & f'(0) = 0 \\ f''(x) = -\cos x & f''(0) = -1 \\ f'''(x) = \sin x & f'''(0) = 0 \\ f^{(4)}(x) = \cos x & f^{(4)}(0) = 1 \\ \vdots & \vdots \end{array}$$

Thus, by using the definition of a Maclaurin series we have

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!}$$

You might ask about the relationship between a Maclaurin series and a Maclaurin polynomial. Figure 8.23 shows the function $f(x) = \cos x$ from Example 1 along with some successive Maclaurin polynomials. We use the notation $M_n(x)$ to represent the first $n+1$ terms of the corresponding Maclaurin series. Note that the polynomials are quite close to the function near $x = 0$, but that they become very different as x moves away from the origin.

Figure 8.23 Comparison of $\cos x$ and its Maclaurin polynomials**Example 2** Maximum error when using a Maclaurin polynomial approximation

Find the Maclaurin polynomial $M_5(x)$ for the function $f(x) = e^x$ and use this polynomial to approximate e . Use Taylor's theorem to determine the accuracy of this approximation.

Solution First find the Maclaurin series for $f(x) = e^x$.

$$\begin{aligned} f(x) &= e^x & f(0) &= 1 \\ f'(x) &= e^x & f'(0) &= 1 \\ f''(x) &= e^x & f''(0) &= 1 \\ &\vdots & & \vdots \end{aligned}$$

The Maclaurin series for e^x is

$$e^x = 1 + \frac{1}{1!}x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \cdots = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

Then find $M_5(x)$:

$$M_5(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \frac{x^5}{120} \approx e^x$$

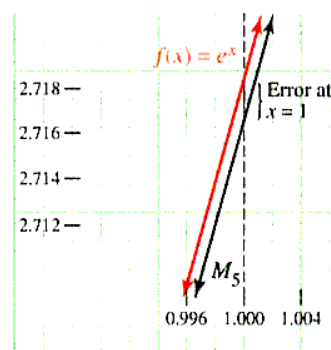
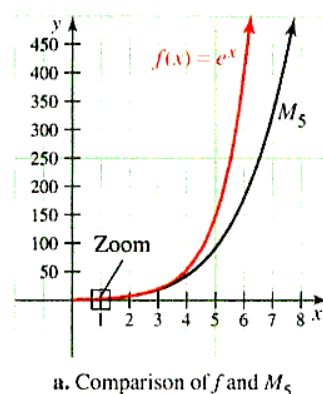
A comparison of f and $M_5(x)$ is shown in Figure 8.24.

Notice from the graphs of these functions that the error seems to increase as x moves away from the origin. To determine the accuracy, we use Taylor's theorem

$$\begin{aligned} e &= 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \frac{1}{120} + R_5(1) \\ &= 2.71\overline{6} + R_5(1) \end{aligned}$$

The remainder term is given by

$$R_5(1) = \frac{e^{\xi}}{(5+1)!}(1)^{5+1}$$

Figure 8.24 Comparison of $f(x) = e^x$ and $M_5(x)$

for some z_5 between 0 and 1. Because $0 < z_5 < 1$, we have $e^{z_5 5} < e$, and since $e < 3$, it follows that

$$R_5(1) < \frac{e}{6!} < \frac{3}{6!} \approx 0.00417$$

Thus, the number e satisfies

$$\begin{aligned} 2.716667 - 0.004167 &< e < 2.716667 + 0.004167 \\ 2.712500 &< e < 2.720834 \end{aligned}$$

Example 3 Taylor series

Find the Taylor series for $f(x) = \ln x$ at $c = 1$.

Solution Note that f is infinitely differentiable at $x = 1$. We find

$$\begin{array}{ll} f(x) = \ln x & f(1) = 0 \\ f'(x) = \frac{1}{x} & f'(1) = 1 \\ f''(x) = -\frac{1}{x^2} & f''(1) = -1 \\ f'''(x) = \frac{2}{x^3} & f'''(1) = 2 \\ f^{(4)}(x) = -\frac{6}{x^4} & f^{(4)}(1) = -6 \\ \vdots & \vdots \\ f^{(k)}(x) = \frac{(-1)^{k+1}(k-1)!}{x^k} & f^{(k)}(1) = (-1)^{k+1}(k-1)! \end{array}$$

Then, use the definition of a Taylor series to write

$$\begin{aligned} \ln x &= 0 + \frac{1}{1!}(x-1) - \frac{1}{2!}(x-1)^2 + \frac{2}{3!}(x-1)^3 - \frac{6}{4!}(x-1)^4 + \cdots \\ &= (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 - \frac{1}{4}(x-1)^4 + \cdots \\ &= \sum_{k=1}^{\infty} \frac{(-1)^{k+1}(x-1)^k}{k} \end{aligned}$$

Suppose f is a function that is infinitely differentiable at c . Now we have two mathematical quantities, f and its Taylor series. There are several possibilities:

1. The Taylor series of f may converge to f on the interval of absolute convergence, $-R < x - c < R$ (that is, $|x - c| < R$).
2. The Taylor series may converge only at $x = c$, in which case it certainly does not represent f on any interval containing c .
3. The Taylor series of f may have a positive radius of convergence (even $R = \infty$), but it may converge to a function g that does not equal f on the interval $|x - c| < R$.

Example 4 A function defined at points where its Taylor series does not converge

Show that the function $\ln x$ is defined at points for which its Taylor series at $c = 1$ does not converge.

Solution We know that $\ln x$ is defined for all $x > 0$. From the previous example, we know the Taylor series for $\ln x$ at $c = 1$ is

$$\sum_{k=1}^{\infty} \frac{(-1)^{k-1}(x-1)^k}{k}$$

We find the interval of convergence for this series centered at $c = 1$ by computing

$$\begin{aligned} L &= \lim_{k \rightarrow \infty} \left| \frac{\frac{(-1)^k(x-1)^{k+1}}{k+1}}{\frac{(-1)^{k-1}(x-1)^k}{k}} \right| \\ &= \lim_{k \rightarrow \infty} \left| \frac{\frac{(-1)^k}{k+1}}{\frac{(-1)^{k-1}}{k}} \right| |x-1| \\ &= \lim_{k \rightarrow \infty} \left(\frac{k}{k+1} \right) |x-1| \\ &= |x-1| \end{aligned}$$

This means that the power series converges absolutely for

$$\begin{aligned} |x-1| &< 1 \\ 0 &< x < 2 \end{aligned}$$

Test the endpoints:

$$\begin{aligned} x = 0: \quad \sum_{k=1}^{\infty} \frac{(-1)^{k-1}(0-1)^k}{k} &= \sum_{k=1}^{\infty} \frac{-1}{k} \text{ diverges} && \text{Opposite of harmonic series} \\ x = 2: \quad \sum_{k=1}^{\infty} \frac{(-1)^{k-1}(2-1)^k}{k} &= \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} \text{ converges} && \text{Alternating harmonic series} \end{aligned}$$

Thus, the series converges only on $(0, 2]$, but the function $\ln x$ is defined for all $x > 0$. The function is compared with the sixth degree Taylor polynomial

$$T_6(x) = \sum_{k=1}^6 \frac{(-1)^{k-1}}{k} (x-1)^k \text{ approximation in Figure 8.25.}$$

The next example introduces a function whose Maclaurin series does not represent that function on *any* interval.

Example 5 A function represented by its Maclaurin series at a single point

Let f be the function defined by

$$f(x) = \begin{cases} e^{-1/x^2} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Show that f is represented by its Maclaurin series only at $x = 0$.

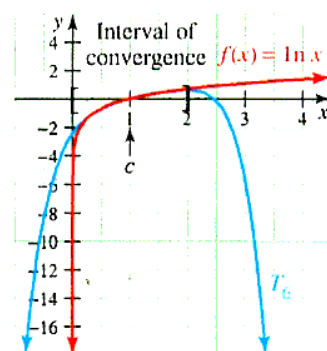


Figure 8.25 Interactive
Comparison of
 $f(x) = \ln x$ and $T_6(x)$

Solution It can be shown that f is infinitely differentiable at 0 and that $f^{(k)}(0) = 0$ for all k . Therefore, f has the Maclaurin series

$$0 + \frac{0}{1!}x + \frac{0}{2!}x^2 + \frac{0}{3!}x^3 + \cdots$$

which converges to the zero function for all x and thus represents $f(x)$ only at $x = 0$. The graph of f and the Maclaurin series are shown in Figure 8.26.

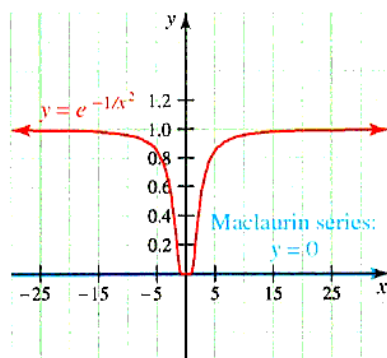


Figure 8.26 Interactive Graph of f and its Maclaurin series at $y = 0$

Operations with Taylor and Maclaurin Series

According to the uniqueness theorem, the Taylor series for f at c is the only power series in $x - c$ that can satisfy $f(x) = \sum_{k=0}^{\infty} a_k(x - c)^k$ for all x in some interval containing c . The coefficients in this series can always be found by substituting into the formula $a_k = \frac{f^{(k)}(c)}{k!}$, but occasionally they can also be found by algebraic manipulation.

In several of the examples that follow, we will use the geometric series equation

$$\frac{1}{1-u} = \sum_{k=0}^{\infty} u^k \quad \text{for } |u| < 1$$

to obtain Maclaurin series for rational functions. The general procedure is illustrated in Example 6.

Example 6 Maclaurin series using substitution in a geometric series

Find the Maclaurin series for $\frac{1}{1+x^2}$.

Solution We want a series of the form $\sum_{k=0}^{\infty} a_k u^k$ that represents $\frac{1}{1+x^2}$ on an interval containing 0. We could proceed directly using the definition of a Maclaurin series, but instead we will modify a known series. In fact, trying to proceed directly quickly becomes cumbersome for this series (try it and you will see). We know that if $|u| < 1$ we can write

$$\frac{1}{1-u} = 1 + u + u^2 + \cdots$$

so by substitution ($u = -x^2$) we have

$$\frac{1}{1+x^2} = \frac{1}{1-(-x^2)} = 1 - x^2 + x^4 - x^6 + \cdots$$

provided $|-x^2| < 1$; that is, $-1 < x < 1$. Hence, by the uniqueness theorem, the desired representation is

$$\frac{1}{1+x^2} = \sum_{k=0}^{\infty} (-1)^k x^{2k} \quad \text{for } -1 < x < 1$$

Example 7 Maclaurin series by modifying a geometric series

Find the Maclaurin series for $f(x) = \frac{5-2x}{3+2x}$ and determine the interval of convergence of this series.

Solution Our strategy is to rewrite $f(x)$ in the form $\frac{1}{1-u}$ of a geometric series:

$$\begin{aligned} \frac{5-2x}{3+2x} &= -1 + \frac{8}{3+2x} && \text{Long division} \\ &= -1 + \frac{\frac{8}{3}}{1 - (-\frac{2}{3}x)} && \text{Write in the form } \frac{1}{1-u}. \\ &= -1 + \sum_{k=0}^{\infty} \frac{8}{3} \left(-\frac{2}{3}x\right)^k \\ &= -1 + \frac{8}{3} - \frac{16}{9}x + \frac{32}{27}x^2 - \frac{64}{81}x^3 + \cdots \end{aligned}$$

The convergence set for $f(x)$ is the same as for the series $\sum_{k=1}^{\infty} (-\frac{2}{3}x)^k$; namely,

$$\begin{aligned} \left|-\frac{2}{3}x\right| &< 1 \\ -\frac{3}{2} &< x < \frac{3}{2} \end{aligned}$$

What we see in these examples is that if one can find a Maclaurin series for a given function by some means, it is *the* Maclaurin series. A similar statement can be made about a Taylor series for a particular choice of c . This is because of the uniqueness theorem for power series representation above.

Example 8 Using a trigonometric identity with a known series

Find the Maclaurin series for **a.** $\cos x^2$ **b.** $\cos^2 x$

Solution

a. In Example 1 we found that

$$\cos u = 1 - \frac{u^2}{2!} + \frac{u^4}{4!} - \frac{u^6}{6!} + \cdots \quad \text{for all } u$$

Therefore, by substituting $u = x^2$, we obtain

$$\begin{aligned} \cos x^2 &= 1 - \frac{(x^2)^2}{2!} + \frac{(x^2)^4}{4!} - \frac{(x^2)^6}{6!} + \cdots && \text{Let } u = x^2. \\ &= 1 - \frac{x^4}{2!} + \frac{x^8}{4!} - \frac{x^{12}}{6!} + \cdots && \text{for all } x \end{aligned}$$

b. For $\cos^2 x$ we could use the definition of a Maclaurin series, but instead we will use a double-angle trigonometric identity.

$$\begin{aligned}
 \cos^2 x &= \frac{1}{2} + \frac{1}{2} \cos 2x \\
 &= \frac{1}{2} + \frac{1}{2} \left[1 - \frac{(2x)^2}{2!} + \frac{(2x)^4}{4!} - \frac{(2x)^6}{6!} + \dots \right] \quad \text{Let } u = 2x. \\
 &= \frac{1}{2} + \frac{1}{2} - \frac{2x^2}{2!} + \frac{2^3 x^4}{4!} - \frac{2^5 x^6}{6!} + \dots \\
 &= 1 - x^2 + \frac{1}{3}x^4 - \frac{2}{45}x^6 + \dots \quad \text{for all } x
 \end{aligned}$$

Example 9 Maclaurin series using substitution

Find the Maclaurin series for $f(x) = \ln\left(\frac{1+x}{1-x}\right)$ and use this series to compute $\ln 2$ correct to five decimal places.

Solution For this function, we first use a property of logarithms:

$$f(x) = \ln\left(\frac{1+x}{1-x}\right) = \ln(1+x) - \ln(1-x)$$

From Example 3, we know

$$\ln x = (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 - \dots$$

so that

$$\begin{aligned}
 \ln(1+x) &= (1+x-1) - \frac{1}{2}(1+x-1)^2 + \frac{1}{3}(1+x-1)^3 - \dots \\
 &= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \\
 \ln(1-x) &= (1-x-1) - \frac{1}{2}(1-x-1)^2 + \frac{1}{3}(1-x-1)^3 - \dots \\
 &= (-x) - \frac{(-x)^2}{2} + \frac{(-x)^3}{3} - \frac{(-x)^4}{4} + \dots \\
 &= -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \dots
 \end{aligned}$$

By subtracting series, we find

$$\begin{aligned}
 f(x) &= \ln(1+x) - \ln(1-x) \\
 &= \left[x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \right] - \left[-x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \dots \right] \\
 &= 2 \left[x + \frac{x^3}{3} + \frac{x^5}{5} + \dots \right]
 \end{aligned}$$

Next, by solving the equation $\frac{1+x}{1-x} = 2$, we find $x = \frac{1}{3}$, so we are looking for $\ln\left(\frac{1+x}{1-x}\right)$ where $x = \frac{1}{3}$. It follows that

$$\begin{aligned}
 \ln 2 &= 2 \left[\frac{1}{3} + \frac{\left(\frac{1}{3}\right)^3}{3} + \frac{\left(\frac{1}{3}\right)^5}{5} + \dots \right] \\
 &= 2 \left[\frac{1}{3} + \frac{1}{3} \left(\frac{1}{3}\right)^3 + \frac{1}{5} \left(\frac{1}{3}\right)^5 + \dots \right]
 \end{aligned}$$

According to the Taylor theorem, we find

$$\ln 2 = \frac{2}{3} + \frac{2}{3} \left(\frac{1}{3}\right)^3 + \frac{2}{5} \left(\frac{1}{3}\right)^5 + \cdots + \frac{2}{2n+1} \left(\frac{1}{3}\right)^{2n+1} + R_n \left(\frac{1}{3}\right)$$

where $R_n \left(\frac{1}{3}\right)$ is the remainder term. To estimate the remainder, we note that $R_n \left(\frac{1}{3}\right)$ is the tail of the infinite series for $\ln 2$. Thus,

$$\begin{aligned} \left| R_n \left(\frac{1}{3}\right) \right| &= \frac{2}{2n+3} \left(\frac{1}{3}\right)^{2n+3} + \frac{2}{2n+5} \left(\frac{1}{3}\right)^{2n+5} + \frac{2}{2n+7} \left(\frac{1}{3}\right)^{2n+7} + \cdots \\ &< \frac{2}{2n+3} \left(\frac{1}{3}\right)^{2n+3} + \frac{2}{2n+3} \left(\frac{1}{9}\right) \left(\frac{1}{3}\right)^{2n+3} + \cdots \\ &= \frac{2}{2n+3} \left(\frac{1}{3}\right)^{2n+3} \left[1 + \frac{1}{9} + \frac{1}{81} + \cdots \right] \end{aligned}$$

Because the geometric series in brackets converges to $\frac{1}{1-\frac{1}{9}} = \frac{9}{8}$, we find that

$$\left| R_n \left(\frac{1}{3}\right) \right| < \frac{2}{2n+3} \left(\frac{9}{8}\right) \left(\frac{1}{3}\right)^{2n+3}$$

In particular, to achieve five-place accuracy, we must make sure the term on the right is less than 0.000005 (six places to account for round-off error). With a calculator we can see that if $n = 4$, we have

$$\frac{2}{2(4)+3} \left(\frac{9}{8}\right) \left(\frac{1}{3}\right)^{11} = 0.0000012$$

Thus, we approximate $\ln 2$ with $n = 4$:

$$\ln 2 \approx T_4 \left(\frac{1}{3}\right) = \frac{2}{3} + \frac{2}{3} \left(\frac{1}{3}\right)^3 + \frac{2}{5} \left(\frac{1}{3}\right)^5 + \frac{2}{7} \left(\frac{1}{3}\right)^7 + \frac{2}{9} \left(\frac{1}{3}\right)^9 \approx 0.6931460$$

which is correct with an error of no more than 0.0000012; therefore,

$$\begin{aligned} 0.6931460 - 0.0000012 &< \ln 2 < 0.6931460 + 0.0000012 \\ 0.6931448 &< \ln 2 < 0.6931472 \end{aligned}$$

Rounded to five-place accuracy, $\ln 2 \approx 0.69315$.

We can check with a calculator (to 10 decimal accuracy): $\ln 2 \approx 0.6931471806$. ■

TECHNOLOGY NOTE: “Why bother with the accuracy part of Example 9?” you might ask. “I can just press $\ln 2$ on my calculator and obtain better accuracy anyway!” The answer is that the accuracy built into calculators and computers uses the same principles illustrated in Example 9. Calculators and computers have not changed the need for accuracy considerations but have simply pushed out the limits of these accuracy arguments. It is only for convenience that we ask for limits of accuracy that are less than what we can obtain with a calculator (or any available machine), but the method is just as valid if we ask for 100- or 1,000-place accuracy. Please see Appendix C, section “Significant Figures” for more details.

Some of the more common power series that we have derived are given in Table 8.2. Before we present this table, there is one last result we should consider. It is a generalization of the binomial theorem that was discovered by Isaac Newton while he was still a student at Cambridge University.

Theorem 8.26 Binomial series

If p is any real number and $-1 < x < 1$, then

$$(1+x)^p = 1 + px + \frac{p(p-1)}{2!}x^2 + \frac{p(p-1)(p-2)}{3!}x^3 + \cdots = \sum_{k=0}^{\infty} \binom{p}{k} x^k$$

where $\binom{p}{k} = \frac{p!}{k!(p-k)!}$ for k and p integers with $p \geq k \geq 0$ (no such shorthand exists for other values of p).

Proof: We begin by showing how the Maclaurin series is found. Let $f(x) = (1+x)^p$.

$$\begin{array}{ll} f(x) = (1+x)^p & f(0) = 1 \\ f'(x) = p(1+x)^{p-1} & f'(0) = p \\ f''(x) = p(p-1)(1+x)^{p-2} & f''(0) = p(p-1) \\ f'''(x) = p(p-1)(p-2)(1+x)^{p-3} & f'''(0) = p(p-1)(p-2) \\ \vdots & \vdots \\ f^{(n)}(x) = p(p-1)\cdots(p-n+1)(1+x)^{p-n} & f^{(n)}(0) = p(p-1)\cdots(p-n+1) \end{array}$$

We see this produces the Maclaurin series. We could use the ratio test to show that this series converges on the interval $(-1, 1)$, but to show that it converges to $(1+x)^p$ requires advanced calculus. ♦

The binomial series converges on the open interval $-1 < x < 1$, but convergence at the endpoints $x = -1$ and $x = 1$ of this interval depends on the exponent p . In particular, it can be shown that if $-1 < p \leq 0$, the series converges at $x = 1$; if $p \geq 0$, the series converges at both $x = 1$ and $x = -1$. Moreover, if p is a nonnegative integer, the series terminates after only a finite number of terms (since $\binom{p}{k} = 0$ for $k > p$) and thus reduces to the ordinary binomial expansion, which, of course, converges for all x .

Table 8.2 Power series for elementary functions*

Name	Series	Interval of Convergence
Exponential series	$e^u = 1 + u + \frac{u^2}{2!} + \frac{u^3}{3!} + \frac{u^4}{4!} + \cdots + \frac{u^k}{k!} + \cdots$ $e^x = e^c + e^c(x-c) + \frac{e^c(x-c)^2}{2!} + \frac{e^c(x-c)^3}{3!} + \cdots + \frac{e^c(x-c)^k}{k!} + \cdots$	$(-\infty, \infty)$
Cosine series	$\cos u = 1 - \frac{u^2}{2!} + \frac{u^4}{4!} - \frac{u^6}{6!} + \cdots + \frac{(-1)^k u^{2k}}{(2k)!} + \cdots$ $\cos x = \cos c - (x-c) \sin c - \frac{(x-c)^2}{2!} \cos c + \frac{(x-c)^3}{3!} \sin c + \cdots$	$(-\infty, \infty)$
Sine series	$\sin u = u - \frac{u^3}{3!} + \frac{u^5}{5!} - \frac{u^7}{7!} + \cdots + \frac{(-1)^k u^{2k+1}}{(2k+1)!} + \cdots$ $\sin x = \sin c + (x-c) \cos c - \frac{(x-c)^2}{2!} \sin c - \frac{(x-c)^3}{3!} \cos c + \cdots$	$(-\infty, \infty)$
Geometric series	$\frac{1}{1-u} = 1 + u + u^2 + u^3 + \cdots + u^k + \cdots$	$(-1, 1)$
Reciprocal series	$\frac{1}{x} = 1 - (x-1) + (x-1)^2 - (x-1)^3 + (x-1)^4 - \cdots$	$(0, 2)$
Logarithmic series	$\ln x = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \cdots + \frac{(-1)^{k-1}(x-1)^k}{k} + \cdots$ $\ln(1+u) = u - \frac{1}{2}u^2 + \frac{1}{3}u^3 - \frac{1}{4}u^4 + \cdots + \frac{(-1)^{k-1}u^k}{k} + \cdots$ $\ln x = \ln c + \frac{x-c}{c} - \frac{(x-c)^2}{2c^2} + \frac{(x-c)^3}{3c^3} - \cdots + \frac{(-1)^{k-1}(x-c)^k}{kc^k} + \cdots$	$(0, 2)$
Inverse tangent series	$\tan^{-1} u = u - \frac{u^3}{3} + \frac{u^5}{5} - \frac{u^7}{7} + \cdots + \frac{(-1)^k u^{2k+1}}{2k+1} + \cdots$	$[-1, 1]$
Inverse sine series	$\sin^{-1} u = u + \frac{u^3}{2 \cdot 3} + \frac{1 \cdot 3 u^5}{2 \cdot 4 \cdot 5} + \frac{1 \cdot 3 \cdot 5 u^7}{2 \cdot 4 \cdot 6 \cdot 7} + \cdots + \frac{1 \cdot 3 \cdot 5 \cdots (2k-3) u^{2k-1}}{2 \cdot 4 \cdot 6 \cdots (2k-2)(2k-1)} + \cdots$	$[-1, 1]$
Binomial series	$(1+u)^p = 1 + pu + \frac{p(p-1)}{2!}u^2 + \frac{p(p-1)(p-2)}{3!}u^3 + \cdots$	

*Taylor series at $x = c$ follows the Maclaurin series for each function.

The following example illustrates how the binomial series theorem can be used.

Example 10 Using the binomial series theorem to obtain a Maclaurin series expansion

Determine the Maclaurin series for $f(x) = \sqrt{9+x}$ and find its interval of convergence.

Solution Write $f(x) = \sqrt{9+x} = (9+x)^{1/2} = 3\left(1+\frac{x}{9}\right)^{1/2}$.

Thus,

$$\begin{aligned}\sqrt{9+x} &= 3\left(1+\frac{x}{9}\right)^{1/2} \\ &= 3\left[1 + \frac{1}{2}\left(\frac{x}{9}\right) + \frac{\frac{1}{2}(\frac{1}{2}-1)}{2!}\left(\frac{x}{9}\right)^2 + \frac{\frac{1}{2}(\frac{1}{2}-1)(\frac{1}{2}-2)}{3!}\left(\frac{x}{9}\right)^3 + \cdots\right] \\ &= 3\left[1 + \frac{1}{18}x - \frac{1}{648}x^2 + \frac{1}{11,664}x^3 - \cdots\right] \\ &= 3 + \frac{1}{6}x - \frac{1}{216}x^2 + \frac{1}{3,888}x^3 - \cdots\end{aligned}$$

Since the exponent $p = \frac{1}{2}$ satisfies $p > 0$, p not an integer, it follows from Theorem 8.26 that the series converges for $|\frac{x}{9}| \leq 1$, that is for $|x| \leq 9$. ■

PROBLEM SET 8.8

Level 1

- **What does this say?** Compare and contrast Maclaurin and Taylor series.
- **What does this say?** Discuss the binomial series theorem.

Find the Maclaurin series for the functions given in Problems 3–26. Assume that a is any constant, and that all the derivatives of all orders exist at $x = 0$.

- | | | |
|-------------------------------|-----------------------------------|------------------------|
| 3. e^{2x} | 4. e^{-x} | 5. e^{x^2} |
| 6. e^{ax} | 7. $\sin x^2$ | 8. $\sin^2 x$ |
| 9. $\sin ax$ | 10. $\cos ax$ | 11. $\cos 2x^2$ |
| 12. $\cos x^3$ | 13. $x^2 \cos x$ | 14. $\sin \frac{x}{2}$ |
| 15. $x^2 + 2x + 1$ | 16. $x^3 - 2x^2 + x - 5$ | |
| 17. xe^x | 18. $e^{-x} + e^{2x}$ | |
| 19. $e^x + \sin x$ | 20. $\sin x + \cos x$ | |
| 21. $\frac{1}{1+4x}$ | 22. $\frac{1}{1-ax}, a \neq 0$ | |
| 23. $\frac{1}{a+x}, a \neq 0$ | 24. $\frac{1}{a^2+x^2}, a \neq 0$ | |
| 25. $\ln(3+x)$ | 26. $\log(1+x)$ | |

 Find the first four terms of the Taylor series of the functions in Problems 27–36 at the given value of c .

27. $f(x) = e^x$ at $c = 1$ 28. $f(x) = \ln x$ at $c = 3$

29. $f(x) = \cos x$ at $c = \frac{\pi}{3}$ 30. $f(x) = \sin x$ at $c = \frac{\pi}{4}$
 31. $f(x) = \tan x$ at $c = 0$ 32. $f(x) = \sqrt{x}$ at $c = 9$
 33. $f(x) = \frac{1}{2-x}$ at $c = 5$ 34. $f(x) = \frac{1}{4-x}$ at $c = -2$
 35. $f(x) = \frac{3}{2x-1}$ at $c = 2$ 36. $f(x) = \frac{5}{3x+2}$ at $c = 2$

Expand each function in Problems 37–42 as a binomial series. Give the interval of convergence of the series.

37. $f(x) = \sqrt{1+x}$ 38. $f(x) = \frac{1}{\sqrt{1+x^2}}$
 39. $f(x) = (1+x)^{2/3}$ 40. $f(x) = (4+x)^{-1/3}$
 41. $f(x) = \frac{x}{\sqrt{1-x^2}}$ 42. $f(x) = \sqrt[4]{2-x}$

Level 2

43. Use the Maclaurin series for e^x and e^{-x} to find the Maclaurin series for $\sinh x = \frac{e^x - e^{-x}}{2}$.
 44. Use the Maclaurin series for e^x and e^{-x} to find the Maclaurin series for $\cosh x = \frac{e^x + e^{-x}}{2}$.
 45. Use the Maclaurin series expansion of e^x to estimate $\sqrt[3]{e}$ with three decimal place accuracy. How many terms are needed?

46. Use the Maclaurin series expansion of e^x to estimate $\sqrt{e}$ with three decimal place accuracy? How many terms are needed?

Find the Maclaurin series for the functions given in Problems 47-48. Hint: Begin by expressing each function as a sum of partial fractions.

47. $f(x) = \frac{1}{x^2 - 3x + 2}$

48. $f(x) = \frac{1}{(x+2)(x^2-1)}$

49. Use an appropriate identity to find the Maclaurin series for $f(x) = \sin x \cos x$.

50. a. It can be shown that

$$\cos 3x = 4 \cos^3 x - 3 \cos x$$

Use this identity to find the Maclaurin series for $\cos^3 x$.

- b. Find the Maclaurin series for $\sin^3 x$.

51. Find the Maclaurin series for

$$\ln \left[\frac{1+2x}{1-3x+2x^2} \right]$$

52. Find the Maclaurin series for

$$\ln [(1+2x)(1+3x)]$$

53. Find the Maclaurin series for

- a. $f(x) = x + \sin x$ and then use this series to find

$$\lim_{x \rightarrow 0} \frac{x + \sin x}{x}$$

- b. Find the Maclaurin series for $g(x) = e^x - 1$ and then use this series to find

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$$

In Problems 54-55, the Maclaurin series of a function appears with its remainder. Verify that the remainder satisfies the given inequality.

54. $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$; $|R_n(x)| \leq \frac{e^p |x|^{n+1}}{(n+1)!}$ where p is some constant such that $0 < p < x$.

55. $\ln(x+1) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{k+1}}{k+1}$ for $0 \leq x < 1$;

$$|R_n(x)| \leq \frac{|x|^{n+1}}{n+1}$$

56. Historical Quest

The two mathematicians most closely associated with power series are Brook Taylor and Colin Maclaurin. The result stated in this section, called Taylor's theorem, was published by Taylor in 1712, but the result itself was discovered by the Scottish mathematician James Gregory



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Brook Taylor
(1685-1731)

(1638-1675) in 1670 and was obtained by Taylor in 1688, after Gregory's death. James Gregory apparently anticipated some of the key calculus discoveries, but died prematurely and never received proper recognition for his work. In 1694, Johann Bernoulli (see [Quest Problem 49, Section 4.5](#)) published a series that was very similar to what we call Taylor's series. In fact, after Taylor published his work in 1714, Bernoulli accused him of plagiarism. In this [Quest](#) we seek to duplicate Bernoulli's work. He started by writing

$$\begin{aligned} n \, dx &= n \, dx + \left(x \, dn - x \frac{dn}{dx} dx \right) \\ &= n \, dx + \left(x \, dn - x \frac{dn}{dx} dx \right) \\ &\quad - \left(\frac{x^2}{2!} \frac{d^2 n}{dx^2} dx - \frac{x^2}{2!} \frac{d^2 n}{dx^2} dx \right) \\ &= n \, dx + \left(x \, dn - x \frac{dn}{dx} dx \right) \\ &\quad - \left(\frac{x^2}{2!} \frac{d^2 n}{dx^2} dx - \frac{x^2}{2!} \frac{d^2 n}{dx^2} dx \right) \\ &\quad + \left(\frac{x^3}{3!} \frac{d^3 n}{dx^3} dx - \frac{x^3}{3!} \frac{d^3 n}{dx^3} dx \right) - \dots \\ &= (n \, dx + x \, dn) \\ &\quad - \left(x \frac{dn}{dx} + \frac{x^2}{2!} \frac{d^2 n}{dx^2} \right) dx \\ &\quad + \left(\frac{x^2}{2!} \frac{d^2 n}{dx^2} + \frac{x^3}{3!} \frac{d^3 n}{dx^3} \right) dx - \dots \\ &= d(nx) - d \left(\frac{x^2}{2!} \frac{dn}{dx} \right) + d \left(\frac{x^3}{3!} \frac{d^2 n}{dx^2} \right) - \dots \end{aligned}$$

Integrate termwise to obtain what is known as Bernoulli's series. Next, see if you can show what angered Bernoulli by deriving Taylor's series from Bernoulli's series. Incidentally, Bernoulli's assertion

was unfounded. The first known explicit statement of the general Taylor series was in *De Quadrature* (1691) by Isaac Newton.

57. Historical Quest

Colin Maclaurin was a mathematical prodigy. He entered the University of Glasgow at the age of 11. At 15 he took his master's degree and gave a remarkable public defense of his thesis on gravitational attraction. At 19 he was elected to the chair of mathematics at the Marischal College in Aberdeen and at 21 published his first important work, *Geometrica organica*. At 27 he became deputy, or assistant, to the professor of mathematics at the University of Edinburgh. There was some difficulty in obtaining a salary to cover his assistantship, and Newton offered to bear the cost personally so that the university could secure the services of such an outstanding young man. In time Maclaurin succeeded Newton and at the age of 44, published the first systematic exposition of Newton's work. In his work, Maclaurin uses what were called at the time fluxions, rather than derivatives. Here is what he concluded: "If the first fluxion of the ordinate, with its fluxions of several subsequent orders vanish, the ordinate is a minimum or maximum, when the number of all those fluxions that vanish is 1, 3, 5, or any odd number. The ordinate is a *minimum*, when the fluxion next to those that vanish is positive; but a *maximum* when the fluxion is negative . . . But if the number of all the fluxions of the first and successive orders that vanish be an even number, the ordinate is then neither a *maximum* nor



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Colin Maclaurin
(1698-1746)

minimum."* What is Maclaurin saying in this quotation?

Level 3

58. The functions $J_0(x)$ and $J_1(x)$ are defined as the following power series:

$$J_0(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(k!)^2 2^{2k}}$$

and

$$J_1(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{k!(k+1)! 2^{2k+1}}$$

- Show that $J_0(x)$ and $J_1(x)$ both converge for all x .
 - Show that $J_0'(x) = -J_1(x)$. These functions are called **Bessel functions of the first kind**. Bessel functions were first used in analyzing Kepler's laws of planetary motion. These functions play an important role in physics and engineering.
59. Show that J_0 , the Bessel function of the first kind defined in Problem 58, satisfies the differential equation

$$x^2 J_0''(x) + x J_0'(x) + x^2 J_0(x) = 0$$

60. **Journal Problem** (*The Pi Mu Epsilon Journal* by Robert C. Gebhardt)[†] For what x does

$$\sum_{n=0}^{\infty} \frac{x^n}{(2n)!}$$

converge, and what is the sum?

*The Historical Development of the Calculus by C. H. Edwards, Jr., Springer-Verlag, 1979, p. 292.

[†]Volume 8, 1984, Problem 586.

CHAPTER 8 REVIEW

There is nothing now which has ever given me any thought or care in algebra except divergent series, which I cannot follow the French in rejecting.

Augustus De Morgan
(New York: 1882-1889), p. 240

Proficiency Examination

Concept Problems

- What is a sequence?
- What is meant by the limit of a sequence?
- Define the convergence and divergence of a sequence.
- Explain each of the following terms:
 - bounded sequence
 - monotonic sequence
 - strictly monotonic sequence
- State the BMCT.
- What is an infinite series?
- Compare or contrast the convergence and divergence of sequences and series.
- What is a telescoping series?
- Describe the harmonic series. Does it converge or diverge?
- Define a geometric series and give its sum.
- State each indicated test.
 - divergence test
 - integral test
 - p -series test
 - direct comparison test
 - limit comparison test
 - zero-infinity limit comparison test
 - ratio test
 - root test
 - alternating series test
- How do you make an error estimate for an alternating series?
- State the absolute convergence test.
- What is meant by absolute and conditional convergence?
- What is the generalized ratio test?
- What is a power series?
 - How do you find the convergence set of a power series?
- What are the radius of convergence and interval of convergence for a power series?
- What is a Taylor polynomial?
 - State Taylor's theorem.
- What are a Taylor series and a Maclaurin series?
- State the binomial series theorem.

Practice Problems

- Find the limit of the sequence $\left\{ \left(1 + \frac{1}{n}\right)^n \right\}$.
- Find the limit of the sequence $\left\{ \frac{e^n}{n!} \right\}$.
 - Test the series $\sum_{k=1}^{\infty} \frac{e^k}{k!}$ for convergence.
 - Discuss the similarities and/or differences of parts **a** and **b**.

In Problems 23-26, test the given series for convergence.

- $\sum_{k=2}^{\infty} \frac{1}{k \ln k}$
- $\sum_{k=1}^{\infty} \frac{\pi^k k!}{k^k}$
- $\sum_{k=2}^{\infty} \frac{1}{(\ln k)^{1/k}}$
- $\sum_{k=1}^{\infty} \frac{3k^2 - k + 1}{(1 - 2k)k}$
- $1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} + \frac{1}{25} - \frac{1}{36} + \frac{1}{49} - \frac{1}{64} + \cdots + \frac{(-1)^{n+1}}{n^2} + \cdots$

28. Find the convergence set for the series

$$1 - 2u + 3u^2 - 4u^3 + \dots$$

29. Find the Maclaurin series for $f(x) = \sin 2x$.

30. Find the Taylor series for $f(x) = \frac{1}{x-3}$ at $c = \frac{1}{2}$.

Supplementary Problems*

Determine whether each sequence in Problems 1-15 converges or diverges. If it converges, find its limit.

1. $\left\{ \frac{(-2)^n}{n^2 + 1} \right\}$
2. $\left\{ \frac{(\ln n)^2}{\sqrt{n}} \right\}$
3. $\left\{ \left(1 - \frac{2}{n} \right)^n \right\}$
4. $\left\{ \frac{e^{0.1n}}{n^5 - 3n + 1} \right\}$
5. $\left\{ \frac{n + (-1)^n}{n} \right\}$
6. $\left\{ \frac{3^n}{3^n + 2^n} \right\}$
7. $\left\{ \frac{5n^4 - n^2 - 700}{3n^4 - 10n^2 + 1} \right\}$
8. $\left\{ \left(1 + \frac{e}{n} \right)^{2n} \right\}$
9. $\{ \sqrt{n+1} - \sqrt{n} \}$
10. $\{ \sqrt{n^4 + 2n^2} - n^2 \}$
11. $\left\{ \frac{\ln n}{n} \right\}$
12. $\{ 1 + (-1)^n \}$
13. $\left\{ \left(1 + \frac{4}{n} \right)^n \right\}$
14. $\{ 5^{2/n} \}$
15. $\left\{ \frac{n^{3/4} \sin n^2}{n + 4} \right\}$

16. Give an alternate proof that the harmonic series diverges by showing that $S_{2^n} > 1 + \frac{n}{2}$ for $n > 1$.

17. Find the sum $\sum_{k=-123,456,788}^{123,456,789} \frac{k}{370,370,367}$

Find the sum of the given convergent geometric or telescoping series in Problems 18-25.

18. $\sum_{k=1}^{\infty} 4 \left(\frac{2}{3} \right)^k$
19. $\sum_{k=1}^{\infty} \left(\frac{e}{3} \right)^k$
20. $\sum_{k=2}^{\infty} \frac{1}{k^2 - 1}$
21. $\sum_{k=1}^{\infty} \frac{1}{4k^2 - 1}$
22. $\sum_{k=0}^{\infty} \left[\left(-\frac{3}{8} \right)^k + \left(\frac{3}{4} \right)^{2k} \right]$
23. $\sum_{k=0}^{\infty} \frac{e^k + 3^{k-1}}{6^{k+1}}$
24. $\sum_{k=1}^{\infty} (-1)^{k+1} \left(\frac{1}{k} + \frac{1}{k+1} \right)$
25. $\sum_{k=1}^{\infty} \frac{k}{(k+1)(k+2)(k+3)}$

Test the series in Problems 26-45 for convergence.

26. $\sum_{k=1}^{\infty} \frac{5^k k!}{k^k}$
27. $\sum_{k=1}^{\infty} \frac{1}{\sqrt{k^2 + 4}}$
28. $\sum_{k=0}^{\infty} \frac{k!}{2^k}$
29. $\sum_{k=1}^{\infty} \frac{1}{2k-1}$
30. $\sum_{k=0}^{\infty} k e^{-k}$
31. $\sum_{k=1}^{\infty} \frac{k^3}{k!}$
32. $\sum_{k=1}^{\infty} \frac{7^k}{k^2}$
33. $\sum_{k=1}^{\infty} \frac{k}{(2k-1)!}$
34. $\sum_{k=0}^{\infty} \frac{k^2}{(k^3 + 1)^2}$
35. $\sum_{k=0}^{\infty} \frac{k^2}{3^k}$
36. $\sum_{k=2}^{\infty} \frac{1}{k(\ln k)^{1.1}}$
37. $\sum_{k=2}^{\infty} \frac{1}{k(\ln k)^2}$
38. $\sum_{k=0}^{\infty} (\sqrt{k^3 + 1} - \sqrt{k^3})$
39. $\sum_{k=1}^{\infty} \frac{1}{\sqrt{k(k+1)}}$
40. $\sum_{k=1}^{\infty} \frac{k-1}{k 2^k}$
41. $\sum_{k=1}^{\infty} \frac{k!}{k^2(k+1)^2}$
42. $\sum_{k=0}^{\infty} \frac{1}{1 + \sqrt{k}}$
43. $\sum_{k=1}^{\infty} \frac{k^2 3^k}{k!}$
44. $\sum_{k=1}^{\infty} \frac{k^2}{\sqrt{k^3 + 2}}$
45. $\sum_{k=1}^{\infty} \frac{k}{k^2 + 1}$

*The supplementary problems are presented in a somewhat random order, not necessarily in order of difficulty.

Determine whether each series given in Problems 46–55 converges conditionally, converges absolutely, or diverges.

46. $\frac{1}{1 \cdot 2} - \frac{1}{3 \cdot 2} + \frac{1}{5 \cdot 2} - \frac{1}{7 \cdot 2} + \cdots$

48. $\frac{3}{2} - \frac{4}{3} + \frac{5}{4} - \frac{6}{5} + \cdots$

50. $\frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \frac{1}{11} + \cdots$

52. $\sum_{k=1}^{\infty} \frac{(-1)^k}{k [\ln(k+1)]^2}$

54. $\sum_{k=1}^{\infty} (-1)^k \tan^{-1} \left(\frac{1}{2k+1} \right)$

47. $\frac{1}{1 \cdot 2} - \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} - \frac{1}{4 \cdot 5} + \cdots$

49. $1 - \frac{1}{3} + \frac{1}{9} - \frac{1}{27} + \frac{1}{81} - \cdots$

51. $-1 + \frac{1}{\sqrt{2}} - \frac{1}{\sqrt[3]{3}} + \frac{1}{\sqrt[4]{4}} - \cdots$

53. $\sum_{k=1}^{\infty} (-1)^k \left(\frac{3k+85}{4k+1} \right)^k$

55. $\sum_{k=1}^{\infty} \frac{(-1)^{k(k+1)/2}}{2^k}$

Find the interval of convergence for each power series in Problems 56–67.

56. $\sum_{k=1}^{\infty} \frac{kx^k}{3^k}$

57. $\sum_{k=1}^{\infty} k(x-1)^k$

58. $\sum_{k=1}^{\infty} \frac{x^k}{k(k+1)}$

59. $\sum_{k=1}^{\infty} \frac{\ln k (x^2)^k}{\sqrt{k}}$

60. $\sum_{k=1}^{\infty} \frac{k^2(x+1)^{2k}}{2^k}$

61. $\sum_{k=1}^{\infty} \frac{(-1)^k x^k}{k^k}$

62. $\sum_{k=1}^{\infty} \frac{(-1)^k (2x-1)^k}{k^2}$

63. $\sum_{k=1}^{\infty} \frac{(x+2)^k}{k \ln(k+1)}$

64. $\sum_{k=1}^{\infty} \frac{k(x-3)^k}{(k+3)!}$

65. $\sum_{k=1}^{\infty} \frac{(3k)!}{k!} x^k$

66. $\sum_{k=1}^{\infty} \frac{(-1)^k x^k}{9k^2 - 1}$

67. $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots + \frac{(-1)^{k+1} x^{2k-1}}{(2k-1)!} + \cdots$

Find the Maclaurin series for each function given in Problems 68–73.

68. $f(x) = x^2 e^{-3x}$

69. $f(x) = x^3 \sin x$

70. $f(x) = 3^x$

71. $f(x) = x^2 + \tan^{-1} x$

72. $f(x) = \frac{5+7x}{1+2x-3x^2}$

73. $f(x) = \frac{11x-1}{2+x-3x^2}$

74. Compute the sum of the series

$$1 - \frac{1}{2} + \frac{1}{2!} \left(\frac{1}{2} \right)^2 - \frac{1}{3!} \left(\frac{1}{2} \right)^3 + \cdots$$

correct to three decimal places.

75. Find a bound on the error incurred by approximating the sum of the series

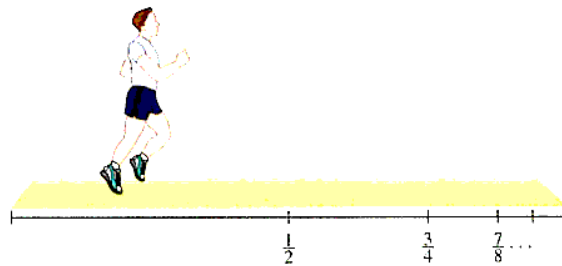
$$1 - \frac{1}{2!} + \frac{2}{3!} - \frac{3}{4!} + \cdots$$

by the sum of the first eight terms.

76. Use geometric series to write $12.342\overline{132}$ as a rational number.

77. Find the first three nonzero terms of the Maclaurin series for $\tan x$ and then use term-by-term integration to find the first three nonzero terms of the Maclaurin series for $\ln |\cos x|$.

78. **Historical Quest** The Greek philosopher Zeno of Elea (495–435 B.C.) presented a number of paradoxes that caused a great deal of concern to the mathematicians of his period. These paradoxes were important in the development of the notion of infinitesimals. Perhaps the most famous is the racecourse paradox, which can be stated as follows: A runner can never reach the end of a race: As he runs the track, he must first run $\frac{1}{2}$ the length of the track, then $\frac{1}{2}$ the remaining distance, so that at this point he has run $\frac{1}{2} + \frac{1}{4} = \frac{3}{4}$ of the length of the track, and $\frac{1}{4}$ remains to be run. After running half this distance, he finds he still has $\frac{1}{8}$ of the track to run, and so on, indefinitely.



Suppose the runner runs at a constant pace and that it takes him T minutes to run the first half of the track. Set up an infinite series that gives the *total* time required to run the track, and verify that the total time is $2T$, as intuition would lead us to expect.

79. Historical Quest

Ramanujan was that rarest of mathematicians, an instinctive genius with virtually no formal training. Like his Hindu predecessor, Bhaskara (see the **Historical Quest** in Problem 92 of the supplementary problems of Chapter 4), Ramanujan had an uncanny instinct for numerical “truth” and conceived of his results in much the same way that a sculptor “sees” a statue in a raw block of stone. In his short life, he initiated new ways of thinking about number theory and made conjectures that are the subject of mathematical inquiry to this day.

A story related by his friend and mentor, the eminent British number theorist G. H. Hardy (1877–1947), serves to illustrate the resonance between Ramanujan’s mind and the concept of number. Ramanujan was ill and Hardy came to visit him in the hospital. At a loss for how to begin the conversation, Hardy idly remarked that he had arrived in a taxi with the “dull” number 1729. Ramanujan immediately grew excited and exclaimed, “No, Hardy, it is a very interesting number, for it is the smallest integer that can be expressed as the sum of cubes in two different ways.”



Karl Smith library

Srinivasa Ramanujan
(1887–1920)

$$1729 = 1^3 + 12^3 = 9^3 + 10^3$$

In a letter from Ramanujan to G. H. Hardy at Cambridge University, Ramanujan stated*

$$1 - 5\left(\frac{1}{2}\right)^3 + 9\left(\frac{1 \cdot 3}{2 \cdot 4}\right)^3 - 13\left(\frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6}\right)^3 + \cdots = \frac{2}{\pi}$$

Hardy spent a great deal of time wondering how this sum could be equal to $2/\pi$. There is an interesting discussion of this identity, as well as biographical information on Ramanujan, in the article “Ramanujan the Phenomenon,” by S. G. Gindikin, *Quantum*, pp. 4–9, April 1998.

a. Find a_k so this sum can be expressed as $1 + \sum_{k=1}^{\infty} a_k$.

b. There is no elementary way to prove this identity. Use technology to check that this formula is valid.

- 80. Historical Quest** Leonhard Euler (1707–1783) was introduced in Chapter 4 Supplementary Problem 93, where we saw that Euler writes $a^\epsilon = 1 + k\epsilon$ for an “infinitely small” number ϵ . In this **Quest** we will see why we define

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$$

by asking you to reconstruct the steps of Euler’s work. Let x be a (finite) number. Euler introduced the “infinitely large” number $N = x/\epsilon$. Explain each step.

$$\begin{aligned} a^x &= a^{N\epsilon} \\ &= (a^\epsilon)^N \\ &= \left(1 + \frac{kx}{N}\right)^N \end{aligned}$$

*Quantum, March/April 1998.

$$\begin{aligned}
 &= 1 + N \left(\frac{kx}{N} \right) + \frac{N(N-1)}{2!} \left(\frac{kx}{N} \right)^2 + \frac{N(N-1)(N-2)}{3!} \left(\frac{kx}{N} \right)^3 + \cdots \\
 &= 1 + kx + \frac{1}{2!} \frac{N(N-1)}{N^2} k^2 x^2 + \frac{1}{3!} \frac{N(N-1)(N-2)}{N^3} k^3 x^3 + \cdots
 \end{aligned}$$

Because N is infinitely large, Euler assumed that

$$1 = \frac{N-1}{N} = \frac{N-2}{N} = \cdots$$

Substitute these values into the given derivation and also substitute e for a to obtain

$$e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots$$

Euler calculated e to 24 places:

$$e \approx 2.71828182845904523536028$$

Show that Euler's derivation is equivalent to $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$.

81. Prove that if $\sum a_k = A$ and $\sum b_k = B$, then $\sum (a_k + b_k) = A + B$.
82. **Think Tank Problem** Either prove or disprove (by finding a counterexample) each of the following statements:
- If $\sum a_k$ and $\sum b_k$ are both divergent series, then $\sum (a_k + b_k)$ is also divergent.
 - If $\sum a_k$ and $\sum b_k$ are both convergent series, then $\sum (a_k + b_k)$ is also convergent.
83. a. Show that $\sum_{k=1}^n (a_k - a_{k+2}) = (a_1 + a_2) - (a_{n+1} + a_{n+2})$
- b. Use part a to show that if $\lim_{k \rightarrow \infty} a_k = A$, then $\sum_{k=1}^{\infty} (a_k - a_{k+2}) = a_1 + a_2 - 2A$
- c. Use part b to evaluate $\sum_{k=1}^{\infty} [k^{1/k} - (k+2)^{1/(k+2)}]$
- d. Evaluate $\sum_{k=2}^{\infty} \frac{1}{k^2 - 1}$.
84. Find the Maclaurin series for $f(x) = \frac{2 \tan x}{1 + \tan^2 x}$
85. Show that

$$\cos x = \cos c - (x - c) \sin c - \frac{(x - c)^2}{2!} \cos c + \frac{(x - c)^3}{3!} \sin c + \cdots$$

86. Show that

$$e^x = e^c + e^c(x - c) + \frac{e^c(x - c)^2}{2!} + \frac{e^c(x - c)^3}{3!} + \cdots + \frac{e^c(x - c)^k}{k!} + \cdots$$

87. Express the integral $\int \sqrt{x^3 + 1} \, dx$ as an infinite series by writing the first four nonzero terms.
Hint: Use the binomial theorem and then integrate term by term.
88. Estimate the value of the integral

$$\int_0^{0.4} \frac{dx}{\sqrt{1+x^3}}$$

with five decimal-place accuracy.

89. Show that if $a > 0$, $a \neq 1$,

$$a^x = \sum_{k=0}^{\infty} \frac{(\ln a)^k}{k!} x^k \quad \text{for all } x$$

90. Show that for $0 < x < 2c$,

$$\frac{1}{x} = \frac{1}{c} - \frac{1}{c^2}(x-c) + \frac{1}{c^3}(x-c)^2 - \frac{1}{c^4}(x-c)^3 + \cdots$$

91. Use series to find $\sin 0.2$ correct to five decimal places.
 92. Use series to find $\ln 1.05$ correct to five decimal places.
 93. Use a Maclaurin series to find a cubic polynomial that approximates e^{2x} for $|x| < 0.001$. Find an upper bound for the error in this approximation.
 94. For what values of x can $\sin x$ be replaced by $x - \frac{x^3}{6}$ if the allowable error is 0.005?
 95. For what values of x can $\cos x$ be replaced by $1 - \frac{x^2}{2}$ if the allowable error is 0.00005?
 96. **Modeling Problem** When we studied carbon dating in Section 5.6, it was pointed out that radiocarbon methods apply only to objects that are not too “old” (roughly 40,000 years). To date older objects, radioactive isotopes other than ^{14}C are often employed. Regardless of the radioactive substance used, it can be shown that the ratio of stable isotope to radioactive isotope at any given time may be modeled by

$$\frac{S(t) - S(0)}{R(t)} + 1 = e^{(\ln 2)t/\lambda}$$

where $R(t)$ is the number of atoms of radioactive material at time t , $S(t)$ is the number of atoms of the stable product of radioactive decay, $S(0)$ is the number of atoms of stable product initially present (at $t = 0$), and λ is the half-life of the radioactive isotope.*

- a. Use the first two terms of the Maclaurin series for e^x to approximate $e^{(\ln 2)t/\lambda}$, and then solve the modeling equation for t .
 b. Suppose a piece of mica is analyzed, and it is found that 5% of the atoms in the rock are radioactive rubidium-87 and 0.04% of the atoms are strontium-87. Assuming that the strontium was produced by decay of the rubidium-87 originally in the rock, what is the approximate age of the sample? Use the result in part a, with $\lambda = 4.86 \times 10^9$ years as the half-life of rubidium-87.
 97. **Putnam Examination Problem** Express

$$\sum_{k=1}^{\infty} \frac{6^k}{(3^{k+1} - 2^{k+1})(3^k - 2^k)}$$

as a rational number.

98. **Putnam Examination Problem** Find the sum of the convergent alternating series

$$\sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{3n-2}$$

99. **Putnam Examination Problem** For positive real x , let

$$B_n(x) = 1^x + 2^x + 3^x + \cdots + n^x$$

Test for convergence of the series

$$\sum_{k=2}^{\infty} \frac{B_k(\log_k 2)}{(k \log_2 k)^2}$$

*This model is discussed in detail in the article, “How Old Is the Earth?” Paul J. Campbell, *UMAP Modules 1992: Tools for Teaching*, Lexington, MA: Consortium for Mathematics and Its Applications, Inc. 1993, pp. 105-137.

CHAPTER 8 GROUP RESEARCH PROJECT*

Working in small groups is typical of most work environments, and this book seeks to develop skills with group activities. We present a group project at the end of each chapter. These projects are to be done in groups of three or four students.

Elastic Tightrope Project



Suppose a squirrel starts at one end of a 100-m tightrope and runs toward the other end at a speed of 1 m/s. At the end of every minute, the tightrope stretches uniformly and instantaneously, increasing its length by 100 meters each time. Thus, after the first minute, the critter has traveled 60 meters and the length of the tightrope is 200 m. After the second minute, the squirrel has traveled another 60 meters and the tightrope has become 300 m long, and so on.

Does the squirrel ever reach the other end? If so, how long does it take?

Extended paper for further study. Your paper is not limited to the following questions but should include these concerns. Suppose it is you standing on the elastic rope b meters from the one end and c meters from the other end. Then suppose the entire rope stretches uniformly, increasing its length by d meters. Only the tightrope stretches; units of length and time remain constant. How far are you from each end? You might also try to generalize the tightrope problem.

*Adapted from a group project by David Pengelley, New Mexico State University, Las Cruces.

Cumulative Review Problems—Chapters 6-8

- What does this say? Describe a procedure for using integration to
 - compute work done by a variable force
 - fluid (or hydrostatic) force
- What does this say? Describe integration by parts.
- What does this say? Describe the method of partial fractions.
- What does this say? Describe a geometric series. For what values does such a series converge? What is the sum of a convergent geometric series?
- What does this say? Describe a procedure for determining the convergence set of the power series $\sum a_k x^k$.
- What does this say? Outline a procedure for determining the convergence of $\sum a_k$ for $a_k > 0$.

Evaluate each of the integrals in Problems 7-20.

- | | | | |
|--|---|---|--|
| 7. $\int \ln \sqrt{x} \, dx$ | 8. $\int \sin^2 x \cos^3 x \, dx$ | 9. $\int \frac{\tan^{-1} x}{1+x^2} \, dx$ | 10. $\int \sec^{3/2} x \tan x \, dx$ |
| 11. $\int \sqrt{4-x^2} \, dx$ | 12. $\int \frac{1-x^2}{1+x^2} \, dx$ | 13. $\int \frac{(3+2\sin t)}{\cos t} \, dt$ | 14. $\int \frac{e^{1-\sqrt{x}} \, dx}{\sqrt{x}}$ |
| 15. $\int \frac{dx}{x^2+x+1}$ | 16. $\int x^3 \sin x^2 \, dx$ | 17. $\int \cos^{-1}(-x) \, dx$ | 18. $\int \frac{(2x+1) \, dx}{\sqrt{4x-x^2-2}}$ |
| 19. $\int \frac{e^{-x} - e^x}{e^{2x} + e^{-2x} + 2} \, dx$ | 20. $\int \frac{dx}{(x+\frac{1}{2})\sqrt{4x^2+4x}}$ | | |

In Problems 21-24, either evaluate the given improper integral or show that it diverges.

- | | | | |
|-------------------------------------|-----------------------------------|--------------------------------------|--|
| 21. $\int_0^\infty x e^{-2x} \, dx$ | 22. $\int_0^{\pi/2} \sec x \, dx$ | 23. $\int_0^\infty x^2 e^{-x} \, dx$ | 24. $\int_0^\infty x^3 e^{-x^2} \, dx$ |
|-------------------------------------|-----------------------------------|--------------------------------------|--|
25. Let $f(x) = \frac{x^2 + 17x - 8}{(2x+3)(x-1)^2}$
- Express $f(x)$ as a partial fraction sum.
 - Find $\int f(x) \, dx$.
 - Either evaluate the improper integral $\int_2^\infty f(x) \, dx$ or show it diverges.

26. Evaluate $\lim_{x \rightarrow \infty} \frac{\sinh^{-1} x}{\cosh^{-1} x}$

In Problems 27-32, test the given series for convergence.

- | | | |
|--|---|--|
| 27. $\sum_{k=0}^\infty \frac{k^2 3^{k+1}}{4^k}$ | 28. $\sum_{k=1}^\infty \frac{k+3k^{3/2}}{5k^2-k+3}$ | 29. $\sum_{k=1}^\infty \frac{(-1)^{k+1} k}{k+1}$ |
| 30. $\sum_{k=1}^\infty \left(\frac{2k+5}{3k-2} \right)^k$ | 31. $\sum_{k=1}^\infty k e^{-k^2}$ | 32. $\sum_{k=1}^\infty \frac{k^2}{k!}$ |

In Problems 33 and 34, find the sum of the given convergent series.

- | | |
|--|--|
| 33. $\sum_{k=1}^\infty 5 \left(\frac{-2}{3} \right)^{2k-1}$ | 34. $\sum_{k=1}^\infty \frac{2}{(k+1)(k+3)}$ |
|--|--|

In Problems 35 and 36, find the convergence set of the given power series.

- | | |
|---|--|
| 35. $\sum_{k=1}^\infty \frac{x^k}{k^3}$ | 36. $\sum_{k=1}^\infty \frac{(-x)^k}{k+1}$ |
|---|--|

In Problems 37 and 38, solve each of the first-order linear differential equations.

37. $y' - \frac{y}{x} = \frac{x}{1+x^2}$

38. $y' + (\tan x)y = \tan x$, where $y = 3$ when $x = 0$

Find the Maclaurin series for the functions in Problems 39–41.

39. $f(x) = \sin x^2$

40. $f(x) = \sqrt{e^x}$

41. $f(x) = \frac{3x-2}{5+2x}$

42. A triangular fin of a manta ray submerged in seawater ($\rho g = 64.0 \text{ lb/ft}^3$) to a depth of 12 ft, as shown in Figure 8.27 (that is, the top of the fin is 12 ft below the surface).

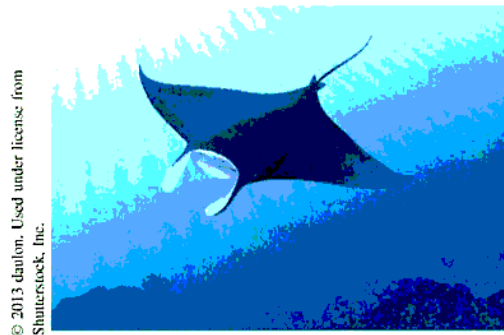


Figure 8.28 Submerged manta ray

Find the fluid force on the triangular fin if we assume the sides of the fin are 5 ft and the base is 6 ft.

43. A tank has the shape of a circular cone with height 10 ft and top radius 4 ft. If the tank is filled with water ($\rho g = 62.4 \text{ lb/ft}^3$) to a depth of 6 ft, how much work is done (rounded to the nearest ft-lb) in pumping all the water in the tank to a height 2 ft above the top?
44. Find the arc length of the curve $y = \frac{3}{2}x^{2/3}$ on the interval $[\frac{1}{8}, 8]$.
45. Let R be the region bounded by the x -axis, the curve $y = \frac{1}{\sqrt{x+1}}$, $x = 3$, and $x = 8$. Find the volume of the solid generated when R is revolved about:
- the x -axis
 - the y -axis
 - the line $y = -1$
46. Find both the exact and approximate volume of the solid generated when the region under the curve

$$y^2 = x^2 \left(\frac{4-x}{4+x} \right)$$

on the interval $[0, 4]$ is revolved about the x -axis.

47. Find the area of the region bounded by the graphs of

$$y = \frac{2x}{\sqrt{x^2 + 9}}$$

and $y = 0$ from $x = 0$ to $x = 4$.

48. Evaluate

$$\int \sqrt{1 + \cos x} \, dx$$

Hint: Use the identity

$$\cos x = 2 \cos^2 \frac{x}{2} - 1$$

49. Show that

$$\sum_{k=0}^{\infty} \frac{1}{(a+k)(a+k+1)} = \frac{1}{a}$$

if $a > 0$.

50. Evaluate $\sum_{k=0}^{\infty} \frac{1}{(2+k)^2 + 2 + k}$.
51. If $\sum_{k=0}^{\infty} a_k^2 = \sum_{k=0}^{\infty} b_k^2 = 4$ and $\sum_{k=0}^{\infty} a_k b_k = 3$, what is $\sum_{k=0}^{\infty} (a_k - b_k)^2$?
52. **Think Tank Problem** Let $\sum a_k^2$ be a series that converges. Either prove that $\sum a_k$ must diverge or find a counterexample.
53. **Think Tank Problem** Let $\sum a_k^2$ be a series that converges. Either prove that $\sum a_k$ must also converge or find a counterexample.
54. Find the average value (to the nearest hundredth) of the function $f(x) = x \sin^3 x^2$ between $x = 0$ and $x = 1$.
55. An object moves along the x -axis in such a way that its velocity at time t is $v(t) = \sin t + \sin^2 t \cos^3 t$. Find the distance moved by the object between times $t = 0$ and $t = \frac{\pi}{3}$.
56. A ball is dropped from a height of 20 ft. Each time it bounces, it rises to $\frac{3}{4}$ of its previous height. What is the total distance traveled by the ball?
57. What is the volume of the solid obtained when the region bounded by $y = x\sqrt{9-x^2}$ and $y = 0$ is revolved about the x -axis?
58. Find the length of the curve $y = \ln(\sec x)$ from $x = 0$ to $x = \frac{\pi}{4}$.
59. Find the centroid (to the nearest hundredth) of the region bounded by the curve $y = x^2 e^{-x}$ and the x -axis, between $x = 0$ and $x = 1$.
60. A tank contains 15 lb of salt dissolved in 70 gallons of water. Suppose four gallons of brine containing 3 lb/gal of salt run into the tank every minute and the mixture (kept uniform by stirring) runs out of the tank at the rate of 2 gal/min.
- Set up and solve a differential equation for the amount of salt $S(t)$ in the tank at time t .
 - Suppose the capacity of the tank is 100 gal. How much salt is in the tank at the time it just begins to overflow?